



Affirmed Networks System Administration Guide

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Inside ...

- Initial system settings
- Managing user accounts
- Managing the system
- System security, IPsec, Security policy
- Managing clusters and nodes
- IP protocols and routing

Affirmed Networks Inc.

35 Nagog Park
Acton, MA 01720
USA

Tel: +1-978-268-0800

www.affirmednetworks.com

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Preface

This guide describes an overview of the CLI objects and commands required for the initial system setting, managing user accounts, managing the system, and managing clusters and nodes. This guides also describes the CLI objects and commands for configuring system security and IP protocols.

In this guide, objects and commands are listed in alphabetical order.

For information on accessing the Command Line Interface (CLI), see the *Affirmed Networks CLI Reference Guide*. For information on statistics, see *Acuitas Performance Statistics*.

Note: *The Affirmed Networks Mobile Content Cloud™ (MCC) is the mobile services software solution that operates on virtual platforms such as blade systems.*

Audience

This guide is intended for network operators (sometimes referred to as Users or Operators in this guide), installation team staff, and other qualified service personnel responsible for provisioning and monitoring the system. The audience should be familiar with Long Term Evolution (LTE), 3G, or 4G networks.

Release Information

See the Release Notes for important information, requirements, limitations, and restrictions that apply to this release.

How to Use This Guide

Read the following chapters and appendices in the order in which they appear:

Chapter/Appendix	Describes
Chapter 1, Initial System Settings	Initial systems settings for the MCC such as running the initialization script and ping.
Chapter 2, User Account Management	User account management objects and commands such as user-access, password settings, and tacacs-plus.
Chapter 3, Managing the System	The objects and commands used to manage the system such as restore-config, image-management, online-upgrade, and logging.
Chapter 4, System Security	The system security objects and commands such as policy tunnels and IPsec.
Chapter 5, Managing Clusters and Nodes	The cluster configuration objects and CLI commands used for virtual node management such as building-block-strategy, personalities, and DPI settings.
Chapter 6, IP Protocols and Routing	The BGP and BFD IP protocols and routing objects and commands.
Chapter 7, Monitoring the System	How to maintain and monitor the system using commands such as save-tech-support-info and file-management.

Features in this Release

This release includes the following new features:

Note: For a complete list of the features supported in this release, see the software release notes. In addition to the information in this guide, see the additional Affirmed Networks documentation listed in the Affirmed Networks Documentation Library. For a comprehensive list of terms and acronyms used in Affirmed Networks documentation, see Affirmed Networks Terms and Acronyms.

Feature	Description																																																								
Added user account password security improvements. See Password Policies (page 2-3) .	Added user account password security improvements in this release. Use the following command to configure password policies: <pre>an3000-mcm-slot7cpu1# user-access password-policies</pre> Use the following command to display the current password policies by domain: <pre>an3000-mcm-slot7cpu1# show user-access password-policies default domain policy</pre> <table><tr><td>Numeric Complexity</td><td>: 0</td><td>range 0 - 6</td><td></td></tr><tr><td>Uppercase Complexity</td><td>: 1</td><td>range 0 - 6</td><td></td></tr><tr><td>Lowercase Complexity</td><td>: 1</td><td>range 0 - 6</td><td></td></tr><tr><td>Non-alphanumeric Complexity</td><td>: 1</td><td>range 0 - 6</td><td></td></tr><tr><td>Limit on sequential characters</td><td>: 0</td><td>range 0 - 64</td><td></td></tr><tr><td>Limit on repeated characters</td><td>: 0</td><td>range 0 - 64</td><td></td></tr><tr><td>Allow username in password</td><td>: 1</td><td>range 0 - 1</td><td>[0=no 1=yes]</td></tr><tr><td>Minimum Password Length</td><td>: 8</td><td>range 0 - 32</td><td></td></tr><tr><td>Password Dictionary</td><td>: 0</td><td>range 0 - 2</td><td>[0=no 1=default 2=custom]</td></tr><tr><td>Previous Password History</td><td>: 3</td><td>range 0 - 30</td><td></td></tr><tr><td>Lockout Failure Limit</td><td>: 5</td><td>range 0 - 6</td><td></td></tr><tr><td>Lockout Duration</td><td>: 10</td><td>range 0 - 60</td><td>[minutes]</td></tr><tr><td>Minimum Password Age</td><td>: 0</td><td>range 0 - 7</td><td>[day(s)]</td></tr><tr><td>Maximum Password Age</td><td>: 0</td><td>range 0 - 365</td><td>[day(s)]</td></tr></table> admin domain policy (unset values use defaults)	Numeric Complexity	: 0	range 0 - 6		Uppercase Complexity	: 1	range 0 - 6		Lowercase Complexity	: 1	range 0 - 6		Non-alphanumeric Complexity	: 1	range 0 - 6		Limit on sequential characters	: 0	range 0 - 64		Limit on repeated characters	: 0	range 0 - 64		Allow username in password	: 1	range 0 - 1	[0=no 1=yes]	Minimum Password Length	: 8	range 0 - 32		Password Dictionary	: 0	range 0 - 2	[0=no 1=default 2=custom]	Previous Password History	: 3	range 0 - 30		Lockout Failure Limit	: 5	range 0 - 6		Lockout Duration	: 10	range 0 - 60	[minutes]	Minimum Password Age	: 0	range 0 - 7	[day(s)]	Maximum Password Age	: 0	range 0 - 365	[day(s)]
Numeric Complexity	: 0	range 0 - 6																																																							
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Lowercase Complexity	: 1	range 0 - 6																																																							
Non-alphanumeric Complexity	: 1	range 0 - 6																																																							
Limit on sequential characters	: 0	range 0 - 64																																																							
Limit on repeated characters	: 0	range 0 - 64																																																							
Allow username in password	: 1	range 0 - 1	[0=no 1=yes]																																																						
Minimum Password Length	: 8	range 0 - 32																																																							
Password Dictionary	: 0	range 0 - 2	[0=no 1=default 2=custom]																																																						
Previous Password History	: 3	range 0 - 30																																																							
Lockout Failure Limit	: 5	range 0 - 6																																																							
Lockout Duration	: 10	range 0 - 60	[minutes]																																																						
Minimum Password Age	: 0	range 0 - 7	[day(s)]																																																						
Maximum Password Age	: 0	range 0 - 365	[day(s)]																																																						
Increase MTU and MRU on Sgi to 2000 bytes	Increase MTU and MRU on SGI to 2000 bytes to help with IPv6 fragmentation issues.																																																								

Feature	Description
Updated DPI script to validate user-defined signatures and use image management techniques to load the packages dynamically See data-path dpi (page 5-19) and the <i>Affirmed Networks DPI Script User Guide</i>.	The DPI script enables a Network Operator to add or update classification support for applications and protocols without requiring an MCC software update or restart. The DPI script validates user-defined signatures and increases system performance by using image management techniques to load the packages dynamically. Configure the dpisript parameters as follows: <pre>an3000-mcm-slot7cpu1(config)# data-path dpi dpiscript</pre> ■ script-classification – Set to enabled or disabled (default). In the disabled state, user-defined signatures are loaded on the system but the classification/detection phase does not consider these signatures. ■ http-classification – Set to enabled or disabled for dpiscript http support. This parameter takes effect only when script-classification is enabled. ■ https-classification – Set to enabled or disabled for dpiscript https support. This parameter takes effect only when script-classification is enabled. See the <i>Affirmed Networks DPI Script User Guide</i> for information on configuring and creating a DPI script and collecting and analyzing pcaps.
Added the ISM VMNAME on which the session has landed See node (page 5-12).	Use the instructions in node (page 5-12) to configure the vm-name. If the vm-name is not configured, MCC uses a generic name. For example: <pre>an3000-mcm-slot7cpu1(config)# cluster <id> node <id> type vm-name</pre> This feature also adds the ISM VMNAME in the header. For example: <pre>an3000-mcm-slot7cpu1(config)# service-construct http-header-enrichment name request add-header header-string name value</pre> Possible completions: <pre><String, may include variable such as IMSI, MSISDN, HTTPVERSION, UEIPADDR, ORIGINALAPN, IMEI, ULICGI, RATTYPER, GWIPADDR, PLMNID, CHARGINGCHAR, TIMESTAMP, SUBSCRIBERID, VMNAME, example: \$(MSISDN)></pre>
Added enhancements to the existing global firewall function. See infrastructure logging sys-logs (page 3-29).	This feature adds the ability to provide syslog for the following scenarios: ■ Summary of count of connections dropped per session due to incoming connections exceeding the max-connections threshold configured in workflow data-profile. ■ Summary of count of packets dropped due to malformed IP/TCP header. ■ Summary of count of flows dropped due to flow rate exceeding flow-rate-limit threshold configured in security policy rule. ■ Summary of count of packets dropped due to packet rate exceeding the packet-rate-limit threshold configured in security policy rule. The following limitations apply to this feature: ■ Upon SSM crash, firewall records that were not sent to the MCM are lost. ■ Upon MCM crash, firewall records that were not sent to the syslog server are lost.

Feature	Description
Added a configurable max-packet-size parameter to the cluster configuration. See Table 5-2 on page 5-5. Also see the <i>Affirmed Networks Gateway Administration Guide</i>.	Added a configurable max-packet-size parameter to the cluster configuration to support an increase in the MTU/MRU value for all IP interfaces. The default is 2048. This value must be at least as large as the MTU/MRU value you want to use for the IP interfaces. You must reboot the cluster after modifying this value. <pre>an3000-mcm-slot7cpu1(config)# cluster <id> max-packet-size (<unsignedInt, 2048 .. 9216>):</pre> Previously, the maximum MTU/MRU value was 1800 bytes. With this release, MCC supports an MTU/MRU as large as 9100 bytes. The feature allows Operators to configure the MTU/MRU according to network requirements and enables the MCC to support IP packets greater than 1800 bytes without triggering IP fragmentation. After configuring the max-packet-size value, configure the interface MTU/MRU value. The default is 1500. See the <i>Affirmed Networks Gateway Administration Guide</i> for details. <pre>an3000-mcm-slot7cpu1(config)# network-context <name> ip-interface <name> mtu (<unsignedInt, 576 .. 9100>):</pre>
Allow Martian routes to be advertised by BGP See ip-protocols bgp router (page 6-12).	Martian addresses are addresses reserved by IANA for special use. The following addresses are reserved for private networks and treated as Martian routes. 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16, 169.254.0.0/16 This feature allows BGP to advertise Martian routes created for UE Pools and loopbacks. You can use route maps to suppress Martian routes.
Support for configurable Tethering Quality Index (QI) See data-path dpi (page 5-19).	In previous MCC releases, the Quality Index that rates the confidence level of DPI-detected tethering used hard coded values. This feature supports configurable values. A connection is marked as tethered only if the reported Quality Index (Confidence Index) is greater than or equal to the detection algorithm QI value configured using the following command: <pre>data-path dpi tethering-quality-index value <QI> [default = 70]</pre>

Feature	Description
Enhanced tethering detection logic using GQUIC user agent fingerprinting See data-path dpi (page 5-19).	Enhanced tethering detection logic using GQUIC user agent fingerprinting. Added the following parameter to enable running the tethering detection algorithm using GQUIC user agent fingerprinting: <pre>an3000-mcm-slot7cpu1(config)# data-path dpi tethering-detection-algorithm gquic-user-agent <enable disable></pre> By default, this parameter is disabled. Changes to this setting take effect upon next reboot. Added the following parameter to configure the weight percentage for the GQUIC user agent quality index: <pre>an3000-mcm-slot7cpu1(config)# data-path dpi tethering-quality-index-result-weight percentage gquic-user-agent <percentage></pre> Added the following statistics to indicate the number of tethered flows detected by GQUIC user agent fingerprinting: <pre>an3000-mcm-slot7cpu1# show subscriber pdn-session <session-id> an3000-mcm-slot7cpu1# show zone default deep-packet-inspection statistics grid-level tethering <id> an3000-mcm-slot7cpu1# show zone default gateway statistics [cluster-level grid-level node-level] call-performance apn <id> an3000-mcm-slot7cpu1# show zone default gateway statistics [cluster-level grid-level node-level] call-performance pgw <id></pre> tethered-flows-detected-using-gquic-user-agent In the FLOW EDR, added the GQUIC user agent value to the tetheringAlgorithm field: ■ tetheringAlgorithm <string> {ttl, userAgent, tcpipfingerprinting, gquic-user-agent}

Conventions

This guide uses the following conventions, when applicable:

Description	Convention	Examples
Command names, keywords, dialog boxes, buttons, icons, menus	Times New Roman bold font	Use configure_ha to configure... Enter y . To continue, press Enter .
Selecting a menu item	Menu => Option	Select Security => Group Management .
Variable parameters for which you supply values	<i><courier bold italic blue font></i>	configure_dr convert slave --masterhost <host IP>
User input	Courier bold blue font	cd /opt/Affirmed/NMS/bin
Screen messages, system output, sample source code	Courier regular blue font	Checking for License...
Required keywords and arguments in square brackets	[]	cluster [id] node [id] reboot
Optional keywords and arguments in curly brackets	{ }	{cdr-file-name}
Keywords separated by the pipe symbol. Requires you to specify only one item from the set.	 	Is this correct? [y n]
Keywords representing a single element in angle brackets	< >	<host name>
Asterisk after a CLI object indicates a required object.	*	name*
Variables, book and chapter titles, file path names, new terms, and emphasized text	<i>Times New Roman italic font</i>	See the <i>Acuitas User's Guide</i> .

Tip: Helpful suggestions.

Note: Additional information that may apply to the subject text.



CAUTION: Proceed carefully to avoid possible equipment damage or data loss.

How to Obtain Product Documentation

Documentation is available for currently supported product releases. Documentation is available online in Adobe PDF format. You can view PDFs online using the Adobe Reader[®]. To download the latest version of the Adobe Reader software from the Adobe web site, go to <http://get.adobe.com/reader/>

Product documentation is available from the Affirmed Networks SFTP server. Contact your sales representative or the TAC for instructions on accessing the SFTP server.

See the *Affirmed Networks Documentation Library* for a complete list of Affirmed Networks product documentation.

Technical Assistance Center

Affirmed Networks Technical Assistance Center (TAC) provides technical support 24 hours a day, 7 days a week (24 x 7) for any customer with an maintenance contract.

To contact the TAC:



E-mail: prodsupport@affirmednetworks.com



24 x 7 Telephone support:

Local: 1-978-268-0871

Toll free: 1-888-283-2838

Initial System Settings

This sections describes the initial systems settings for the MCC.

Before you begin, complete the steps in [Running the Installation and Initialization Script \(page 1-1\)](#). Affirmed Networks recommends setting the default idle timeout and paginate settings. Setting the prompt is optional.

Running the Installation and Initialization Script

After you power on the system and establish a console connection, the pre-commission prompt appears. Run the installation and initialization script. See the *Affirmed Networks CLI Reference Guide* for details on establishing a console connection.

Before you begin, record values for the following information. The system prompts you for this information during the script.

Prompt	Value
Local management IP address for MCM 7	
Primary management IP address for MCM 7	
Primary management prefix for MCM 7	
Primary management VLAN identifier for MCM 7	
Primary management gateway IP address for MCM 7	
Primary management IP address for MCM 8	
Primary management prefix for MCM 8	
Primary management VLAN identifier for MCM 8	
Primary management gateway IP address for MCM 8	

Example Output

The following example shows the sample output for the installation and initialization script.

Note: After you enter the required information, data is pushed to the peer MCM20 and the system reboots. The CLI login prompt appears on the console on the primary MCM20. Enter your **username** and **password**.

Note: The Admin user can create additional user accounts.

Note: In the event that you need to change the commissioning parameters, see rollback in [Table 3-1 on page 3-2](#). After using the rollback command, the system prompts you to reboot. Contact the TAC prior to performing a rollback.



CAUTION: The following procedures should only be performed by the TAC or an authorized Affirmed representative.

If the CLI is not accessible and the system does not appear to be starting after an upgrade, Affirmed Networks recommends rolling back to the previous image as follows:

- 1 Log in to both MCM consoles as the **maint** user.
- 2 Issue the **tos stop** command on both systems.
- 3 Run the `/user/bin/rollback.py` script on both systems and follow the instructions.

If the CLI does not recover after performing the rollback procedure, contact the TAC.

```

=====
***AN3000 Install and Initialization***
=====

***AN3000 Install and Initialization***

Current timezone is : UTC
Current system time is: Mon Sep 28 16:00:57 2015

Do you want to modify the tinezone and time? <y/N> N

Enter logical management IP (aaa.bbb.ccc.ddd):
10.192.2.200

Enter primary management IP for MCM-7 (aaa.bbb.ccc.ddd):
10.192.2.207

Enter primary management prefix for MCM-7 (example 24):
20

Enter primary management VLAN identifier for MCM-7 (0 if none):
0

Enter management gateway IP for MCM-7 (aaa.bbb.ccc.ddd, 0 if none):
10.192.0.1

Enter primary management IP for MCM-8 (aaa.bbb.ccc.ddd):
10.192.2.208

Enter primary management prefix for MCM-8 (example 24):
10.192.0.1
Invalid prefix. Please re-enter.

Enter primary management prefix for MCM-8 (example 24):
20

Enter primary management VLAN identifier for MCM-8 (0 if none):
0

Enter management gateway IP for MCM-8 (aaa.bbb.ccc.ddd, 0 if none):
10.192.0.1

Enable password aging? <Y/n> Y

Enable root user ssh access (strongly recommend no)? <y/N> N

Would you like to generate new ssh host keys (recommend yes)? <Y/n> Y
ssh-keygen: generating new host keys: RSA1 RSA DSA ECDSA

Enable multiple simultaneous logins per username? <y/N> 09/28/15 16:01:48.37: mcmremote.py: slot 7 cpu 1: local security files
changed - relinked to /tosdb/linux l'/ssh/ssh_host_key.pub', '/ssh/ssh_host_key', '/ssh/ssh_host_rsa_key.pub', '/ssh/ssh_host_rs
_key', '/ssh/ssh_host_ecdsa_key.pub', '/ssh/ssh_host_ecdsa_key', '/ssh/ssh_host_dsa_key.pub', '/ssh/ssh_host_dsa_key' }
N

PROCESSING CONFIG
Building /etc/platform/an3000.conf...
Building XML files
Building networkcontext.xml with mgmtipintf data ...
Updating /tosdb/conf/cdb/networkcontext.xml...
Building chassis.xml ...
Perform Upgrade Cleanup ...

Broadcast Message from root@an3000-m
(/dev/pts/1) at 16:01 ...

Rebooting in 5sec...

Building Mgmt IP script programMgmtIp.sh...
UpdateOnly is not set
Updating XML init files and clearing CDB
Rebooting chassis

Broadcast Message from root@an3000-m
(/dev/pts/1) at 16:01 ...

Rebooting in 5sec...

Broadcast Message from root@an3000-m
(/dev/pts/1) at 16:01 ...

Rebooting in 4sec...

```

```
Broadcast Message from root@an3000-n
(/dev/pts/1) at 16:01 ...

Rebooting in 3sec...

Broadcast Message from root@an3000-n
(/dev/pts/1) at 16:01 ...

Rebooting in 3sec...

Broadcast Message from root@an3000-n
(/dev/pts/1) at 16:01 ...

Rebooting in 2sec...

Broadcast Message from root@an3000-n
(/dev/pts/1) at 16:01 ...

Rebooting in 2sec...

Broadcast Message from root@an3000-n
(/dev/pts/1) at 16:01 ...

Rebooting in 1sec...

Broadcast Message from root@an3000-n
(/dev/pts/1) at 16:01 ...

Rebooting in 1sec...

Rebooting peer

The system is going down for reboot NOW!lot7cpu1 (pts/1) (Mon Sep 28 16:02:01
16:02:01.93 failed: ssh: connect to host 169.254.1.81 port 2020: No route to host
Rebooting local
INIT: Switching to runlevel: 6
INIT: Sending processes the TERM signal
```

date

Configures the system date and time. This command is executed in config mode.

license

Use the following CLI commands to activate and configure licensing. Enter the cluster IP address and enter the IP address of the managing device.

```
an3000-mcm-slot7cpu1# license
```

Possible completions:

```
add-authorized-mcc Add an MCC to the PLS authorized list
```

```
add-certificate      Add License Certificate
```

```
get-pls-fingerprint  Get the PLS fingerprint
```

```
management-discovery Force management discovery of PLS
```

```
restart-pls          Restart the PLS
```

```
revoke-certificate Revoke License Certificate
```

Use the following CLI commands to add the IP address of the PLS server and configure redundancy mode. This command is executed in config mode.

```
an3000-mcm-slot7cpu1(config)# license server ?
```

Possible completions:

IP Address of PLS Server

```
an3000-mcm-slot7cpu1(config-server-<ipaddress># ?
```

Possible completions:

```
port                Port used by PLS Server
```

```
redundancy-mode     PLS Redundancy Mode /Backup
```

Use the following CLI commands to display licensing information:

```
an3000-mcm-slot7cpu1# show license
```

Possible completions:

```
lm-status           Display the licensable services status
```

```
pls-status          Display the PLS status
```

```
services-status     Display the licensable services status
```

```
| Output modifiers
```

```
<cr>
```

Licensing Overview

The Affirmed Networks License Management System (LMS) contains three functional entities:

- Entitlement Generation System (EGS)
- Pooling Licensing Server (PLS)
- Affirmed Networks application software to support license enforcement and licensing features.

Affirmed Networks uses a third-party software application (SafeNet/Gemalto) as the EGS to generate entitlements on a per-customer, deployment, or NE basis. The PLS consists of the SafeNet server application running in a VM or bare metal deployment. The PLS manages service licenses for one or more MCCs.

The licensing feature requires you to install the PLS and configure a disaster recovery PLS. The disaster recovery PLS contains temporary licenses and acts as a standby server in the event that the primary server is not available prior to the grace period expiry.

MCC supports binary licenses which are configured as either On (enabled) or Off (disabled) for each feature or function requiring a license.

MCC uses a Time Enforcement Threshold (TE) to enforce all overage limits and issues an alarm or log when a limit is reached. The TE is set at 150%.

License certificates and revocation are generated from Affirmed Networks order administration and applied to the PLS. Contact the TAC for details.

See the *Affirmed Networks Pooling License Certificate Generation and Revocation Guide*.

Licensing Statistics

The Affirmed Networks statistics director collects statistics associated with all binary and metric-based data to determine overage and trigger enforcement actions. See *Acuitas Performance Statistics*.

paginate

Paginate the output of the CLI commands. If you set the paginate parameter to **false**, the system does not display -- More -- at the bottom of the window. Instead, the system streams the output.

For example:

```
an3000-mcm-slot7cpu1# paginate
Possible completions:
false true
an3000-mcm-slot7cpu1# paginate true
Possible completions:
| <cr>
```

ping

Test IPv4 connectivity to a host from a given network-context. The ping command arguments are standard Linux ping command arguments. See the Linux man page for details.

For example:

```
an3000-mcm-slot7cpu1# ping <network-context>
Possible completions:
  NYC management-context test
an3000-mcm-slot7cpu1# ping NYC
Possible completions:
| <cr>
an3000-mcm-slot7cpu1# ping NYC
Usage: ping network-context [-LRUbdnqrvVaA] [-c count] [-i interval]
[-w deadline] [-p pattern] [-s packetsize] [-t ttl] [-I interface or
address] [-M mtu discovery hint] [-S sndbuf] [ -T timestamp option ] [
-Q tos ] [hop1 ...] destination
```

ping-network

Send ICMP ECHO REQ messages on all the network contexts. This command supports the following standard options.

P – pattern

c – count

s – packet size

For example:

```
an3000-mcm-slot7cpu1# ping-network <NWCTX> loopback <LPBK> or  
interface <intf> [OPTIONS] <HOST IPAddress>
```

ping-subscriber

Send ICMP ECHO REQ messages to the subscriber. This command supports the following standard options.

P – pattern

c – count

s – packet size

For example:

```
an3000-mcm-slot7cpu1# ping-subscriber <NWCTX> loopback <LPBK>  
[OPTIONS] <HOST IPAddress>
```

ping6

Test IPv6 connectivity to a host from a given network-context. The ping command arguments are standard Linux ping command arguments. See the Linux man page for details.

For example:

```
an3000-mcm-slot7cpu1# ping6 <network-context>
```

Possible completions:

```
    NYC management-context test
```

```
an3000-mcm-slot7cpu1# ping NYC
```

Possible completions:

```
    | <cr>
```

```
an3000-mcm-slot7cpu1# ping NYC
```

```
Usage: ping6 network-context [-LUdfnqrvVaA] [-c count] [-i interval]  
[-w deadline] [-p pattern] [-s packetsize] [-t ttl] [-I interface] [-M  
mtu discovery hint] [-S sndbuf] [-F flow label] [-Q traffic class]  
[hop1 ...] destination
```


pki

This feature adds a librarian for Public Key Infrastructure (PKI) related objects including certificates, certificate revocation lists, and private keys that are stored in `/tosdb/certs` called PkiDir. The librarian tracks objects under this directory as well as monitors the expiration and revocation status of these objects. The librarian makes this information available to other tos applications that use these objects.

Operators use the pki CLI command to install, remove, and update a pki object.

```
an3000-mcm-slot7cpu1# pki
```

Possible completions:

```
install    Install a pki object onto the disk
remove     Remove a pki object from the disk
ssh        ssh PKI parameters
update     Update a pki object on disk from its http source or altsource
```

The pki director:

- Secures and tracks (via traps) pki objects in defined subdirectories on `/tosdb/certs`
- Provides an internal client to allow tos applications to access pki objects and their status
- Monitors the expiration and revocation of x509 certificates
- Monitors the update available status of x509 certificate revocation lists
- Tracks key files and unreadable files (encrypted keys)
- Attempts to automatically re-download expiring x509 certificates and update available x509 certificate revocation lists when these objects were installed with a specified http source.

Affirmed Networks supports Ed25519 as a public key type. Ed25519 is an elliptic curve signature scheme that offers better security and performance compared with ECDSA and DSA. It is recognized by the pki subsystem and may be used for both user and host keys.

Note: Clients must be running openssh version 6.6 or greater.

install

Operators install the pki object using any of the following sources: http/https, file/public, or cut and paste.

```
an3000-mcm-slot7cpu1# pki install
```

```
altsource    Alternate http source to get the object from
```

```
encrypt      Encrypt the object in the destination
```

```
name         Override the name implied by source/altsource or give name  
if using cut and paste
```

```
object       Allow cut and paste of the object as a quoted text
```

```
source       Scp or http source to get the object from
```

```
type         Pki type to install the object in
```

```
|           Output modifiers
```

From http/https

The install attempts to immediately get the object. If the object does not exist on the http/https server, the file is empty, or the file is flawed, the command fails.

The command succeeds even if the server is unreachable, and MCC reattempts to get the object every hour until it either succeeds or the Operator removes the object. If repeated attempts take longer than 24 hours, MCC generates a periodic trap indicating that the object is still unreachable but the get attempts continue. When in this state, the object is not yet on the disk and is therefore unusable.

From a Local Disk

If the source is a file path under the */public* directory, then MCC moves the file to the chosen type after parsing the file. Moving the object also removes it from the former location. The file install attempts to move the file. If the file is not found, is flawed, or not located in */public*, the command fails.

Using Cut and Paste

The object can be given as a cut and paste quoted string of the certificate/key contents. The quoted string can contain spaces but cannot contain control characters. Replace any newlines with `\n` in the quoted string.

Using scp/sftp

To install an object using scp/sftp, push the object to the */public/certs/public*, */public/certs/private*, or */public/images* directory using your choice of scp/sftp clients. Otherwise, pull the object to one of those directories using **filecommand**. Then, follow the guidelines for local disk.

Object	Description	HTTP/HTTPS Source Guidelines	File/Public Source Guidelines	Cut/Paste Source Guidelines
altsource	Alternate http source to get the object from.	Supply the alternate source in the form of the http address of the file. For example: http://10.6.1.108/example_cert.pem.	Do not specify.	Do not specify.
encrypt	Encrypt the object in the destination.			
name	Override the name implied by source/altsource or give name if using cut and paste.	Optionally, specify a different filename than the http filename. This may be useful if the application expects the name in a certain form or with a certain name extension.	Optionally, specify a different filename.	Specify a name.
object	Allow cut and paste of the object as a quoted text.	Do not specify.	Do not specify.	Must be a quoted string containing no control characters. Substitute /n for any new lines.
source	/public file or http source to get the object from.	Specify a source from which to get the object.		
type	Pki type to install the object in. public private ca ca ca tls-ca tls-server tfd epdg user	Optionally, specify an object type that determines the subdirectory under <i>/tosdb/certs</i> in which the object is stored. MCC retains the public and private types for backward compatibility. The admin or content group users can directly read and write these files through the CLI at <i>/public/certs/public</i> and <i>/public/certs/private</i> . As a result, these files are less secure. Other object types are named for the application likely to look in them for pki objects and are only visible to root users. Affirmed Networks recommends using these objects types. If type is not specified, MCC uses the default public.		Optionally, specify an object type that determines the subdirectory in which <i>/tosdb/certs</i> is stored.

remove

The **pki remove** CLI command removes all records in all states from the named object. Operators use this command to remove an expired or revoked object or to remove a state updating record that will never reach its source. The **type** object is optional.

```
an3000-mcm-slot7cpu1# pki remove
```

name – Name of the pki object to be removed.

type – Pki type to remove the object from. Options include: ca, calea, epdg, host, imsi, private, public, sendd, tfd, tls-ca, tls-server, user

update

The **pki update** CLI command generates an updating state record for an existing object, which was originally installed with an http source, without impacting the existing disk file. When the attempt succeeds, MCC replaces the existing file and removes the updating record.

```
an3000-mcm-slot7cpu1# pki update
```

Note: This command is only valid on pki objects installed using http/https and where their source is known to the system.

name – Name of the object to update

type – Pki type of the object to update. Options include: ca, calea, epdg, host, imsi, private, public, sendd, tfd, tls-ca, tls-server, user

ssh

This section describes an overview of the cluster hierarchy to aid in the configuration of ssh pki. The cluster supports three hierarchies: admin, CALEA, and maint.

The following ssh commands are also described in this section:

- [pki ssh authorizedkey \(page 1-20\)](#)
- [pki ssh clearlearnedhost \(page 1-21\)](#)
- [pki ssh knownhostkey \(page 1-21\)](#)
- [pki ssh localhostcertificate \(page 1-22\)](#)
- [pki ssh principals \(page 1-23\)](#)
- [pki ssh securitylevel \(page 1-24\)](#)
- [pki ssh stricthostkeychecking \(page 1-26\)](#)
- [pki ssh trustedusercakey \(page 1-27\)](#)
- [pki ssh verifyhostkeydns \(page 1-28\)](#)

[Table 1-1](#) describes the cluster hierarchy.

Table 1-1 Cluster Hierarchy

User	Function
admin	■ Manages the collection of all CLI groups (cli-admin, cli-content, cli-user3g, cli-user4g, and cli-readonly) except the cli-calea group, sets the domain policy, sets CLI user attributes. ■ Performs standard administrative functions for these groups such as create/remove new users, sets attributes including password, resets lost user passwords, specifies a user name, group, and password, configure ssh pki associations, enables/disables the use of passwords for ssh login, etc. ■ Locks/unlocks user account, specifies a hybrid CLI /maint user, specifies a non-hybrid CLI only user (default), allows password or pki ssh authentication or only pki ssh authentication, limits user disk access to their home directory or by Linux file permissions, permits or denies binary file execution. ■ Creates cli-admin group users to act as sub-manager of these groups performing these same functions even on each other. <i>Note: Created admin group members are not permitted to act on the master default admin user itself, including performing cli-admin group scope key operations as these would affect admin as a member of the cli-admin group. Created users inherit the setting of password/pki/pkionly of the default user in that group.</i> <i>Note: Passwords - When creating users, if no password is specified, the initial password is set to the username. If password aging is enabled on a cluster during commissioning, the user is forced to change their password upon first login.</i>
CALEA	■ Manages the cli-calea group and its users. ■ Performs standard administrative functions for this group such as create/remove new users, reset lost user passwords, configure ssh pki associations, enable/disable the use of passwords for ssh login, etc.
maint	■ Manages the maint group and its users. ■ Performs the standard administrative functions for this group such as create/remove new users, reset lost user passwords, configure ssh pki associations, enable/disable the use of passwords for ssh login, etc. ■ Also see maint Group Users (page 1-17).
<i>Note: root user: Affirmed Networks strongly recommends that production systems answer the commissioning time security question to disable ssh access to root. (root is still available for local login at the console for use in emergency recovery.) If you do not disable ssh access to root, Affirmed Networks recommends enabling ssh pki protection for this user.</i>	

Classic ssh

Use the following guidelines to configure classic ssh public/private key pair authentication.

- 1 Create an ssh key pair on the machine from which the ssh client will run, using:
ssh-keygen in the ~/.ssh dir
- 2 Create an ssh key pair directly in that directory or move it to the directory.

You can also add a passphrase in the key pair for more security or use **blank** to avoid being prompted during key use.

This release supports the following ssh public key formats:

- ecdsa-sha2-nistp521
- ecdsa-sha2-nistp384
- ecdsa-sha2-nistp256
- ssh-dss
- ssh-rsa

The ssh-keygen options required to generate the ssh public key formats vary depending on the users' client machine. Typically the following is used:

ssh-keygen -t <type> of (rsa, dsa or ecdsa)

The user sends their public key to the system administrator of their user's group (admin, calea, or maint).

- 3 The administrator installs and configures the public key using either an admin group user, calea user, or main user (depending on the user's group) to install the public key as type user.

The administrator uses the following command to add or associate the user with the public key.

pki ssh authorizedkey

The administrator may optionally use **user-access ssh-password-authentication user <user_name>** to require pki for that user on the cluster.

- 4 The user logs in using their private key to authenticate. If the keys use their default names in the `~/.ssh dir`, they are discovered by the ssh client automatically. Otherwise, use the **ssh -i <name>** option to force ssh to find the private key.

Signed ssh Certificate Authentication

This model is easier to administer than the classic model, especially across multiple clusters because only the ca public key(s) (which never expire) are installed on the cluster instead of per-user (or per group) public keys.

***Note:** You can simultaneously configure the signed ssh certificate authentication model and the classic model on the same cluster without interference.*

- 1 The administrator creates a certificate authority key pair. This key pair is not placed in a home director. Instead, the pair is placed in a secure directory to which only the administrator has access.
- 2 Install the public key as a trusted ca. The administrator uses `pki` to install the public key as type `user`. The administrator then uses `pki ssh trustedca add` to register the public key as a ca with either cluster or group scope.
- 3 Create certificates for each user from their personal public key.

The user sends their own public key (as generated in the [Classic ssh \(page 1-15\)](#) example) to the Administrator who uses the secret ca private key and `ssh-keygen` to create a signed certificate. The certificate can contain a validity period. It should also be encoded with a ‘principals’ list of the user name (or user names) for which it is valid.

- 4 The administrator sends the certificate file back to the user. The administrator may optionally use `user-access ssh-password-authentication user <user_name>` to require `pki` for that user on the cluster.

This release accepts the following certificate types:

- `ecdsa-sha2-nistp521-cert-v01@openssh.com`
 - `ecdsa-sha2-nistp384-cert-v01@openssh.com`
 - `ecdsa-sha2-nistp256-cert-v01@openssh.com`
 - `ssh-dss-c ssh-dss-cert-v01@openssh.com`
 - `ssh-rsa-cert-v01@openssh.com`
- 5 The user logs in using their public key and the signed certificate to authenticate.

If the default certificate and key names are used in the `~/.ssh` dir, they will be discovered by the ssh client automatically. Otherwise, the `ssh -i <name> -i <name>-cert.pub` options can be used to force ssh to find the certificate and the user private key on which it is based.

maint Group Users

The **pkissh shell** command provides the master maint user with the same CLI command abilities within the maint user hierarchy as administrators in the admin and calea group.

For example:

```
maint@an3000-mcm-slot7cpu1:~$ pkissh --help
USAGE      : pkissh [options]
```

Configure the options as follows:

Table 1-2 **maint users**

Object	Description
authorized	Add or remove an authorized key from the user or group.
trustedca	Add or remove a trusted user ca key from the cluster.
principal	Add or remove a list of user and/or principal names from the group's or user's authorized principal list.
--user <user>	Specify user name.
--group maint	Specify the group maint.
--add <name>	Add the pki obj user/<name> to either --authorized or --trustedca or when used with --principal it takes a comma separated list of user and/or principal names.
--remove <name>	Remove the pki obj user/<name> from either --authorized or --trustedca or when used with --principal it takes a comma separated list of user and/or principal names.
--show	Can be used alone or combined with the other options to show the pki ssh configuration after any other option specified are applied. This utility allows the <i>maint</i> user to establish or modify the pki ssh configuration for <i>maint</i> group users. This utility directly modifies the pki ssh configuration files. The change is only applied to the sshd servers when that configuration is read by the pkidir task every 30 seconds. For example: <pre>pkissh --show pkissh --authorized --add <pki obj name> [--user <user>] [--show] pkissh --authorized --remove <pki obj name> [--user <user>] [--show] pkissh --trustedca --add <pki obj name> [--group maint] [--show] pkissh --trustedca --remove <pki obj name> [--group maint] [--show] pkissh --principal --add <user/principal list> [--user <user>] [--show] pkissh --principal --remove <user/principal list> [--user <user>] [--show]</pre>

Protection Against Connections to Unknown Servers

Typically, the cluster acts as an sshd server with connected clients. However, the cluster may also be asked to ssh, scp, and sftp to a remote server. Affirmed Networks supports the following three non-mutually exclusive methods of protecting against a connection to unauthorized external servers. The commands associated with these methods are also described in this section.

- [Using Locally Configured Known Host Keys \(page 1-19\)](#)
- [Using an External DNSSEC Server Configured to Verify Host Keys \(page 1-19\)](#)
- [Using Host Certificates on Authorized Servers \(page 1-19\)](#)

Note: You must first configure an authoritative DNS server to support DNS verification. When using DNS verification, the SSHFP records of the servers' preferred key algorithm must be configured in the DNSSEC server. MCC rejects SSHFP records for that host if they are using a different algorithm. DNS tries only the first key received. **Contact your TAC representative for assistance with configuring an authoritative DNS server.**

DNSSEC is a set of extensions to DNS which provide DNS clients (resolvers) with source authentication, data integrity, and authenticated denial of existence. DNSSEC utilizes a series of public/private key combinations to sign information resources and provide a mechanism for the end-user to verify the integrity of the DNS response.

If supported by an authoritative DNS server, a DNS zone can be secured with DNSSEC using a process called zone signing. Signing a zone with DNSSEC adds validation support to a zone without changing the basic mechanism of a DNS query and response.

Validation of DNS responses occurs using digital signatures that are included with DNS responses. These digital signatures are contained in new DNSSEC-related resource records that are generated and added to the zone during zone signing.

Validation of DNS responses is usually done in a network by the DNS cache resolvers or recursive resolvers. However, for this feature, no DNS resolver is installed on MCC. Instead, this feature assumes that the configured external DNS server is DNSSEC-aware and capable of authenticating the DNS responses (in other words, it must be a cache DNS resolver or a recursive DNS resolver).

Using Locally Configured Known Host Keys

- 1 The Administrator must gather the public host key(s) from authorized servers and install the keys in pki as type *host*. Use [pki ssh knownhostkey \(page 1-21\)](#) to associate the keys against a server IP or name wildcard.

***Note:** In order to choose the proper format for the server option of the [pki ssh knownhostkey \(page 1-21\)](#) command, manually ssh to that server and view the [show pki ssh learnedhost \(page 1-32\)](#) output.*

- 2 After all authorized servers are configured with a knownhost entry, set the [pki ssh stricthostkeychecking \(page 1-26\)](#) mode to **yes** to require such an entry to connect.
- 3 Run [pki ssh clearlearnedhost \(page 1-21\)](#) to remove learned host entries that would otherwise also allow a connection.

Using an External DNSSEC Server Configured to Verify Host Keys

- 1 Configure an external DNSSEC server containing signed public host keys for the authorized servers.
- 2 Configure a management context DNS server to point to the DNSSEC server and enable dnsextensions in the same record.
- 3 Set the [pki ssh verifyhostkeydns \(page 1-28\)](#) mode to **yes** to require such an entry to connect. This also forces [pki ssh stricthostkeychecking \(page 1-26\)](#) to **yes**.
- 4 Remove any other unintended configured known host entries as these will allow a connection.
- 5 Run [pki ssh clearlearnedhost \(page 1-21\)](#) to remove learned host entries that would otherwise also allow a connection.

Using Host Certificates on Authorized Servers

- 1 The Administrator must gather the public host key(s) from authorized servers and create certificates from the keys using an **ssh-keygen** key pair generated and kept private by the Administrator as a certificate authority key pair.
- 2 The Administrator configures each authorized server to use the certificate created from its own public host key as its *host certificate*.
- 3 The Administrator must only install the public key of their ca key pair in pki and use [pki ssh knownhostkey add <name> hostca](#) to configure that public ca key to authenticate each authorized server's host certificate. This is the preferred method when dealing with a large network, as adding a new authorized server only requires creating one new certificate from the new server's host key, and adding that host key and configuring the new server to offer that host certificate. The distributed clients already have the ca key and no further action is required.
- 4 When the cluster is acting as an authorized server, a host certificate (built from the cluster's host keys found in */public/hostkeys*) can also be added to the cluster using [pki ssh localhostcertificate \(page 1-22\)](#).

- 5 Set **pki ssh stricthostkeychecking** (page 1-26) mode to **yes** to require the server to supply a certificate to connect.
- 6 Remove any other unintended configured known host entries as these will allow a connection.
- 7 Run **pki ssh clearlearnedhost** (page 1-21) to remove learned host entries that would otherwise also allow a connection.

pki ssh authorizedkey

Use the **pki ssh authorizedkey** command to modify a users' or group of users' authorized keys file(s).

Configure the objects as follows:

Table 1-3 pki ssh authorizedkey

Object	Description
add	The type user pki object to add to the user's authorized_keys file for classic ssh pki.
group	Update all users in this group's authorized_keys files. The group option is a convenience option used to add or remove the same public key from every current user in the group. Sharing a public key in this way is less secure. If a new user(s) is added to the group, you must run this command again to assign the new user keys. Use the group option to add the key to group users who do not already have a key.
remove	The type user pki object to remove from the user's authorized_keys file for classic ssh pki.
user	Update this user's authorized_keys file.

pki ssh clearlearnedhost

Clears the learned host files for all users over which the current user has Administrative authority. Some users such as maint can see learned host data to which they do not have authority to clear. In that case, the command does not fail but that data is unaffected.

Note: Clearing learned host data prevents that data from allowing those hosts to be considered **known** for strict host checking purposes.

For example:

```
an3000-mcm-slot7cpu1# pki ssh clearlearnedhost
result success
reason cleared 1 ssh learned host files accross 6 users
```

pki ssh knownhostkey

Use the **pki ssh knownhostkey** command to modify the cluster known hosts list and add approved remote hosts to the cluster-wide known host file. You must first copy the public host key of the remote host and then install the key in pki as type host.

To locate the correct server name, set strict mode to **no** and manually connect to the server. After connecting, use the [show pki ssh learnedhost \(page 1-32\)](#) command to display the server name. Affirmed Networks recommends using the *server* option if using different key types. The first key type exchanged may not be the correct type, but if it matches the *server* option, it will match against those keys and fail.

Note: Use caution when using broad wildcard names for **server** when receiving mixed host key types. As each host key type is received from the remote server, the system examines all matching **server** entries and compares their keys. If one or more matching **server** entries is found but none of their keys match, then the connection fails without evaluating the next received host key type. If no **server** entries match, then the next host key type sent is evaluated for a possible match. This process continues until a match is found.

Configure the objects as follows:

Table 1-4 pki ssh knownhostkey

Object	Description
add	The type host pki object to add to the cluster known_hosts list.
hostca	The object is a host ca key used to validate received host certificates.
remove	The type host pki object to remove from the known_hosts list.
server	A comma-separated list of wildcard hostnames, IP addresses, and IP addresses modified by ports that identify the known host. The * is a valid wildcard in these names.

pki ssh localhostcertificate

Configure a local host certificate for all incoming ssh connections to allow the cluster to act as the server in this model. This model of remote server authentication is designed to scale in large, changing networks.

Note: Registering a host ca key to recognize a remote host by receiving a certificate signed by that host ca key, even if the host is new is described in [pki ssh knownhostkey \(page 1-21\)](#).

The Administrator must configure the following:

Note: The client requires no new configuration as they recognize the new server as trusted by the host ca key used to sign the certificate.

- 1 Configure the public key of their secret host ca key on all clients as a known host ca key in that client's known host file.
- 2 As new servers are added to the network, scp the public host key from the new server.
- 3 Create a certificate from the public key and the secret host ca key.
- 4 Configure the new server to offer certificates to all incoming ssh connections as its host certificate.
- 5 On the Affirmed Networks system, use **filecommand** (readonly) to scp the cluster public keys out of the new */public/hostkeys* directory.

Configure the objects as follows:

Table 1-5 pki ssh localhostcertificate

Object	Description
add	The type host pki object to add as the cluster host certificate
remove	The type host pki object to remove as the cluster host certificate

pki ssh principals

Use the **pki ssh principals** command to add or remove a list of names to or from the user's or groups' authorized principals file. For more information about authorized principals, refer to the *openssh* documentation.

The pki ssh option supports two levels of authorized principals configuration; per-group list and per-user list.

per-group list – Allows for an easier to administer shared principal list. This method poses a security risk by allowing multiple user certificates to share multiple user accounts. To manage group principals, specify the **group <group>** option. In addition to certificate aliases, you must include any *usernames* you wish those aliases to access in the group.

per-user list – Allows complete control in permitting aliases by the user or allowing a certificate for one user account to access another. Specify **user <user>** to manage the per-users principals lists.

Note: *If you choose to use both per-user and per-group principal lists, then the list for a given user account becomes its own list, including all of the entries in the group list **only if** the user account is in the group list.*

For ease of administration, every user account implicitly has itself in a hidden per-user principals list (that is, every user always accepts a certificate containing its own username).

Configure the objects as follows:

Table 1-6 pki ssh principals

Object	Description
add	Add the comma separated users or names to the users's or group's authorized_principals file. Use double quotes if the list if it contains non-alphanumeric characters.
group	The group whose authorized_principals file is to be modified.
remove	Remove the comma-separated list of users or names from the users's or group's authorized_keys file.
user	The user whose authorized_principals file is to be modified.

pki ssh securitylevel

The cluster can operate at either the compatible or maximum ssh security level.

***Note:** Connecting to or from other systems may require those systems to be running openssh 6.6 or greater.*

You can change the security level on a running cluster and it only affects new connections. In a typical configuration, the security level should be set to **maximum**. If legacy systems are unable to interface with the cluster, relax the security level to **compatible** until you can perform an upgrade on the legacy system.

In a cluster with the security level set to **maximum**, any public keys stored in pki that are unsupported are marked as 'weak' and are hidden from any pki ssh configuration.

MCC supports the following MAC algorithms at the **maximum** pki ssh security level:

- hmac-ripemd160
- hmac-sha2-512
- hmac-sha2-256

If the security level is reverted to **compatible**, the ‘weak’ designation disappears and any keys previously configured for use in pki ssh reappear in the configuration.

Note: The dsa public key type is considered ‘weak’ at **maximum** security level.



WARNING: In a cluster configured with **maximum** security level, the cluster does not gracefully sync nodes that are running an earlier release, powered off, and added back to the node.

Configure the objects as follows:

Table 1-7 pki ssh securitylevel

Object	Description
compatible	All current and legacy public key, cipher, and macs are supported. Using this security level, there are no issues connecting to or being connected to by other equipment.
maximum	Supports only those public key, cipher, and macs that are currently considered as secure. Older and weaker algorithms are not supported.

pki ssh stricthostkeychecking

Set the ssh host checking mode. A server is considered recognized if any of the following are true:

- The server has a key configured in the cluster-wide known hosts file.
- The server was visited by this user when the stricthostkeychecking object was configured to **no** and therefore had an entry added to that user's learned hosts file.
- The server is configured to offer a host certificate that was signed by a host ca key configured in the cluster-wide known hosts file.
- Verify DNS is set to **yes** and an external DNSSEC server has been configured to return expected host keys and that key matches the key returned by the remote server.

Configure the objects as follows:

Table 1-8 pki ssh stricthostkeychecking

Object	Description
ask	Causes the cluster to fail ssh connection to any unrecognized remote server and the user is prompted to continue the connection.
no	Causes the cluster to fail ssh connection to any unrecognized remote server.
yes	A remote server is only rejected if there is a previously learned host key but the key returned by the remote server has changed.

pki ssh trustedusercakey

Use the **pki ssh trustedusercakey** command to add or remove a ca public key from the second form of ssh pki authentication.

Cluster scope ca keys are passed to ssh in the *trusted_user_ca_keys* file while group scope ca keys are passed to ssh in each user in the group's *authorized_keys* file as option cert-authority lines.

Configure the objects as follows:

Table 1-9 pki ssh trustedusercakey

Object	Description
add	The type user pki object to add to the <i>trusted_user_ca_keys</i> file. As new users are added to a group with trusted ca keys, they inherit access to the same ca keys.
group	The scope of the add or remove is either group scope or if no group then cluster scope. If a group is specified, the trusted ca key is only available to authenticate users in that group. The use is entirely within the single user hierarchy and is never blocked from being changed by the hierarchy administrator. If no group is specified, then the ca key is available to all groups. Note: This may pose a security risk as it has the potential for a ca install on the group hierarchy to be used to access users in another group hierarchy. To avoid this, after the cluster administrator sets the cluster wide ca, the administrator should login as admin, maint, and calea. In each case, add a meaningless name to that user's principals list. This causes the pki dir subsystem to view the cluster-wide list as 'used' by all three hierarchies, and therefore not allow any of the hierarchies to further add or remove to/from the cluster-wide trusted user ca list.
remove	The type of user pki object to remove from the <i>trusted_user_ca_keys</i> file.

pki ssh verifyhostkeydns

Set the ssh verify host keys using dns mode.

Note: You must first configure network-context <name> and specify the DNS IP address prior to enabling DNS extensions.

For example:

```
an3000-mcm-slot7cpu1(config)# network-context management-context dns
server-address 10.33.13.24 use-extension-dns ?
```

Possible completions:

```
disabled enabled
```

```
an3000-mcm-slot7cpu1(config)# network-context management-context dns
server-address 10.33.13.24 use-extension-dns enabled
```

```
an3000-mcm-slot7cpu1(config-server-address-10.33.13.24)# commit
```

Commit complete.

```
an3000-mcm-slot7cpu1# show running-config network-context
management-context
```

```
network-context management-context
security disabled
number-of-security-services 1
vrf-lightweight false
tunnel-fragmentation-type inner
block-gtpu-error-pkt false
control-dscp-value dscp_0_be
mgmt-ip-interface 120 7 1
  ip-address 10.64.14.20
  prefix 20
  vlan-tag 0
  gateway-ip 10.64.0.1
!
mgmt-ip-interface 120 8 1
  ip-address 10.64.13.20
  prefix 20
  vlan-tag 0
  gateway-ip 10.64.0.1
!
logical-mgmt-ip-address 0.0.0.0
!
dns server-address 10.33.13.24
  use-extension-dns enabled
```

Configure the objects as follows:

Table 1-10 pki ssh verifyhostkeydns

Object	Description
ask	When the ssh client is used with a host name in addition to getting an IP address, it attempts to get the expected host key of that server from the configured external DNSSEC server. If it fails, the connection is rejected. If the host key sent by DNSSEC does not match the host key offered by that server, the system displays a warning and the user is prompted to continue to connect to the host. A management DNS server must be configured and management DNS extensions must be enabled. Setting this option to ask forces pki ssh stricthostkeychecking (page 1-26) to act as ask as well.
yes	When the ssh client is used with a host name in addition to getting an IP address, it attempts to get the expected host key of that server from the configured external DNSSEC server. If it fails or if the host key sent by DNSSEC does not match the host key offered by that server, the connection fails. A management DNS server must be configured and management DNS extensions must be enabled. Setting this option to yes forces pki ssh stricthostkeychecking (page 1-26) to act as yes as well.
no	The ssh client does not attempt to use any configured DNS server to validate host keys.

resolved

Resolve conflicts. This command is executed in config mode.

revert

Copy the running configuration to the candidate configuration. The revert command resets the current candidate configuration. This command is executed in config mode.

set-default-idle-timeout

Set the default idle-timeout (in minutes) for any future sessions. This value defaults to 30 minutes.

For example:

```
an3000-mcm-slot7cpu1# set-default-idle-timeout
Enter default idle timeout (in minutes): 10
Set default idle timeout to 10 minutes? (Y/n):y
```

set-prompt

Set the CLI prompt. The following special options are supported when setting the CLI prompt:

Note: The '\' character is only supported for the following prompt options.

\d	date in "YYYY-MM-DD" format (for example, "1978-06-10")
\h	hostname up to the first '.'
\H	hostname
\t	current time in 24-hour HH:MM:SS format
\T	current time in 12-hour HH:MM:SS format
\@	current time in 12-hour am/pm format
\u	username of the current user

sftp

SFTP a file to/from a host.

summary

Use the **summary** command to create an aggregate address to replace a series of more specific addresses. The metric used to advertise the summary is the smallest metric of the more specific routes. This command helps to reduce the size of the routing table.

Configure MCC to advertise one external route as an aggregate for all loopbacks or UE sub-pools that are covered by the address.

Configure the objects as follows:

Table 1-11 summary

Object	Description
ip-prefix	IP prefix designated for a range of addresses, including the prefix length. Specify the ip-prefix as an IP address. Specify the length as a number from 1 to 31. For example, 49.1.1.0/24
not-advertise	(optional) When set to false, suppresses routes that match the specified prefix/length pair.
tag	(optional) Specify the tag value that is used in the summary route being advertised. The range is 1 to 65535.

Show Commands

Display the following:

- [show pki objects \(page 1-32\)](#)
- [show pki ssh host \(page 1-32\)](#)
- [show pki ssh learnedhost \(page 1-32\)](#)
- [show pki ssh mode \(page 1-33\)](#)
- [show pki ssh user \(page 1-33\)](#)

show pki objects

Displays all records in all states available to the system. State **ready** and state **invalid** records represent files on the disk. State **updating** and state **pending** records represent queued attempts to get or update the objects to the cluster. The **modified** field represents the modify data for disk objects and the time the updating/pending record was created for those records.

show pki ssh host

Displays all configured known host keys and host ca keys, as well as the configured local host key certificate, if any. For example:

```
an3000-mcm-slot7cpu1# show pki ssh host
result success
reason
localhostcert          user3g-id_dsa-cert.pub
```

show pki ssh learnedhost

Displays the hosts learned by various users because they did not exist in configured known hosts when that user visited them. In general, when `stricthostkeychecking` is configured to **yes**, then no new hosts can be learned. The strict setting can be overridden by any user who has direct access to the `ssh`, `scp`, and `sftp` command line. For example:

```
an3000-mcm-slot7cpu1# show pki ssh learnedhost
result success
reason
cli-admin    admin          10.1.1.123
showed 1 ssh learned host entries accross 6 users
```


show pki ssh mode

Displays the current values of strict host, verify dns, and security level. It also lists the ciphers, MACs, and public key types supported by sshd and ssh at this security level. For example:

```
an3000-mcm-slot8cpu1# show pki ssh mode
result success
reason
Verify Host w/DNS      no
Strict Host Checking   no
Security level         maximum
    PubkeyTypes
ssh-ed25519-cert-v01@openssh.com,ecdsa-sha2-nistp521-cert-v01@openssh.
com,

ecdsa-sha2-nistp384-cert-v01@openssh.com,ecdsa-sha2-nistp256-cert-v01@
openssh.com,

ssh-rsa-cert-v01@openssh.com,ssh-ed25519,ecdsa-sha2-nistp521,ecdsa-sha
2-nistp384,

                                ecdsa-sha2-nistp256,ssh-rsa
    Ciphers                aes256-ctr,aes192-ctr,aes128-ctr
    MACs                    hmac-ripemd160
    KexAlgorithms
curve25519-sha256@libssh.org,diffie-hellman-group-exchange-sha256,
diffie-hellman-group14-sha1
```

show pki ssh user

Displays pki ssh configured users. For example:

```
an3000-mcm-slot7cpu1# show pki ssh user
Possible completions:
  all cli-admin cli-content cli-readonly cli-user3g cli-user4g |
<cr>
```


User Account Management

This section describes how to manage user accounts, set, change, and reset passwords, and delete users.

logout

Log out a user.

user-access

The MCC supports the following admin user commands used to manage user accounts: add-user, change-password, delete-user, and reset-password. Local non-admin users are only permitted to change their own password.

Configure the objects as follows:

Table 2-1 user-access

Object	Description										
add-user	Add a new user and password. Choose a group for the new user. Options include: admin, admin/maint, content, readonly, user3g, and user4g. The admin group and the maint group support a hybrid user. As a hybrid user, remote TACACS+ users can log in to both the CLI and the Linux shell using the group admin-maint in the TACACS+ custom AV pair in the initial authorization. <table><tr><td><u>User</u></td><td><u>Privilege</u></td></tr><tr><td>admin</td><td>All</td></tr><tr><td>user3g</td><td>3G services</td></tr><tr><td>user4g</td><td>4G services</td></tr><tr><td>readonly</td><td>Read-only access</td></tr></table>	<u>User</u>	<u>Privilege</u>	admin	All	user3g	3G services	user4g	4G services	readonly	Read-only access
<u>User</u>	<u>Privilege</u>										
admin	All										
user3g	3G services										
user4g	4G services										
readonly	Read-only access										

Table 2-1 user-access

Object	Description
change-password	The <i>installation and initialization script</i> prompts you to enable password aging for all user accounts. When you enable password aging, users are prompted to change their own password on the first login and again after 45 days. The system requires a complex password. You must enter your existing password first and then your new password. Note: When you change a password, you must enter a password that differs from the last three password entries. Note: To disable password aging, run the <i>installation and initialization script</i> and answer n at the <i>Enable password aging? <y/n></i> prompt. Note: A login failure results in a two-second pause before the system prompts the user for another password. After five consecutive failed login attempts, the system locks the account for ten minutes. You can change the admin user password using the CLI and enter the same password at the Acuitas Edit Network Element screen. See the <i>Acuitas User's Guide</i> for details. Note: If you change the password on the network element, you must also change that account information in Acuitas.
change-sftp-dir	Change the directory in which sftp/scp starts for this user. To change the default SFTP directory of user accounts for which you have administrator privileges, see <code>reset-sftp-dir</code> .
delete-user	Choose the user to be deleted.
password-policies	Set/update password policies for the current domain. See Password Policies (page 2-3) . These policies apply when changing your own password. The admin user can override these policies on a per-domain basis.
prompt-acknowledgment	Enables users to disable CLI confirmation prompts. This object applies only to the user that issues the command, and only disables prompts for the current session. It does not affect other users with the same username. Confirmation prompts are enabled by default.
reset-password	Reset the password for this user. This is a recovery utility intended for admin users. This command enables admin users to reset another user's password to a simple password. If password aging is enabled, the user is required to change their password to a complex password on the next login.

Table 2-1 user-access

Object	Description
reset-sftp-dir	Allows account administrators (admin group members and the default CALEA user) to change the default SFTP directory of user accounts for which they have administrator privileges, including read-only users. Administrators must set the SFTP directory to either the user's "home" (default) directory or a subdirectory under the <code>/public/*</code> directory. <pre>an3000-mcm-slot7cpu1# user-access reset-sftp-dir user <username></pre> The system prompts: <pre>Enter sftp directory:</pre> Enter the path to set the default SFTP directory. Security Note: Configuring an SFTP directory for a read-only user enables SFTP/SCP for that user. Typically, read-only users whose SFTP directory is set to "home" are not permitted to use SFTP/SCP.
restrict-execute-permission	Enable/disable execution of binaries from the homedir
restrict-file-access	Enable/disable restricting file access beyond the homedir
ssh-password-authentication	Enable/disable the use of passwords for ssh on a per-user basis. Disabling ssh password authentication forces the user to authenticate ssh with pki. When new users are created, they inherit this setting from the lead user (admin, content, user3g, user4g, readonly, calea, main) in their group.
update-dictionary	Set/update custom password exclusion dictionary.

Password Policies

Use the following command to configure password policies:

```
an3000-mcm-slot7cpu1# user-access password-policies
```

Note: The following password domain policies apply only when you change your own password.

Note: If you perform a rollback, the systems uses the password that you used prior to reverting to an older release.

Note: When you change your password, you must supply both the existing password and new password.

Enter password minimum length – The default is 8 characters. The range is 6-32 characters and you must use a minimum of 6 characters. Press Enter to skip this parameter.

Enable/Disable/Skip password aging – Set to enable or disable or press Enter to skip this parameter.

Configure password character sets complexity – Set to yes or no or press Enter to skip this parameter. MCC supports digits (0), uppercase (1), lowercase (1), and special characters (non-alphanumeric) (1). The range is 0-6. Enter the minimum number of required lowercase, uppercase, and special characters and required digits.

Passwords must be at least 6 characters (up to 32 characters) and must include uppercase and lowercase characters and at least one special character. Passwords may also include numbers. Special characters include ([!@#\$%^&*])

Note: The exclamation point (!) is a special character used to comment out the rest of the CLI command. If a CLI command requires the use of a special character, Operators should add \ before ! to mask this behavior. For example, \!password. See the Affirmed Networks CLI Reference Guide for more information on navigating the CLI and keyboard shortcuts.

Enter the number of saved password changes before a password can be reused – This value indicates the number of unique passwords that must be associated with a user account before a password can be reused. A value of 0 disables this setting. The default is 3 passwords and the range is 0–30 passwords.

Enable/Disable/Skip password dictionary – Set to enable or disable or press Enter to skip this parameter. If set to disable, dictionary exclusion processing is disabled. If set to enable, enter the name of the customer password exclusion dictionary. Include additional customer dictionary entries: Yes or No

Configure logout parameters – Set to yes or no or press Enter to skip this parameter.

- Enter the number of allowed password authentication failures before locking an account. MCC denies access if the amount of consecutive failures exceeds the configured value. A value of 0 disables this setting. The default is 5 password failures and the range is 0-6 password failures.
- Enter the lock duration which indicates the number of minutes to wait before unlocking an account. MCC allows user access after the configured number of minutes expires after a user has been locked out. The default is 10 minutes and the range is 0-60 minutes. A value of 0 indicates that the account is locked out and remains locked out.
- Set the logout failure limit which indicates the number of times the system allows a user to be locked out. A value of 0 indicates that the user is never locked out. The range is 1-6 attempts.

Configure misalliance options. Set to yes or no or press Enter to skip this parameter.

- Allow new passwords with user name. Set to yes or no or press Enter to skip this parameter. If set to no, MCC rejects the new password if the password contains the name of the user in straight or reversed form.
- Enter the maximum number of consecutive characters in a new password. A value of 0 indicates no limit to the number of consecutive characters. The range is 0-64 characters.

Enter the maximum number of days before a password change is required (if aging is enabled) – The maximum password age defaults to 45 days. A value of 0 indicates no maximum. The range is 0-365 days.

Enter the minimum number of days before a password change is allowed – The minimum password age defaults to 0 days. The range is 0-7 days.

Enter the maximum number of sequential characters in a new password. MCC rejects the password if it contains a monotonic character sequence longer than N. The default of 0 disables this verification. The range is 0-64. Examples of monotonic sequences include '12345' or 'fedcb'. Note that most passwords will not pass the simplicity check, unless the sequence is only a minor part of the password.

Note: The exclamation point (!) is a special character used to comment out the rest of the CLI command. If a CLI command requires the use of a special character, Operators should add \ before ! to mask this behavior. For example, \!password. See the Affirmed Networks CLI Reference Guide for more information on navigating the CLI and keyboard shortcuts.

Requirement to Change Password on Acuitas Server

If you change the password on the system, you must also change that account information on the Acuitas server that manages that system.

To change the login password for the system on the Acuitas server:

- 1** Right-click the network element and click **Edit Element**.
- 2** In the CLI & NETCONF section, enter the username and new password.
- 3** In the SFTP section, enter the username and new password.
- 4** Click **Save**.
- 5** At the confirmation prompt, click **OK**. User password complexities are now homogenized for Acuitas and the MCC CLI.

user-access aaa admin login-banner

The login-banner enables Operators to configure a customized login banner. This banner may contain a maximum of 4000 characters and is entered in multi-line mode and terminated with ctrl-D. The system displays the banner prior to displaying the password prompt.

user-access aaa tacacs-plus

The MCC provides TACACS+ service for remote authentication, authorization, and accounting.

The TACACS+ user authorization and accounting services feature allows Operators to use Terminal Access Controller Access Control System Plus (TACACS+) as an external authentication, authorization, and accounting server to manage user access to the MCC. When users log in to the MCC, an authentication request is sent to the TACACS+ server and the server either accepts or rejects the request based on information in the user database.

Note: The remote **maint** user must be logged in to the primary MCM when logging in to a secondary MCM with a **maint** account that is authenticated by a remote TACACS server, rather than with a local **maint** account. If the remote **maint** user is not logged in to the primary MCM, the remote **maint** user is allowed to log in to the shell, but the session will close within a matter of seconds.

In addition, if the remote **maint** user was previously used on the primary MCM, and that account is used to log in to the secondary MCM as it is booting but prior to the MCMs being in sync, the user session may be terminated after the MCM sync completes. If that occurs, reestablish the session.

Upon successful authentication, the MCC sends an authorization request to which the server responds with the user's group assignment (used to determine the commands to which the user has access). The authorization information propagated from the TACACS+ server is via a custom attribute-value-pair in the form of `group=<user_group>`, where `user_group` is defined in the CLI. The MCC requires a customer AV pair be returned in the initial authorization request after successful authentication. The `group=abc` AV pair is used to place users into certain privilege levels for both local and remote authorization. The predefined MCC groups are `admin`, `readonly`, and `calea`. After the session is established, the system sends user accounting messages to the TACACS+ server, including session termination.

The MCC supports the following general TACACS+ features and functions when the admin-state is set to enable:

- Configuration of up to eight TACACS+ servers.
 - Network elements are capable of specifying individual timeouts for each configured Access Control Server.
 - A global timeout value may be specified as a default when individual timeout values are not configurable for each Access Control Server.
- TACACS+ server connection failure switchover after a fixed number of attempts.
- A local authentication mechanism as fallback if no configured TACACS+ servers are accessible. The MCC supports the TACACS+ method as the primary method. When that method is unavailable, MCC uses local authentication.

Configure the objects as follows:

Table 2-2 user-access aaa tacacs-plus

Object	Description
admin-state	Operators can globally enable or disable the TACACS+ service.
accounting	Each accounting command is sent to the TACACS+ server to be logged. See accounting (page 2-11). ■ mode – Operators can configure remote or local mode and filter based on commands or groups. ■ remote-exclude – Exclude groups or command from remote accounting. – commands – Exclude commands from remote accounting. Options include: configure or show. – groups – Exclude groups from remote accounting. Options include: admin, calea, content, readonly, user3g, user4g. <i>Note: This section describes the accounting records that are generated as part of the access being provided to the device (that is, device logins) and commands being executed on the device. These accounting records are required to track and audit changes and access to the NE. For information on accounting records that are created for usage or billing, see the Affirmed Networks Charging Data Records Specification.</i>

Table 2-2 user-access aaa tacacs-plus

Object	Description
authentication	The authentication commands enables Operators to set the authentication mode for local users when TACACS+ is enabled. The local user is the default and is used only as a backup mechanism in the event that the TACACS+ server is unreachable. Remote accounts are created and managed by the TACACS+ server. See authentication (page 2-11). ■ allow-local Set local users authentication mode. If allow-local is set to fallback, then local users are permitted to log in to the MCC if the TACACS+ server is configured and unreachable. A local account can be duplicated on the TACACS+ server but authentication is performed and decided by the remote server if it is reachable. If the remote server rejects the authentication, then the login fails. If allow-local is set to always, then local authentication is always attempted if the remote authentication fails. – cli-users – CLI users local authentication mode. The default is fallback. – maint-users – Maint users local authentication mode. The default is always.
authorization	Every CLI command issued by a user is authorized by the TACACS+ server and only allowed to complete if the server accepts the command. This command enables Operators to set the authorization mode to either local or remote when TACACS+ is enabled. The remote-exclude command filter displays show commands or config commands from remote authorization as well as filtering based on user group. The user may experience slower performance when remote authorization is enabled, therefore Affirmed Networks recommends disabling remote authorization when loading large configurations. See authorization (page 2-11). ■ mode – Operators can configure remote or local mode and filter based on commands or groups. ■ remote-exclude – Exclude groups or command from remote authorization. – commands – Exclude commands from remote authorization. Options include: configure or show. – groups – Exclude groups from remote authorization. Options include: admin, calea, content, readonly, user3g, user4g.
console-mode	Operators can configure authentication requests to be forwarded to the ACS by access group or port. For example, the CLI enables Operators to configure the console port for local authentication and network ports for TACACS+.

Table 2-2 user-access aaa tacacs-plus

Object	Description
failure-limit	The failure-limit command limits the number of failed attempts (unreachable) before the server is considered down or unreachable. If a server is declared unreachable, the MCC does not attempt another AAA operation until the reattempt time has expired (see timers). Maximum number of failures between 1 and 100 before a server is skipped automatically.
server	Select a priority for the server and then configure additional objects. The MCC supports up to eight TACACS+ servers. Each server is attempted in a round-robin sequence. The servers are indexed based on a priority between 1 (highest) and 8 (lowest). ■ authen-type – Authentication is performed by ASCII or PAP (default). ■ ip-address – Each server requires the IP address of the remote server. ■ port – Port used by TACACS+ Server (49 is the default). ■ secret-key – For security, TACACS+ AAA packets are encrypted based on a configured secret key that is agreed upon by the client and the server. Operators can configure a per-server timeout used to determine the number of seconds that the TACACS+ client (MCC) waits before moving to the next server on the list. The MCC tracks the server failures and after a server crosses the threshold, the MCC marks the server as <i>down</i> and does not retry that server until the reattempt time has elapsed.
timers	The MCC supports the following two global timers for TACACS+ service: ■ retry-primary-aaa – time (in seconds) to wait before the primary AAA method (TACACS+ if configured) is reattempted if it is considered down. MCC uses local authentication as a fallback until the timer expires. The TACACS+ service is considered down if none of the remote servers are reachable and the wait time is intended to prevent network thrashing until the issue is resolved. The MCC tries all the remote servers in a round-robin sequence based on priority. ■ retry-server – time (in seconds) before an individual timer is retried if it is down. After a server crosses the failure limit, it is considered down and is removed from the list of servers. The server is not retired until the reattempt time has elapsed.

accounting

The MCC supports the following accounting features and functions:

- NE forwards all accounting to the ACS to be recorded.
- Operators can enable and disable accounting requests to the ACS to be recorded.
- Operators can enable and disable which accounting requests are forwarded to the ACS based on privilege, commands, and user group.
- NE supports local accounting as a fallback option.

authentication

The MCC supports the following authentication features and functions:

- NE forwards all authentication request to the ACS for validation.
- By default, the MCC support local authentication. Operators can specify local authentication as a fallback option if TACACS+ is not enabled or the server is unreachable.
- Supports a validation mechanism that forces the user to re-authenticate prior to escalating the user's privilege. The authentication request to escalate the user's privilege is then forwarded to the ACS for validation.
- Supports the `server_msg` option in the TACACS+ protocol to allow the ACS to present messages to the user.

authorization

The MCC supports the following authorization features and functions:

- NE forwards all commands to be executed on a device to the Access Control Server to be authorized.
- MCC uses local authentication if the MCC is unable to forward commands to the ACS.
- Operators can enable and disable which authorization requests are to be validated based on privilege, command, and user group.
- NE validates all commands that have the ability to update or change the state of the network element or its configuration.
- NE supports the ability to disable remote authorization and use local authorization as a fallback option.

- NE validates the specified command(s) and all options and arguments prior to execution.
- NE supports the expansion of all commands, options, and arguments to their full names.
- Abbreviated form of command, options, or arguments are never sent to the ACS.
- Aliased commands are translated to their respective command(s) prior to being forwarded to the ACS.
- For command chaining (that is, multiple commands on a single command line) each command, options, and arguments are sent as a separate requests to the ACS.

who

Display currently logged on users. An asterisk appears next to the user who executed the command.

```
an3000-mcm-slot7cpu1# who
```

```
Possible completions:
```

```
| <cr>
```

```
an3000-mcm-slot7cpu1# who
```

Session	User	Context	From	Proto	Date	Mode
16	admin	cli	127.0.0.1	ssh	11:43:38	config-terminal
*14	admin	cli	172.16.128.190	ssh	11:38:48	operational
10	admin	netconf	10.6.1.72	ssh	11:04:05	operational

Show Commands

MCC supports the following user account management show commands. For specific details and output, see the CLI and *Acuitas Performance Statistics*.

show pki summary

This shows the complete pki ssh configuration for the current user hierarchy. It does not display configuration for other heirachies.

Use this command as both any admin group user and the master calea user and use the bash command pkissh –show as the default maint user (and as root if it is not ssh disable on the cluster) for a complete cluster view of pki ssh configuration.

```
an3000-mcm-slot7cpu1# show pki ssh summary
```

```
pki ssh summary
```

GROUP	USER	DESCRIPTION
all		trustedusercakeys user_ecdsa_ca.pub
cli-admin		trustedusercakeys user_dsa_ca.pub
cli-admin	admin	authorizedprincipals (tom),sid
cli-admin	emsadmin	authorizedkeys pass-id_dsa.pub pass-id_rsa.pub
pass-id_ecdsa.pub		
cli-admin	sid	groupprincipals
cli-content		trustedusercakeys user_rsa_ca.pub
cli-content	content	authorizedprincipals (frank),(fred),admin
cli-readonly		
cli-readonly	guest	
cli-user3g		
cli-user3g	user3g	
cli-user4g		
cli-user4g	user4g	

show user-access

Display management groups and users. The system displays information based on server priority. The output displays the following information:

- *consecutive-failures* – Count used to indicate how many failures the server has experienced.
- *credential-failures* – Count used to indicate how many authentication failures the server has experienced.
- *latest-failure* – Date of the last failure.
- *next-server-retry* – Time in the future when the server will be retried after a failure occurs.
- *oper-state* – Indicates the status (up/down) of the server.
- *server* – TACACS+ server IP address.

aaa tacacs-plus

consecutive-failures – consecutive failure count

credential-failures – total credential failures

latest-failure – time of latest failure

next-server-retry – earliest time to retry server

oper-state – server operational state

server – TACACS+ server IP

show subscriber count

Show the count selected subscribers.

show subscriber dynamic-pcc-rule

Display dynamic PCC Rule Information (key: <pdn-session-id> <bearer-id> <pcc>).

show subscriber pdn-bearer

Display PDN Bearer Level Information (key: <pdn-session-id> <bearer-id>). This command displays dynamic-pcc-rule based on the data received from the C-plane.

```
an3000-mcm-slot7cpu1# show subscriber pdn-bearer
```

```
dynamic pcc rules: { pdn-session-id bearer-id pcc-rule-name }  
301064 6 HttpRule
```



```

packet-filter 0
  direction      bidirectional
  network-ip
  ue-port-start  0
  ue-port-end     0
  nw-port-start   80
  nw-port-end     80
  protocol        255

```

show subscriber pdn-flow

Display PDN flow level information (key: <pdn-session-id>).

show subscriber pdn-service-data-flow

Display PDN service data flow level information (key: <pdn-session-id> <service-type> <service-flow-id> <service-rule-id>).

show subscriber pdn-session

To determine the roaming zone, use the following show command:

```

an3000-mcm-slot7cpu1#show subscriber pdn-session <pdn-session-id>
vsa-roam-zone International

```

To display the number of uplink and downlink packets and bytes dropped due to new application/groups of applications aggregate bandwidth limit:

```

an3000-mcm-slot7cpu1#show subscriber pdn-session <id>

```

To display crbn profile and VSA data, use the following command:

```

an3000-mcm-slot7cpu1# show subscriber pdn-session 208903 crbn-profiles
crbn-profiles CRBN-PROF

```

```

an3000-mcm-slot7cpu1# show subscriber pdn-session 208903
vsa-cf                                     ""
vsa-entitlement                           ""
vsa-paymenttype                           PR
vsa-subid                                 05005551051
vsa-httpolicy                             ABR
vsa-midial                                ""
vsa-resellerid                             MVNO-1
vsa-subclassid                             ""
vsa-sponsor-url                           ""
vsa-account-status                         ""

```

```

vsa-bd-policy          ""
vsa-mms-capable        ""
vsa-roam-zone          ""

```

show subscriber service-profile summary

Subscriber Profiles (SPs) are objects containing mobile user specific information. SPs are used to store Toll Free Data (TFD) manifests and are also used to store future per-subscriber parameters such as user preferences, content feature eligibility, optimization opt-outs, content-filtering subscription status, etc.

SPs contain:

- subscriber key (for example, session ID or MSISDN)
- a list of TFD manifests (each with manifest ID + expiration time + URL regexes)

SPs are stored in the Subscriber Profile Table (SPTBL) on the ASMs / CSMs.

Subscriber service profiles are replicated to all ASMs to support redundancy of toll free data manifests.

Use the **show subscriber service-profile** command to display information from the SPS database. The **show summary** command displays an overview of all subscriber MSISDN, IMSI and the number of manifests that subscriber is holding. For example:

```
#show subscriber service-profile summary
```

MSISDN	IMSI	Manifest Total
917567665558999	17567547897899	1
917567665558779	17567547898499	1

The show detail command displays a detailed view of subscriber information including the details in the manifest by the given MSISDN. For example:

```
an3000-mcm-slot7cpu1# show subscriber service-profile detail  
917567665558999
```

```
subscriber service-profile detail 917567665558999
```

```
imsi      17567547897899
```

```
Manifest #1:
```

```
key:          affirmed  
id:           affirmed_manifest_id  
sponsor-id:   Affirmed Networks  
callback-url: http://httpbin.org/post  
version:      2.0  
validity-time: Thu Jun  2 19:17:02 2016(GMT)
```

```
Application #1:
```

```
application-id: affirmed_1  
mf-action:      TF  
url-list:  
    http://vas-webserver1:80/*
```

```
Application #2:
```

```
application-id: affirmed_2  
mf-action:      TF  
url-list:  
    connect://vas-webserver1.affirmed.local:443/*  
    https://vas-webserver1-tf.affirmed.local:443/*  
    http://vas-manifestserver.affirmed.local:8080/*  
    https://vas-webserver1.affirmed.local:443/*
```

```
Manifest #2:
```

```
key:          Patriots  
id:           Patriots id  
sponsor-id:   New England Patriots  
callback-url: http://bourg.net/callback  
version:      1.0  
validity-time: Thu Jun  2 19:26:24 2016(GMT)
```

```
Applications #1:
```

```
application-id: Patriots App  
mf-action:      TF  
url-list:  
    http://www.patriots.com:80/*
```


Managing the System

This section describes the objects and commands used to manage the system. Specifically, this section describes how to save the current configuration, restore a saved configuration, configure a neighbor discovery profile, and prepare for and perform an upgrade. This section also describes how to generate and view logs.

image-management

This section includes information on preparing for and performing the upgrade and recovering from a failed online upgrade. This section also includes information on upgrading the DPI signature. Contact the TAC for assistance with these procedures.

Configure the objects as follows:

Table 3-1 image-management

Object	Description
onlineupgrade	Perform an online upgrade of the node to the specified version (from displayAvailableAdminVersions). See onlineupgrade (page 3-6) .
prepare	Prepare the cluster for upgrade to the specified version (from displayAvailableAdminVersions). See Preparing for the Upgrade (page 3-7) .
releasemodifier	Change the release modifier version applied to the current release version. See show running-config infrastructure (page 3-49) . ■ <i>version</i> – Version to apply to the current release. ■ <i>rollback</i> – Modifier type to which to toggle back to the last version.
removeAdminVersion	Remove a version from the list of availableAdminVersions.
reset	Reset the upgrade engine after a failure in order to retry.
rollback	■ Rollback the node to the previous version (show image-management release-modifier-status rollback-version). See show image-management release-modifier-status (page 3-50).
force	Specify that the operation wait to complete until both MCMs have caught up (rather than wait for all nodes to catch up).
upgrade	Upgrade the cluster to the specified version (from displayAvailableAdminVersions). See upgrade (page 3-6) .

Upgrading the DPI Signature

This section describes how to upgrade the DPI Signature version on the MCC.

To upgrade the DPI Signature:

- 1 Copy the DPI Signature file (*dpisig_<date>.tgz*) to the */public/images* folder on MCM.

- 2 Verify that the *dpisig_<date>* version is available.

```
an3000-mcm-slot7cpu1# show image-management available-admin-versions
result success
reason Image management action accepted
dpisig_2019-05-24
```

- 3 Check the current *dpisig* release.

```
an3000-mcm-slot7cpu1# show image-management release-modifier-status
image-management release-modifier-status
RELEASE          RELEASE OPERATIONAL
ROLE             VERSION  MODIFIER  VERSION  ROLLBACK VERSION  ORIGINAL
VERSION
-----
rollback      8.4.1.0-53.REL  dpisig    dpisig_2019-05-10  dpisig_2019-06-13-1
dpisig_2018-12-14
rollback      8.4.1.0-53.REL  maint     maint_2017-06-12   NotAvailable
maint_2017-06-12
rollback      8.4.1.0-53.REL  plugin    plugin_2019-03-25  NotAvailable
plugin_2015-09-09
operational   9.2.0.0-1108.REL dpisig    dpisig_2019-05-10  NotAvailable
dpisig_2019-03-22
operational   9.2.0.0-1108.REL maint     maint_2017-06-12   NotAvailable
maint_2017-06-12
operational   9.2.0.0-1108.REL plugin    plugin_2019-03-25  NotAvailable
plugin_2019-03-25
```

- 4 Start the DPI signature upgrade:

```
an3000-mcm-slot7cpu1# image-management releasemodifier version
dpisig_<date>
result success
reason Image management action accepted
NoActivationInProgress
an3000-mcm-slot7cpu1#
```

- 5 Check the upgrade progress.

```
an3000-mcm-slot7cpu1# show image-management progress
result success
reason Image management action accepted
09:21:58.02->Modifier upgrade requested
09:22:01.02->Started: Validate Release Modifier dpisig_2019-05-24
```

- 6 After the upgrade process completes, verify that no failures exist. Your output may differ slightly, based on the release you are running.

```
an3000-mcm-slot7cpu1# show image-management last-activation-result
result success
reason Image management action accepted
09:21:58.02->Modifier upgrade requested
09:22:01.02->Started: Validate Release Modifier dpisig_2019-05-24
09:22:07.06-> dpisig_2019-05-24.tgz package is valid
09:22:07.06-> dpisig_2019-05-24.tgz is compatible with the current
release
09:22:07.06-> dpisig_2019-05-24.tgz dpiAppProtoDefines.xml is
compatible with the current configuration
09:22:07.06->Completed: Validate Release Modifier dpisig_2019-05-24 ok
09:22:10.06->Started: Upgrade Release Modifier dpisig_2019-05-24
09:22:10.06->Completed: Upgrade Release Modifier dpisig_2019-05-24 ok
09:22:13.07->Started: Apply Release Modifier
09:23:10.08->Apply Release Modifier waiting for 3 of 3 cpus :
14-v-csm-1-1, 14-v-ssm-5-1, 14-v-mcm-7-1
09:23:23.65->14-v-mcm-7-1: applied dpisig_2019-05-24 to
9.2.0.0-1108.REL
09:23:33.83->14-v-ssm-5-1: applied dpisig_2019-05-24 to
9.2.0.0-1108.REL
09:23:35.17->14-v-csm-1-1: applied dpisig_2019-05-24 to
9.2.0.0-1108.REL
09:23:43.09->Completed: Apply Release Modifier ok
an3000-mcm-slot7cpu1#
```

- 7 Verify that *dpisig_<date>* is the operational version.

```
an3000-mcm-slot7cpu1# show image-management release-modifier-status
```

```
image-management release-modifier-status
RELEASE          RELEASE OPERATIONAL
ROLE             VERSION  MODIFIER  VERSION  ROLLBACK VERSION  ORIGINAL
VERSION
-----
rollback 8.4.1.0-53.REL dpisig dpisig_2019-05-10 dpisig_2019-06-13-1
dpisig_2018-12-14
rollback 8.4.1.0-53.REL maint maint_2017-06-12 NotAvailable
maint_2017-06-12
rollback 8.4.1.0-53.REL plugin plugin_2019-03-25 NotAvailable
plugin_2015-09-09
operational 9.2.0.0-1108.REL dpisig dpisig_2019-05-24 dpisig_2019-05-10
dpisig_2019-03-22
operational 9.2.0.0-1108.REL maint maint_2017-06-12 NotAvailable
maint_2017-06-12
operational 9.2.0.0-1108.REL plugin plugin_2019-03-25 NotAvailable
plugin_2019-03-25
an3000-mcm-slot7cpu1#
```


- 8 Verify that the **DETECTION LIBRARY VERSION** is updated with the new version <YY.MM.DD> for all the nodes. Your output may differ slightly, based on the release you are running.

Note: *If the DETECTION LIBRARY VERSION is not updated with the new date, the SSM/ISM/WSM nodes may require a reboot. Refer to the DPI Classification Release Notes to see if a node reboot is required.*

```
an3000-mcm-slot7cpu1# show ip-data-path dpi dpi-image-version
ip-data-path dpi dpi-image-version
      DETECTION                                CDC
      LIBRARY                                LIBRARY
DECODER
ID  VERSION  DETECTION LIBRARY MD5SUM          VERSION  CDC DETECTION LIBRARY MD5SUM
VERSION  DPIXML MD5SUM
-----
5  19.05.24  fdf1259d94fc4b578a3a63ec761591e6  1.0.2  93f0b9c06fc05d393d3d4e8e5c4a8ef8
4748a6b70a6049f4e6a630a869060820
an3000-mcm-slot7cpu1#
```

Rollback Procedure

This section describes how to rollback to the previous or original dpisig versions.

To rollback to the previous / original dpisig version:

- 1 Execute the following commands, as applicable.
 - To rollback to the previous dpisig version:


```
image-management releasemodifier rollback dpisig
```
 - To rollback to the original dpisig version:


```
image-management releasemodifier original dpisig
```
- 2 Follow [step 5](#) to [step 8 on page 3-5](#) to verify the status.
- 3 Reboot the SSM/ISM/WSM nodes displayed in the output to apply the rollback/original dpisig version. For example:

```
an3000-mcm-slot7cpu1# show ip-data-path dpi dpi-image-version
```

onlineupgrade

The system supports online software upgrade, which moves the software image to the new release while the NE continues to provide service. Contact the TAC for assistance with this procedure.

```
an3000-mcm-slot7cpu1(config)# image-management onlineupgrade version <version>
```

- *continue-on-payload-timeout* – Continue even when payload nodes fail to fully reach oper-state up. This object impacts service. If any tasks hang during the OSU of a payload node, use the command line option **continue-on-payload-timeout** to continue with the OSU. This option enables the OSU to continue even if a TOS task hangs during a payload OSU. If a task hang occurs, the OSU continues but service is impacted.
- *drain* – Drain timeout in seconds. Allows tasks to drain of work before being replaced. It can reduce the disruption caused by replacement, but may significantly extend the time to complete the upgrade (especially if strategy is taskreset).
- *strategy* – Reset strategy to control how many tasks are replaced at a time: cardreset, tosreset, taskreset.

Online software upgrade automatically selects a least-impact reset strategy based on the difference between the current and new release:

- *cardreset* – If the kernel/rootfs dependencies have changed, the system performs a card reset to update the kernel/rootfs.
 - *taskreset* – If only one or more task's dependencies have changed, the system restarts only those tasks. The other tasks are rebased.
 - *tosreset* – If a task's dependencies are unchanged, the task is notified to use the new release's directory without being restarted.
- *version* – Version to upgrade to.

upgrade

Prior to performing the upgrade, verify the following conditions are met:

- Node is redundant
- Node is operationally up (all tasks Operationally up)
- Data model of the new release is identical or a superset of the current release

Note: Prior to performing the upgrade, contact the TAC to verify the supported upgrade paths.

The upgrade process involves two steps:

- *Preparing for the upgrade* – Open the image files and copy them to the system. See [Preparing for the Upgrade \(page 3-7\)](#).
- *Performing the upgrade* – Perform the upgrade steps to trigger the upgrade. The node reboots after the upgrade completes. See [Performing the Upgrade \(page 3-8\)](#)

Preparing for the Upgrade

The optional prepare command prepares the chassis for upgrade. Affirmed Networks recommends preparing a release outside of a maintenance window. Options include:

- *force* – Specify that the operation wait to complete until both MCMs have caught up (rather than wait for all nodes to catch up)
- *version* – Specify the version to prepare.

To prepare for upgrade:

- 1 Copy the release file (*an-<version>.tgz*) to */public/images*.
- 2 Enter the following command to confirm that the images appear as an available image:

```
an3000-mcm-slot7cpu1# show image-management available-admin-versions
result success
reason Image management action accepted
4.2.2.1-32.REL
an3000-mcm-slot7cpu1#
```
- 3 Enter the following command to prepare for the upgrade:

```
an3000-mcm-slot7cpu1(config)# image-management prepare version
<version>
```
- 4 Enter the following command to observe the prepare progress:

```
an3000-mcm-slot7cpu1(config)# show image-management upgrade-progress
```
- 5 Enter the following command to check the progress of the completed or failed prepare:

```
an3000-mcm-slot7cpu1(config)# show image-management
last-activation-result
```
- 6 Enter the following command to list the prepared version:

```
an3000-mcm-slot7cpu1# show image-management status
image-management status
```

STATE	ID	SLOT	CPU	ADMIN VERSION	OPERATIONAL VERSION	ACTIVATION STATUS	ROLLBACK VERSION	PREPARED VERSION
active	109	7	1	4.2.1.0-47.REL	4.2.1.0-47.REL	Idle	NotAvailable	4.2.2.1-32.REL

```
an3000-mcm-slot7cpu1#
```

Performing the Upgrade

Note: Check the disk space prior to performing an upgrade.

The upgrade command moves the software image to the new release. Options include:

- **version** – Version to upgrade to.
- **force** – Specify that the operation wait to complete until both MCMs have caught up (rather than wait for all nodes to catch up)

To perform the upgrade:

- 1 Copy the release file (an-<version>.tgz) to /public/images.
- 2 Enter the following command to perform the upgrade:

```
an3000-mcm-slot7cpu1(config)# image-management upgrade version <version>
```
- 3 Enter the following command to check the progress of the completed or failed upgrade:

```
an3000-mcm-slot7cpu1(config)# show image-management last-activation-result
```

Note: In the event that an upgrade fails, enter the **show image-management upgrade-progress** CLI command. If you encounter output similar to the following, it may indicate that the disk is full.

```
12:48:09.66->1-mm-7-1: error preparing tos 4.2.2.0-192.REL
Contact the TAC for further upgrade assistance.
```

- 4 Enter the following command to list the new version in the Admin Version and Operational Version columns and the rollback version in the Rollback Version column:

```
an3000-mcm-slot7cpu1# show image-management status
image-management status
```

STATE	ID	SLOT	CPU	ADMIN VERSION	OPERATIONAL VERSION	ACTIVATION STATUS	ROLLBACK VERSION	PREPARED VERSION
active	109	7	1	4.2.2.1-32.REL	4.2.2.1-32.REL	Idle		4.2.1.0-47.REL
NotAvailable								

```
an3000-mcm-slot7cpu1#
```

Virtual Example

```
an3000-mcm-slot8cpu1# show cluster summary
```

```
cluster
```

	SLOT	CPU			ADMIN	HA	CPU					
ID	NUMBER	NUMBER	TYPE	PERSONALITY	MODEL	STATE	STATE	STATE	CPU UPTIME	RELEASE		
14	2	1	v-csm DCM	vCSM	enabled	active	up		D:000 H:00 M:37 S:48	6.6.0.3-985.REL.ENG		
3	1	1	v-csm DCM	vCSM	enabled	active	up		D:000 H:00 M:37 S:48	6.6.0.3-985.REL.ENG		
5	1	1	v-ssm WSM	vSSM	enabled	active	up		D:000 H:00 M:37 S:47	6.6.0.3-985.REL.ENG		
6	1	1	v-ssm IOM	vSSM	enabled	active	up		D:000 H:00 M:38 S:02	6.6.0.3-985.REL.ENG		
7	1	1	v-mcm MCM	vMCM	enabled	secondary	up		D:000 H:00 M:16 S:02	6.6.0.3-985.REL.ENG		
8	1	1	v-mcm MCM	vMCM	enabled	primary	up		D:000 H:00 M:24 S:20	6.6.0.3-985.REL.ENG		
9	1	1	v-csm CCM	vCSM	enabled	active	up		D:000 H:00 M:37 S:49	6.6.0.3-985.REL.ENG		
128	1	1	v-csm ASM	vCSM	enabled	active	up		D:000 H:00 M:37 S:47			

Recovering From a Failed Online Upgrade

If the **show image-management status** command lists Activation Status **Failed**, you can recover from the failure.

```
an3000-mcm-slot7cpu1(config)# show image-management status
```

NAME	ADMIN VERSION	OPERATIONAL VERSION	ACTIVATION STATUS
ROLLBACK VERSION	PREPARED VERSION		
ImDir	1.1.0.0-7617.18.REL	1.1.0.0-7617.18.REL	Failed
	Unavailable	1.1.0.0-8617.78.REL	

If the failure occurred during validation, the Prepared Version column lists the new release. If the failure occurred post-validation, the Rollback Version column lists the previous release.

To recover the failure:

- 1 Enter the following command to reset the upgrade engine:

```
an3000-mcm-slot7cpu1(config)# image-management reset
```
- 2 In the case of a post-validation failure, enter the following command to perform a rollback to restore the system to the original state:

```
an3000-mcm-slot7cpu1(config)# image-management rollback
```

infrastructure dynamic-scale

Configure the dynamic scaling configuration for the ASM, DCM, and WSM.

Configure the objects as follows:

Table 3-2 infrastructure dynamic-scale

Object	Description
ASM	■ disk-size – Configure per ASM node disk size in GB. ■ node-connection-capacity – Configure per ASM node connection limit. ■ node-throughput-capacity – Configure per ASM node Throughput (in Mbps) limit.
DCM	node-session-capacity – Configure per DCM node session limit.
WSM	node-session-capacity – Configure per WSM node session limit.

infrastructure export-ip mgmtip

Associate the loopback-ip and network-context with the export object.

Configure the objects as follows:

Table 3-3 infrastructure export-ip mgmtip

Object	Description
loopback-ip*	Name of loopback IP address to be used for export scp/sftp.
network-context*	Network context of loopback IP address.

This section describes how to configure a separate network-context and loopback-ip for access to export records through a North-South interface. The export-ip mgmtip object supports a separate network-context and loopback-ip to enable file management export users to access MCC export files without having to connect through the standard management-context, mgmt-ip address.

You can enable and disable the network-context to act as a Secure Copy Protocol (SCP) or SFTP login port. Connections are made through the loopback-ip address on the north-south subnet using port 2221, for example:

- `sftp -P 2221 admin@<loopback-ip>`
- `scp -P 2221 admin@<loopback-ip>:/* .`

Note: Some export transfers may experience lower performance rates using this port.

The infrastructure file-management rule requires that both an export-rule and export-user be defined for export users to access the network-context address through SCP/SFTP. See [infrastructure file-management profile \(page 3-12\)](#).

Note: Depending on the loopback-ip address you choose for the export-ip network-context, you may need to create a default route on the client machine to forward traffic intended for the loopback-ip to the network-context's interface-ip address. The system will recognize this address as the correct route.

Use the following commands as an example:

- 1 Create a loopback-ip on the network context:

```
an3000-mcm-slot7cpu1(config)# network-context <network-context name>
loopback-ip <loopback-ip-name> ip-address <ip-address>
```
- 2 Associate the loopback-ip with the export object:

```
an3000-mcm-slot7cpu1(config)# infrastructure export-ip mgmtip
network-context <network-context name> loopback-ip <loopback-ip-name>
```
- 3 Configure the export rule to apply when exporting the file to the ./export directory.

```
an3000-mcm-slot7cpu1(config)# infrastructure file-management profile
<type> <filename-prefix> export-rule
[exportrule-default|exportrule-gy-custom1|exportrule-gz-custom1|export
rule-none]
```

See [infrastructure file-management profile \(page 3-12\)](#) for details.
- 4 Specify the user that has read/write access to the files in the ./export directory (depending on the file type).

```
an3000-mcm-slot7cpu1(config)# infrastructure file-management profile
<type> <filename-prefix> export-user <string>
```

infrastructure file-management profile

Operators can create a local file in which to store CDRs. Typically, ASN.1 encoded CDR files are delivered over the Ga interface. Configure a profile for file upload and purging. Configure parameters for transferring and rotating the file. Operators can configure the following profile types: faults, datarecord, eventlog, and other. Also see [infrastructure logging event-log policy \(page 3-18\)](#).

Local data records can be transferred to a billing server in either push or pull mode. To configure data record export rules and user permissions, see the *Affirmed Networks Charging Data Records Specification*.

MCC supports Packet Forwarding Control Protocol (PFCP) logging to capture Sx Node Related messages and Sx Session Related messages sent/received by the PGW-C, SGW-C, PGW-U and SGW-U in the Evolved Packet Core (EPC) Control and User Plane Separation (CUPS) architecture. The MCC captures the packets and stores them as pcapng capture files in *var/log/eventlog* on both the control plane function (CPF) and user plane function (UPF). You can view these files in Wireshark.

Configure the objects in config mode as follows:

Table 3-4 infrastructure file-management profile

Object	Description
compress-enabled	Should this file be compressed (zipped) when it is closed? ■ Compressed data record files are stored in <i>/public/log/datarecord.ink</i> ■ Compressed event logs are stored in <i>/var/log/eventlog</i>.
compress-format	If compress enabled, in what format should file be compressed? ■ gzip – (default) data records are written to a file. After the file is closed it is gzipped. ■ raw – data records are written in zlib raw format. The zlib raw format provides better performance compared with the gzip format because the records are compressed on the source prior to sending them to the agent for writing. Affirmed Networks recommends using the zlib raw format if you expect the system to generate data records at a rate higher than 10K data records per second. See Limitations (page 3-16).
dest-port	Port of file sftp server
dest-url	URL/IpAddress for file sftp server

Table 3-4 infrastructure file-management profile

Object	Description
export-rule	What rules to apply when exporting file to <code>./export</code> directory. Also see the <i>Affirmed Networks Charging Data Records Specification</i> for details. Options include: ■ <code>exportrule-default</code> – The system exports files to the <code>./export</code> directory and applies purge rules for this file profile to all files in the directory. ■ <code>exportrule-gy-custom1</code> – The system exports files to the <code>/public/logs/datarecord.lnk/gy/export/AP_primary</code> directory according to the <code>custom1</code> format configured for CCR-T records. ■ <code>exportrule-gz-custom1</code> – The system exports files to the <code>/public/logs/datarecord.lnk/gz/export</code> directory according to the <code>custom1</code> format configured for Gz/Rf SGW and PGW CDRs. ■ <code>exportrule-none</code> – Specifies that there is no export rule for this file profile. In this case, the MCC sends (pushes) the files to the remote SFTP server based on the upload parameters configured for the file profile.
export-user	Give this user read/write access to files exported.
file-extension	Optional file extension to be appended to file-name.
file-name-format	File name format (default or custom). Enables Operators to customize the file naming convention for local CDRs (ASN.1 only). This object only applies to the datarecord profile. ■ <code>default</code> – The MCC uses the default file naming convention for local CDR files: <code><filename-prefix><YYYYMMDDHHMMSS>_<123>.csv</code> ■ <code>custom1</code> – The MCC uses the following custom file naming convention for local CDR files (ASN.1 only): <code><filename-prefix>_MM_DD_YYYY_hh_mm_ss_<#records>_file<123>.u</code> where: <code>filename-prefix</code> – Indicates the prefix of the filename in which to store CDRs. <code>MM_DD_YYYY_hh_mm_ss</code> – Represents the timestamp. <code>#records</code> – Represents the number of records stored in the file, starting with 0. <code>123</code> – Represents the file sequence number. The number of leading zeros prepended to the file sequence number depends on the value configured for the <code>sequence-field-width</code>. For example, if the <code>sequence-field-width</code> is set to 6, the sequence number will start with 000001. Note: In the rare case of an uncontrolled MCM switchover caused by a crash in the <code>DataRecordAgentMCM</code> process or an MCM kernel crash, the currently open CDR files will remain with a record count of 0, indicating that the number of records could not be determined when the file was closed.

Table 3-4 infrastructure file-management profile

Object	Description
file-rotate-empty	Set to enabled/disabled to indicate if files should be rotated when empty. ■ enabled – Empty CDR/HTDR files are rotated based on the configured file-rotate-frequency interval. ■ disabled – (default) Empty CDR/HTDR files are not rotated.
file-rotate-frequency	The file will be rotated only if the current file has not reached the maximum size or maximum number of records within the specified amount of time (as configured by file-rotate-frequency). For example, if the file-rotate-frequency is set to 1 hour and within that hour, the file reaches the maximum number of records, the file is rotated immediately and is not rotated upon reaching the 1 hour time interval. Options include: 1min, 5mins, 15mins, 30mins, 1hour (default), 2hours, 3hours, 6hour, 12hours, 1day, 7days, 14days, 21days, 30days, 1year.
file-rotate-num-records	Files should be rotated once the file has these many records.
file-rotate-size-bytes	Files should be rotated once they reach size [bytes], max size 1GB. 1 .. 1073741824(10485760)
file-transfer-enabled	Is file transfer enabled for this file?
notify-thr	At what percentage utilization should a notification be sent?
passwd	Password to be used to log into the file sftp server
prefix*	Prefix for file name.
private-information	Optional private information to be appended to file-name.
purge-enabled	Is file purge/cleanup enabled for this file?
purge-file-older-than	Purge files older than X time. Options include: 1min, 5mins, 15min, 30mins, 1hour, 2hours, 3hours, 6hour, 12hours, 1day, 7days, 14days, 21days, 30days, 1year.
purge-low-thr	At what percentage utilization should transferring/purging of files stop? 0 .. 100(0)
purge-thr	At what percentage utilization should files be transferred/purged? 0 .. 100(100)

Table 3-4 infrastructure file-management profile

Object	Description
sequence-field-width	Defines the number of leading zeros to prepend to the filename sequence number. Specify a value between 0 (default) and 10. See Limitations (page 3-16). To facilitate file retrieval and processing, MCC automatically appends a sequence number to the filename to indicate the file. MCC uses the following file naming convention: <filename><YYYYMMDDHHMMSS>_<123>.csv where YYYYMMDDHHMMSS represents the timestamp and 123 represents the file sequence number. For example, <i>pgwcd20161114163821_123.csv</i>. The following examples show how the filename would appear depending on the value configured for the sequence-field-width. sequence-field-width = 0 – <i>pgwcd20161114163821_123.csv</i> sequence-field-width = 3 – <i>pgwcd20161114163821_123.csv</i> sequence-field-width = 7 – <i>pgwcd20161114163821_0000123.csv</i> sequence-field-width = 10 – <i>pgwcd20161114163821_0000000123.csv</i>
type*	What type of file is this? Options include <code>eventlog datarecord other fault ramdatarecord storage ext-storage ramdisk-storage psm tech-support gz-record gy-record </code> <code>ramdatarecord</code> – To direct CDR/NBR onto RAMDISK, configure a file-management profile of type <code>ramdatarecord</code> for file upload and purging. For example: infrastructure file-management profile ramdatarecord <profile name> where <profile name> is used as the prefix prepended to all NBR files associated with the profile. For NBR records, configure a port NBR interface service profile of type <code>ramdatarecord</code>. For example: service-construct interface port-nbr-service <profile name> local nbr-file-type ramdatarecord nbr-file-name
upload-dir-path	At what path on the sftp server should the files be uploaded.
upload-frequency	How often should the files be uploaded? Options include: 1min, 5mins, 15mins, 30mins, 1hour, 2hours, 3hours, 6hour, 12hours, 1day, 7days, 14days, 21days, 30days, 1year.
upload-offset-secs	At what hour should the file upload start? 0 .. 86400(0)
username	Username to be used to log into the file sftp server

Limitations

If you select the **compress-format** raw, Operators must use the Affirmed Networks uncompress utility to uncompress the data records. Affirmed Networks recommends copying the data records to a local server/directory to which you have write permissions. You must be running Linux 2.6.22 or higher to use this utility.

The utility is located on the MCC as follows:

```
/tos/<release_number>/opt/thoroughbred/bin/affirmedUnzip.
```

Copy this utility to the location on your data records server where the compressed data records are being sftped. You can run this utility offline.

For example:

```
maint@an3000-mcm-slot7cpu1:~# affirmedUnzip
linux: affirmedUnzip <file | directory> <file | directory> ...
```

This utility uncompresses files using the ZLIB RAW option (files with .zraw extension). For example:

```
-rw-r--r-- 1 maint maint 10486201 Mar 13 12:42
bob20150313123001_132.csv.zraw
```

When a directory is entered, the utility uncompress all *.zraw files in that directory. Use a wildcard in the file name to uncompress multiple files matching a certain file name.

For example:

```
linux:~# affirmedUnzip
/var/log/datarecord/test/testcdr20150311162909_2244.csv.zraw
Uncompressing testcdr20150311162909_2244.csv.zraw
..... SUCCEEDED
.(6.0.0.0-458.REL)maint@an3000-mcm-slot7cpu1:~#
```

infrastructure logging

The MCC logs any configuration changes made to the system. By default, these CLI audit logs are captured to a log file located on the system's disk. You can set the hour and minute at which to upload the log file. You can also set the upload frequency and file rotation frequency in the following increments:

- 5 minutes
- 15 minutes
- 30 minutes
- 1 hour
- 6 hours
- 1 day

***Note:** Affirmed Networks recommends using the default log file format, binary. If you change the log file format to ASCII, some logs may be dropped if the system becomes overloaded. Only set the log file format to ASCII when the anticipated log volume is low.*

Operators can configure the system to stream the CLI audit logs to a remote syslog server, using the following steps:

- 1 Configure a log policy to capture only the CLI audit logs to the system's syslog facility.
- 2 Set the sink type to **printf** so that the log is written out to the /var/log/messages (that is, syslog) file.
- 3 Configure the syslog facility to stream the logs to a remote server.

The following is an example of a log policy called CLIAudit.

Note: Two filters (*component = Nml* and *componentlogid = 9*) are required so that only the CLI commands executed are logged.

```
an3000-mcm-slot7cpu1# show running-config infrastructure logging
CLIAudit
infrastructure logging event-log policy CLIAudit
log-type debug
filter-field component
    filter-value Nml
    exclude false
!
!
filter-field componentlogid
    filter-value 9
    exclude false
!
!
sinks CLIAuditSink
    type printf
!
```

infrastructure logging event-log policy

Using the **infrastructure logging** CLI command, Operators can monitor the system and generate event logs and system logs ([page 3-29](#)).

MCC maintains event logs that show a history of the activity on the system. Event logs are useful for monitoring the health of the cluster and the events that have occurred.

By default, the system displays event logs in binary format and stores them as *.bin files. Optionally, you can configure the system to display logs in ASCII format and store them as *.txt files (see [Changing the Log File Format to ASCII \(Optional\)](#) on [page 3-27](#)). MCC appends PID (Process ID) or LWPID (Lightweight Process ID) to the log file to indicate the process/thread that generated the log.

Note: Each *sink* must refer to a unique **infrastructure file-management profile** prefix. The prefix must be unique across all event-log policies.

The logs are stored in `/var/log/eventlog`. Event logs can also be written to the system log files (see [infrastructure logging sys-logs](#) on [page 3-29](#)) and stored in `/var/log/messages*`. To view the log files, you must first parse the files (see [Parsing Binary Log Files](#) on [page 3-26](#)).

The MCC framework disallows a badly behaving client to flood the system with events. MCC uses rate control to prevent clients from overwhelming the server.

The MCC infrastructure supports the parsing and bottlenecking of traps to prevent surges and is designed to distribute throttling overhead and filter duplicate events. MCC limits the number of event types that can be forwarded simultaneously and limits the number of events forwarded within a configured time.

Message drop rates are logged in `/var/log/messages` at the Debug level.

Configure the field, service, component, value, protocol, and severity on which to filter. In addition, configure the RADIUS, gtp, diameter, and gtp interface on which to filter. Configure logging sinks.

For example:

Simple Entitlement Example:

```
2015-09-18T10:50:57.692196098+0000 Sev=6 [Content] [LocalHttpServer]
[ID=00] [00015d010001206d] [18768 21286]
[HttpProxy_93_1_1_8_HttpProxy_8] "[IMSI:35712345678993 FlowID:0
SocketID:33 ReqID:177] Request Serviced Successfully:
MSISDN[9986543259858993] EntitlementMode[simple]
ServicesEntitledTo[GTETH GVC GVC1 MHS1 MHSM TETH TOGGL TGLSB TGLSI ]
QueriedService[facetime] ResultCode[200]"
```

Secure Entitlement Example:

```
2015-09-18T11:25:57.569351334+0000 Sev=6 [Content] [LocalHttpServer]
[ID=00] [00015d0100012067] [18941 21284]
[HttpProxy_93_1_1_2_HttpProxy_2] "[IMSI:35712345678993 FlowID:0
SocketID:33 ReqID:192] Request Serviced Successfully:
MSISDN[9986543259858993] EntitlementMode[secure]
ServicesEntitledTo[GTETH GVC GVC1 MHS1 MHSM TETH TOGGL TGLSB TGLSI ]
QueriedService/Result [tethering/6100] [facetime/6100] "
```

Configure the objects as follows:

Table 3-5 infrastructure logging event-log policy

Object	Description
admin-state	Enable/disable the admin state.
log-type	Operators can create an event log policy log-type as debug, protocol, record-debug, or record-protocol. For example: ■ log-type debug filter-field (page 3-21) – captures various events in the system and filters logs. ■ log-type protocol protocol-logging-filter-field (page 3-23) – captures control and data packets coming in and going out of the system.
filter-field	Configure field on which to filter: apn, component, componentlogid, cpu, gwname, httpreqtype, imei, imsi, ipaddress, macaddress, msisdn, nai, npuflowid, npunum, npuslot, portnumber, service, severity, slot, teid, vpn
filter-value	Configure the filter value.
exclude	Indicate if logs with this value should be filtered.
sink	Logging Sinks

Entitlement Transaction Logs

MCC records the following information for each entitlement transaction in an audit log:

- MSISDN
- Entitlement mode (simple/secure)
- Services to which the user is entitled (data from the PCRF server)
- *Queried service(s)* – URL for simple entitlement and the name of the service extracted from the json for secure entitlement.
- *Result for the queried service* – 200/403 response for simple entitlement or 6100,6101 or 6102 for secure entitlement.

Transaction Logs for WAP 2.0 Gateway

The MCC supports transaction logs for WAP 2.0 Gateway. This feature supports Event Data Records (EDRs) to provide a concise log of an HTTP/WAP transaction. This log contains the results of transactions that may be helpful to the Operator in gathering data used to analyze the trend in content traffic. This log can also be used by the Affirmed Networks TAC to debug issues. This feature supports the following EDRs:

- MMS (for MMS transactions)
- General (for HTTP/WAP transactions)

log-type debug filter-field

The log-type **debug** captures various events in the system and filters logs based on the following:

```
an3000-mcm-slot7cpu1(config)# infrastructure logging event-log policy
default log-type debug filter-field
```

Possible completions:

apn	component	componentlogid	cpu
gwname	httpreqtype	imei	imsi
ipaddress	macaddress	msisdn	nai
npufLOWid	npunum	npuslot	portnumber
service	severity	slot	teid
vpn			

log-type debug filter-field filter-value

Configure the value on which to filter the filter-field.

```
an3000-mcm-slot7cpu1(config)# infrastructure logging event-log policy
default filter-field component filter-value
```

Possible completions:

AnalyticsProxy	AvUtil	BFDAgent
BGPAgent	CALEA	CacheLib
CacheMgr	CatEng	ContentCommon
ContentFilter	CrashAgent	Diameter
Dispatcher	DnsLib	DnsResolver
EdrAgent	FIBAgent	FIREWALL
FileMgmt	FilterLib	FlowAgent
FlowMgr	Fm	FpAgent
FpGwHdlr	FpNml	GRD
GTP	GaAgt	GeneralEdr
GeoRedDirector	GtppTxRx	GwCommon
GwSd	GwStatsFw	HTTPProxy

Hm	HttpRequestor	Im
ImsMgr	L2tpManager	LDAPAgent
LicMgr	LicMgrAgt	LocalHttpServer
Logging	MMESim	Mms
MmsEdr	NRA	NasMgmt
NatServer	NetAgent	NetDir
NetKintf	NetSfm	NetSvc
Nml	Nmla	NpuAgent
NpuMon	NsmAgent	OspfAgent
PLA	PathMgmt	PerfStatMgmt
RFC822	Radius	SBM
SGW	SLB	SecDir
SecPolicyAgt	SecPolicyMgr	SecPolicyMgrESPKey
SecPolicyMgrIKEv1Key	SecPolicyMgrIKEv2Key	SessionLearner
SmtpProxy	SpsLib	TLSLib
TcpProxy	TranscodeMgr	UIM
UaProfileLib	UdpProxy	WapGw
WebOpt	WfDir	WfNmlLog
ZipLib	scheduler	test

log-type protocol protocol-logging-filter-field

Protocol log files gather control packets for a specific protocol (such as gtp, RADIUS, etc.) or a configured interface (accounting or authentication). For example:

```
infrastructure logging event-log policy default log-type protocol
protocol-logging-filter-field protocol filter-value
```

Possible completions:

```
diameter gtp gtpv0 gtpv1 gtpv2 radius
```

These packets can be filtered as follows:

```
an3000-mcm-slot7cpu1(config)# infrastructure logging event-log policy
default log-type protocol protocol-logging-filter-field
```

Possible completions:

destination-ip-address	destination-portnumber
imei	imsi
ip-address-pair	ip-protocol
ipaddress	msisdn
packet-direction	packet-type
protocol	source-ip-address
source-portnumber	

Protocol Logging for GiGW for Server and Proxy Mode

Use the following example to configure protocol logging for GiGW for server and proxy mode:

Configure protocol logging filter-field values:

```
an3000-mcm-slot7cpu1(config)# infrastructure logging event-log policy
<policy Name> log-type protocol protocol-logging-filter-field
```

Possible completions:

```
destination-ip-address destination-portnumber imei imsi
ip-address-pair ip-protocol ipaddress msisdn packet-direction
packet-type protocol source-ip-address source-portnumber
```

Configure IMSI as filter-field value:

```
an3000-mcm-slot7cpu1(config)# infrastructure logging event-log policy
<policy Name> log-type protocol protocol-logging-filter-field imsi
filter-value <imsi Value>
```

Configure IMEI /MSISDN as filter-value for protocol logging:

```
an3000-mcm-slot7cpu1(config)# infrastructure logging event-log policy
<policy name> log-type protocol protocol-logging-filter-field
imei/msisdn filter-value <imei /msisdn value>
```

Configure source-ip-address, destination-ip-address, source-portnumber, and destination-portnumber filter-value for Protocol Logging:

```
an3000-mcm-slot7cpu1(config)# infrastructure logging event-log policy
<Policy Name> log-type protocol protocol-logging-filter
destination-ip-address/
destination-portnumber/source-ip-address/source-portnumber
filter-value <value >
```

Note: While configuring these filters, you must also configure protocol-filter-value as follows:

```
an3000-mcm-slot7cpu1(config)# infrastructure logging event-log
policy <Policy Name> log-type protocol protocol-logging-filter-field
protocol filter-value radius
```

Configure packet-direction-filter-value:

```
an3000-mcm-slot7cpu1(config)# infrastructure logging event-log policy
<policy Name> log-type protocol protocol-logging-filter-field
packet-direction filter-value
```

Possible completions:

inbound outbound

Configure packet-type-filter-value:

```
an3000-mcm-slot7cpu1(config)# infrastructure logging event-log policy
<policy Name> log-type protocol protocol-logging-filter-field
packet-type filter-value
```

Possible completions:

ctrl data

Configure sink file attributes:

No packets are collected for a provided IMSI or other filter fields (for example, destination-ip).

```
an3000-mcm-slot7cpu1(config)# infrastructure logging event-log policy
<policy name> log-type protocol sinks <name> type file file-name <file
Name>file-format pcapng
```

Viewing Protocol Logs with Packet Analyzers

MCC performs protocol logging using application software that receives only Application Layer Packets (not Layer3 or Layer4 packet header information). To view protocol logs with packet analyzers, Affirmed Networks recommends that you configure the analyzer to ignore Layer3 and Layer4 header fields, otherwise the tool may incorrectly analyze those layers.

For example, in Wireshark:

- 1 Select **Edit > Preferences > Protocol**.
- 2 Select an individual protocol, such as TCP.
- 3 Unmark the checkbox that specifies to analyze that data.
- 4 Click **Apply**.
- 5 Repeat [step 2](#) through [step 4](#) for the other protocols.

Parsing Binary Log Files

To view log files, enter the following command to parse the binary files:

```
an3000-mcm-slot7cpu1# show logs logfile <filename>
```

The following output shows an example. The command parses a log file and displays the content. Since the output extends beyond the screen size, you must press the spacebar to display more output or enter **q** to quit.

```
show logs logfile tos20120125155222.bin
```

```
/var/log/eventlog/tos20120125155222.bin
```

```
Parsing Binary log file /var/log/eventlog/tos20120125155222.bin
```

```
Successfully parsed log file /var/log/eventlog/tos20120125155222.bin
```

```
2012-01-25T15:52:20.141425810-0500 Sev= 5 [GwCommon][GwStatsFw][
0][GwStatsAgent_17_1_1_GwStatsAgent_0] [GwStats]: GwStatsAgent[grid:1
chassis:17 slot:1]::GwStatsAgent::Received Get Stats Response from
GwStatsServiceAgent[slot:1 pid:73706]. TransactionId:1811
```

```
2012-01-25T15:52:20.141442201-0500 Sev= 5 [GwCommon][GwStatsFw][
0][GwStatsAgent_17_1_1_GwStatsAgent_0] [GwStats]: GwStatsAgent[grid:1
chassis:17 slot:1]::GwStatsAgent::Received Get Stats Response from
GwStatsServiceAgent[slot:1 pid:73720]. TransactionId:1811
```

```
2012-01-25T15:52:20.141458978-0500 Sev= 5 [GwCommon][GwStatsFw][
0][GwStatsAgent_17_1_1_GwStatsAgent_0] [GwStats]: GwStatsAgent[grid:1
chassis:17 slot:1]::GwStatsAgent::Received Get Stats Response from
GwStatsServiceAgent[slot:1 pid:73716]. TransactionId:1811
```

```
2012-01-25T15:52:20.141535881-0500 Sev= 5 [GwCommon][GwStatsFw][
0][GwStatsAgent_17_1_1_GwStatsAgent_0] [GwStats]: GwStatsAgent[grid:1
chassis:17 slot:1]::GwStatsAgent::Received Get Stats Response from
GwStatsServiceAgent[slot:1 pid:73723]. TransactionId:1811
```

```
2012-01-25T15:52:20.141560050-0500 Sev= 5 [GwCommon][GwStatsFw][
0][GwStatsAgent_17_1_1_GwStatsAgent_0] [GwStats]: GwStatsAgent[grid:1
chassis:17 slot:1]::GwStatsAgent::Received Get Stats Response from
GwStatsServiceAgent[slot:1 pid:73721]. TransactionId:1811
```

```
2012-01-25T15:52:20.141577281-0500 Sev= 5 [GwCommon][GwStatsFw][
0][GwStatsAgent_17_1_1_GwStatsAgent_0] [GwStats]: GwStatsAgent[grid:1
chassis:17 slot:1]::GwStatsAgent::Received Get Stats Response from
GwStatsServiceAgent[slot:1 pid:73709]. TransactionId:1811
```

```
.....
```

Changing the Log File Format to ASCII (Optional)

Note: Affirmed Networks recommends using the default log file format, binary. If you change the log file format to ASCII, some logs may be dropped if the system becomes overloaded. Only set the log file format to ASCII when the anticipated log volume is low.

To change the log file format to ASCII, enter the following command:

```
an3000-mcm-slot7cpu1(config)# infrastructure logging event-log policy
<your_policy_name> sinks <your_sink_name> file-format ascii
```

The following output shows an example of configuring the default and critical event log files to write to files in ASCII format.

```
an3000-mcm-slot7cpu1(config)# infrastructure logging event-log policy
default sinks filesink file-format asciii
an3000-mcm-slot7cpu1(config-sinks-filesink)# commit
Commit complete.
an3000-mcm-slot7cpu1(config)# infrastructure logging event-log policy
critical sinks filesink file-format asciii
an3000-mcm-slot7cpu1(config-sinks-filesink)# commit
Commit complete.
```

Streaming Audit Logs to a Remote Syslog Server

Use the instructions in this section to stream audit logs to a remote/syslog server.

- 1 Configure a log policy to capture only the CLI audit logs to the system's syslog facility.
- 2 Set the sink type to **printf** so that the log is written out to the `/var/log/messages` (that is, syslog) file.
- 3 Configure the syslog facility to stream the audit logs to a remote server. See [Steaming System Logs to a Remote Server \(page 3-31\)](#).

The following is an example of a log policy named `CLIAudit`.

Note: Two filters (`component = Nm1` and `componentlogid = 9`) are required so that only the CLI commands executed are logged.

```
an3000-mcm-slot7cpu1# show running-config infrastructure logging
CLIAudit
infrastructure logging event-log policy CLIAudit
log-type debug
filter-field component
    filter-value Nm1
    exclude false
!
!
filter-field componentlogid
    filter-value 9
    exclude false
!
!
sinks CLIAuditSink
    type printf
!
!
```


infrastructure logging sys-logs

Stream sys-logs to remote server. System log files are generated by the Linux subsystem, kernel, and other services/daemons running on the system. System logs are stored in `/var/log/messages*`.

Configure the objects as follows:

Table 3-6 infrastructure logging sys-logs

Object	Description
facilitylevel	Indicates the syslog facility level used to filter logs to a remote server. A facility code (and keyword) specifies the type of program that is logging the message. Messages with different facilities are handled differently. Locally used facilities include: local1, local2, local3, local4, local5, local6, and local7.
ip*	Remote server ip address
port	Remote server port-
protocol	tcp or udp protocol
stream-server	Syslog remote server sink. See Configuring the MCC rsyslog Client (page 3-32) .
hostname-prefix	String used to tag syslog entries. See Tagging syslog Messages (page 3-33) .

MCC generates the syslog only if the event-log policy is configured to allow generation of firewall events. The syslog is generated for all flows that are dropped or rejected due to MCC firewall functionality. MCC also generates the syslog for a newly created flow that is permitted based on the configured security policy rules.

Syslog supports the following features and functions:

- Provide syslogs of all flows that are permitted, dropped, or rejected (based on the configured action in the security policy rule).
- Send the syslogs to a remote server.
- Ability to enable/disable syslogs.

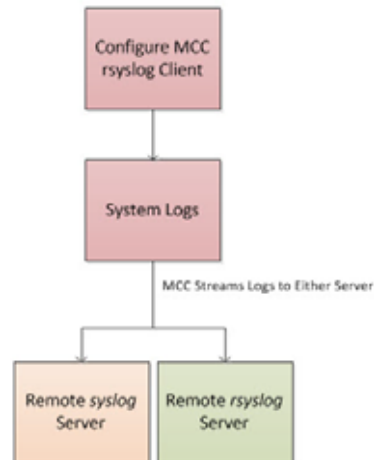
The following limitations apply to this feature:

- Upon an SSM crash, firewall records that were not sent to the MCM are lost.
- Upon an MCM crash, firewall records that were not sent to the syslog server are lost.
- Syslogs sent by the MCC per second may not be proportional to the number of new flows arriving at the MCC per second.

Steaming System Logs to a Remote Server

The MCC is capable of streaming system log files to a remote *rsyslog* server or remote *syslog* server. The MCC uses the *rsyslog* client (<http://www.rsyslog.com>) as its system logging daemon as shown in Figure 3-1.

Figure 3-1 Streaming System Log Files



To stream system logs:

- 1 Regardless of whether you are streaming logs to a remote *rsyslog* or remote *syslog* server, you must first configure the MCC *rsyslog* client (see [Configuring the MCC rsyslog Client](#) on [page 3-32](#)).
- 2 *Optionally*, If you are streaming logs to a remote *rsyslog* server, continue with [Configuring the conf File on the Remote rsyslog Server \(Optional\)](#) ([page 3-32](#)).

Configuring the MCC rsyslog Client

Before you begin, determine the IP address and the IP port of the remote rsyslog daemon used to receive the logs. The default IP port for rsyslogd is 514.

To configure the MCC rsyslog client:

- 1 Enter the following command to stream logs to a remote logging server:

```
an3000-mcm-slot7cpu1(config)# infrastructure logging sys-log
stream-server <ip> port <port> protocol <udp|tcp>
commit
```

where **<ip>** is the IP address and **<port>** is the IP port number for the remote logging server.

- If you are streaming logs to a remote *rsyslog* server, you can use UDP or TCP.
- If you are streaming logs to a remote *syslog* server, you must use UDP.

The system starts to stream logs to a remote server. To add multiple remote logging servers, repeat [step 1](#). The system stream logs to each configured remote server.

- 2 Enter the following command to stop logs from streaming to a remote logging server:

```
an3000-mcm-slot7cpu1(config)# no infrastructure logging sys-log
stream-server <ip> port <port> protocol <udp|tcp>
commit
```

Configuring the conf File on the Remote rsyslog Server (Optional)

If you are streaming logs to a remote *rsyslog* server, use the instructions in this section to configure the *rsyslog* conf file on the remote server in order to distinguish files from different clients. Affirmed Networks recommends using a custom template used to configure the logs with the IP address of the sender/client. The following is an example of a custom template:

- 1 Enter the following command to append “**FROMHOST**” in front of the raw log message:

```
$template myFormat,"%FROMHOST% %rawmsg%\n"
```

For example, **FROMHOST** may be an IP address or hostname so it can perform DNS lookup.

- 2 Enter the following command to set the default format for logs written to the file:

```
$ActionFileDefaultTemplate myFormat
```

Note: You can store logs from different clients on different log files. See <http://www.rsyslog.com/storing-messages-from-a-remote-system-into-a-specific-file/> for more information.

Tagging syslog Messages

This feature is useful in identifying the source of messages on a syslog server when multiple clusters and VMs are in use. This feature allows tagging of syslog messages between the timestamp and hostname fields. Assigning a mnemonic string to the new **hostname-prefix<string>** can be used to help identify the VM and cluster to determine the origin of the messages. For example, note the placement of the tag **<string>** in the following message:

```
2017-12-14T20:13:36.805797+00:00 <string>an3000-mcm-slot7cpu1
tgtscheduler: [GRDApp] Chassis audit completed - no work to do
```

To configure this feature, specify the string using the **hostname-prefix<string>** parameter as shown:

```
infrastructure logging sys-log hostname-prefix<string>
```

Generating syslogs for the Global Firewall

Enable the generation of syslog for firewall events using the existing event-log policy. Firewall events are filtered for inclusion in the syslog using the component **FIREWALL**. Firewall events, up to a certain severity, are filtered for inclusion in the syslog using the **severity** filter-field. Firewall events are streamed as syslog to a remote syslog server by configuring a syslog-sink.

Use following steps to generate syslogs for the global firewall:

- 1 Enable firewall functionality for a network context by enabling **security** on that network context.

```
network-context INTERNET
  security enabled
  number-of-security-services 1
  pass-local-ike-packets true
  security-geo-redundancy-group
```

- 2 Configure a set of security policy rules in the network context.

Flows matching the security policy rule with the highest **priority** (lowest value = highest priority) are permitted, dropped, or rejected depending on the **action** (pass, drop, reject) configured for the security policy rule.

```
security policy rule INTERNET SEC1 //Rule to allow traffic from
IP pools in LOCALPOOL(UE Pool) to 10.53.1.90-10.53.1.255
  priority 100
  action pass
  source-ip-pool user-equipment-pool LOCALPOOL
  destination-packet-filter DEST_OS_FILTER
  protocol 255
  ip-start 10.53.1.90
```

```
ip-end 10.53.1.255
```

- 3 Configure the filter-value **FIREWALL** to filter firewall events for inclusion in syslogs.

```
an3000-mcm-slot8cpu1# show running-config infrastructure logging  
sys-log
```

```
Mon May 20 02:53:44.399 UTC-04:00
```

```
infrastructure logging sys-log hostname-prefix ""
```

```
infrastructure logging sys-log stream-server 10.64.9.94
```

```
port 514
```

```
protocol tcp
```

```
facilitylevel local1
```

```
!
```

```
an3000-mcm-slot8cpu1# show running-config infrastructure logging  
event-log policy firewall
```

```
Mon May 20 02:57:12.590 UTC-04:00
```

```
infrastructure logging event-log policy firewall
```

```
admin-state enabled
```

```
log-type debug
```

```
filter-field component
```

```
filter-value FIREWALL
```

```
exclude false
```

```
!
```

```
!
```

```
sinks server
```

```
type stream-server
```

```
sinkserver 10.64.9.94
```

```
!
```

- 4 Configure the filter-field **severity** to filter firewall events up to a certain severity for inclusion in syslogs.
- 5 Configure the stream-server to stream syslogs to a remote server and link it with the event log policy as one of the sinks.

Generating syslogs of Dropped Flows Due to Max Connection

New connections for a particular UE session are dropped if concurrent connection count is greater than **max-connections**. This syslog flow is logged as severity 5(Notice). Syslog is generated every 60s if flows are dropped since the syslog was last generated.

Configure **max-connections** as follows:

```
workflow data-profile DATAPROF
  max-connections          1024
  force-packet-drop        false
  subscriber-firewall       NAT_POLICY
  charging                 PREPAID
  priority-evaluation-mode  hierarchical
  dpi-pre-detection-max-pkts 0
  dpi-pre-detection-max-bytes 0
  first-packet-over-volume-quota allow
  metering-service AN_METERING
!
```

Sample syslog

```
Sep 11 04:24:03 an3000-114-6-ssm TOSFP_PT_DPDK_PROD[1211]:
2019-09-11T04:24:03.333000795-0400 Sev=4 [FIREWALL] [FIREWALL] [ID=03]
[0001720600011fdc] [01211 01550] [FPexecMaint_114_6_1_FPexecMaint_1]
"FLOW_DROP: cluster-id=114 slot-id=6 Session Id=0 drop-reason=session
connection limit exceeded num-drops=40"session-id is the PGW
session-id when operating in SAEGW mode.
```

drop-count is the count of dropped flows since the syslog was last generated.

Generating syslogs of Dropped Flows Due to Malformed TCP/IP Header

Packets with a malformed TCP/IP header arriving over a network-context are dropped by enabling **header-sanity-check** on that network-context. This syslog flow is logged as severity 5(Notice). Syslog is generated every 60s if packets are dropped since the syslog was last generated.

For example:

```
an3000-mcm-slot7cpu1(config)# network-context <name>
header-sanity-check [true|false]
```

By default, the option is disabled. It can be enabled only when security is enabled:

```
an3000-mcm-slot7cpu1(config)# network-context <name>
security admin-state enabled
```

When header-sanity-check is enabled, MCC validates TCP and IP header fields (checks for malformed packets) and drops packets that do not concur.

Sample syslog

```
Sep 2 07:56:20 an3000-114-5-ssm TOSFP_PT_DPDK_PROD[8163]:
2019-09-02T07:56:20.050818021-0400 Sev=4 [FIREWALL] [FIREWALL] [ID=03]
[0001720500011fd7] [08163 08533] [TosfpFlowAgt_114_5_1_TosfpFlowAgt_0]
" IOM_DROP: cluster-id=114 slot-id=5 network-context=DOS
IfName=DOS-INGRESS drop-reason=packet-header-validation-failure
drop-count=50000"
```

drop-count is the count of dropped packets since the syslog was last generated.

Generating syslogs of Dropped Flows Due to Flow Rate Limiting

New flows matching the security policy rule with the highest priority (lowest value = highest priority) are dropped if the flow rate exceeds the **flow-rate-limit** configured in the security policy rule. This syslog flow is logged as severity 5(Notice). Syslog is generated every 60s if flows are dropped since the syslog was last generated.

```
an3000-mcm-slot7cpu1(config)# security policy rule <name>
```

```
flow-rate-limit Number of flow creations matching this rule allowed
per second, default value 0 indicates unlimited
```

Sample syslog

```
Sep 12 01:57:20 an3000-114-5-ssm TOSFP_PT_DPDK_PROD[12973]:
2019-09-12T01:57:20.824208717-0400 Sev=4 [FIREWALL] [FIREWALL] [ID=03]
[0001720500011fe0] [12973 13963] [TosfpFlowAgt_114_5_1_TosfpFlowAgt_0]
"SECM DROP: cluster-id=114 slot-id=5 interface name=DOS-INGRESS
network-context=DOS rule-name=INGRESS
drop-Reason=flow-rate-limit-exceeded ingress-pkt-dropped=91
egress-pkt-dropped=0"
```


ingress-pkt-dropped is the count of packets dropped while ingressing the interface since the syslog was last generated.

egress-pkt-dropped is the count of packets dropped while egressing the interface since the syslog was last generated.

Generating syslogs of Dropped Flows Due to Packet Rate Limiting

Packets matching the security policy rule with the highest priority (lowest value = highest priority) are dropped if the packet rate exceeds the **packet-rate-limit** configured in the security policy rule. Syslog is generated every 60s if packets are dropped since the syslog was last generated.

```
an3000-mcm-slot7cpu1(config)# security policy rule <name>
packet-rate-limit Number of packets matching this rule allowed per
second, default value 0 indicates unlimited
```

Sample syslog

```
Sep 5 08:42:33 an3000-114-5-ssm TOSFP_PT_DPDK_PROD[8198]:
2019-09-05T08:42:33.622735924-0400 Sev=4 [FIREWALL] [FIREWALL] [ID=03]
[0001720500011fe0] [08198 09257] [TosfpFlowAgt_114_5_1_TosfpFlowAgt_0]
"SECM DROP: cluster-id=114 slot-id=5 interface name=DOS-INGRESS
network-context=DOS rule-name=INGRESS
drop-Reason=packet-rate-limit-exceeded ingress-pkt-dropped=4000
egress-pkt-dropped=0"
```

ingress-pkt-dropped is the count of packets dropped while ingressing the interface since the syslog was last generated.

egress-pkt-dropped is the count of packets dropped while egressing the interface since the syslog was last generated.

infrastructure ntp-server

Configure the IP address of the NTP server. This feature enables Operators to add/delete multiple NTP servers and specify the server(s) as an ipv4 or ipv6 address.

Modifying (adding or deleting) the list of configured NTP servers using the CLI causes the NTP daemon to restart which may result in a brief assertion of the *anNtpSyncError* alarm. The alarm automatically clears when the system is in sync.

Note: If you delete the last configured NTP server, the system asserts the *anNtpSyncError* alarm and this alarm does not clear.

```
an3000-mcm-slot8cpu1(config)# infrastructure ntp-server ipv4 x.x.x.x
```

infrastructure oom

Configure the TOS Cgroup OOM functionality.

The MCC implements its own “out-of-memory killer” functionality to detect a leaking process and restart the process so that the leaked memory is recovered. The benefit of this feature is that the rest of the system is not affected by the leaking process. The MCC subscribes to memory overload events that are generated by the kernel. These events are generated when memory utilization reaches minor, major, and critical threshold levels. At each of these levels, the MCC performs accounting of the memory utilized by the processes. At the critical level, the MCC runs its OOM algorithm to locate the leaking process(es) and then restart(s) that process(es).

MCC collects logs and debug information such as memory profiling and memory statistics in order to identify potential leaks. MCC also generates the [anBladeOver load](#) alarm for each major, minor, critical threshold crossing event. See the *Affirmed Networks SNMP MIB and Alarm Reference Guide* for details on alarms.

Configure the objects as follows:

Table 3-7 infrastructure oom

Object	Description
admin	Enable (default) or disable the TOS Cgroup OOM killer.
faults	Enable or disable faults when Process is killed by TOS OOM killer.
memory-profile	Enable or disable capture of memory-profile when MCC reaches memory overload.
tos-cgroup-mem-limit	Set memory limit for tos cgroup. memory-limit – Memory limit as a percentage of total RAM available for linux applications personality –

Fault Dump File

It may be helpful to maintain a file of potential fault dumps to be used by the Affirmed Networks TAC for troubleshooting. In the event of a failure, the system is designed to recover, restart processes, re-establish connections and communications, and possibly switch to redundant Shelf Manager Cards. The failure information maintained in the fault dump file is very useful to Affirmed Networks Engineering and TAC. This information is used to determine if the issue is a known problem and whether it can be reproduced so a fix can be included in a future build. Always report new failure information to the TAC.

Save the fault files to a safe storage location. Monitor the storage location and if necessary, periodically delete excess fault files. Fault files on the vMCC are saved in /public/faults by default.

To enable fault generation, enter:

```
an3000-mcm-slot7cpu1# infrastructure oom faults enabled
```

To display a list of fault files and cause of the file, enter:

```
an3000-mcm-slot7cpu1# show cluster [integer] node fault-summary
```

```
an3000-mcm-slot7cpu1# show cluster [integer] node fault-detail
```

To clear the fault file, enter:

```
an3000-mcm-slot7cpu1# cluster [integer] clear-faults
```

Note: Affirmed Networks recommends sending the fault file to the TAC (prodsupport@affirmednetworks.com) for diagnosis and resolution.

infrastructure peer-overload-control

Configure a profile for peer overload control.

The peer-overload-control function detects overload on peer servers and any potential network issues on the access network to peer servers. Specifically, this feature detects any potential transient failure condition by reducing the message load on the peer interface. After the peer link recovers, the system increases the messaging rate to its original rate. This feature also detects overload on peer servers by enforcing a maximum messaging rate.

The system monitors the health status (up or down) and processing capacity of peer nodes using different signaling interfaces. Operators can configure policing rates for each interface peer. Interfaces may be associated with a peer-overload-control profile that encapsulates the configuration and threshold for each congestion state.

Each peer node is categorized into different states of congestion based on the size of the outstanding transaction queue and/or failure rates. The system issues an alarm when an interface becomes congested. An interface is declared congested based a failure response with the following status: **System Failure**, **System Busy**, or **Remote Peer not Responding**.

During congestion, the messaging rate is reduced to avoid overloading the peer. Signaling messages (for real-time interfaces such as S5/S8 or RADIUS Authentication) are prioritized to maintain system consistency. For example, the SGW gives higher priority to GTP Delete Session request messages towards the PGW to avoid a stale session on the PGW.

For offline interfaces (for example, Gz and GTP'), the system buffers CDR messages (both in memory and on disk) to be delivered in order to avoid discards. These buffered messages are retransmitted and the signaling rate is restored to the sustained rate after the congestion state clears. The messaging rate is increased gradually to ensure a sustainable rate. Operators can configure the maximum messaging limit based on the capability of the peer server.

This section describes the peer-overload-control options for a pacer. The pacer lowers the rate of requests being made to the peer under overload conditions and supports priority levels of messages/requests. The pacer also enables the display of statistics for overload conditions and success/drop counts. The pacer issues an alarms when it is in an overload state and clears the alarm when the overload state resolves.

Configure the objects in config mode as follows:

Table 3-8 infrastructure peer-overload-control

Option	Description
decrement-rate	When the pacer is decreasing the outgoing request rate, it decrements the rate based on a value between 0 ..100 (10).
failure-rate	The rate the pacer used to determine whether it should increase or decrease its next intervals outgoing rate. For example, if the failure-rate value is set to 20%, the pacer determines the failure rate for the last interval as follows: ■ If the failure rate is > 20%, the pacer reduces the outgoing request for the next upcoming interval. ■ If the failure rate is <20%, the pacer increases the outgoing request for the next upcoming interval.
high-priory-decrement-rate	When the pacer is decreasing the outgoing request rate, it decrements the rate based on a value between 0 ..100 (10) for high-priority messages.
high-priority-increment-rate	When the pacer is increasing the outgoing request rate, it increments the rate based on a value between 0 ..100 (5) for high-priority messages.
increment-rate	When the pacer is increasing the outgoing request rate, it increments the rate based on a value between 0 ..100 (5).

Table 3-8 infrastructure peer-overload-control

Option	Description
low-priory-decrement-rate	When the pacer is decreasing the outgoing request rate, it decrements the rate based on a value between 0 ..100 (10) for low-priority messages.
low-priority-increment-rate	When the pacer is increasing the outgoing request rate, it increments the rate based on a value between 0 ..100 (5) for low-priority messages.
max-supported	Maximum number of requests the peer can handle per second.
medium-priority-decrement-rate	When the pacer is decreasing the outgoing request rate, it decrements the rate based on a value between 0 ..100 (10) for medium-priority messages.
medium-priority-increment-rate	When the pacer is increasing the outgoing request rate, it increments the rate based on a value between 0 ..100 (5) for medium-priority messages.
time-interval	The length of each interval in milliseconds. At the end of the interval, the pacer adjusts its outgoing rates. See Adaptive Algorithm for Pacer time-interval (page 3-42) for additional details.

Adaptive Algorithm for Pacer time-interval

For each interval, the pacer uses the following rates to compute values:

cur-supported-rate – Maximum number of requests the pacer allows for the current interval ($\text{cur-supported-rate} * \text{max-supported} *.01$).

high-priority-cur-supported-rate – Maximum number of high-priority requests the pacer allows for the current interval ($\text{high-priority-cur-supported-rate} * \text{max-supported} *.01$).

medium-priority-cur-supported-rate – Maximum number of medium-priority requests the pacer allows for the current interval ($\text{medium-priority-cur-supported-rate} * \text{max-supported} *.01$).

low-priority-cur-supported-rate – Maximum number of low-priority requests the pacer allows for the current interval ($\text{low-priority-cur-supported-rate} * \text{max-supported} *.01$).

The pacer initially starts with a value of 100% for each of the *cur-supported-rates*. At the end of the interval, the pacer determines the failure rate for the completed interval. Based on the failure rate and the failure-rate configuration, the pacer decides whether it should increment or decrement the outgoing rate using the **increase-rate* and **decrease-rate* configurations to determine the new **cur-supported-rates* for the next upcoming interval. Internally, the pacer maintains a minimum-rate (5%) and the **cur-supported-rates* never drops below this minimum-rate value. This is required in order for the pacer to constantly make requests to the peer to get feedback. Otherwise, if the minimum-rate value were set to 0, no further requests would be made and the pacer would be in a constant loop.

Peer Overload Control Configuration Example

The example in this section uses the following values:

failure-rate	20
increment-rate	5
decrement-rate	10
high-priority-increment-rate	5
high-priory-decrement-rate	10
medium-priority-increment-rate	10
medium-priority-decrement-rate	20
low-priority-increment-rate	20
low-priory-decrement-rate	30
max-supported	

For the first interval, the pacer starts with the following cur-supported-rates:

Rate Value	1st Interval
last-interval-failure-rate	N/A
cur-supported-rate	100
high-priority-cur-supported-rate	100
medium-priority-cur-supported-rate	100
low-priority-cur-supported-rate	100

In this example, the pacer encountered 25% failures in the 1st interval. Since that value is greater than the configured failure-rate value of 20%, the pacer decreases the rates using the *decrease-rate values. For high-priority, the rate is decreased by 10%, for medium-priority the rate is decreased by 20% and so on. The following table shows the *cur-supported-rates for the 2nd interval.

Rate Value	1st Interval	2nd Interval
last-interval-failure-rate	N/A	25
cur-supported-rate	100	90 (rate decreased by 10% of 100)
high-priority-cur-supported-rate	100	90 (rate decreased by 10% of 100)
medium-priority-cur-supported-rate	100	80 (rate decreased by 20% of 100)
low-priority-cur-supported-rate	100	70 (rate decreased by 30% of 100)

In this example, the pacer encounters 15% failures in the 2nd interval. Since that value is less than the configured failure-rate, the pacer increases the rates using the *increase-rate values. The following table shows the *cur-supported-rates for the 3rd interval.

Rate Value	1st Interval	2nd Interval	3rd Interval
last-interval-failure-rate	N/A	25	15
cur-supported-rate	100	90	95 (rate increased by 5% of 100)
high-priority-cur-supported-rate	100	90	95 (rate increased by 5% of 100)
medium-priority-cur-supported-rate	100	80	90 (rate increased by 10% of 100)
low-priority-cur-supported-rate	100	70	90 (rate increased by 20% of 100)

infrastructure ssh-host-keygen

Generate keys for use by SSH protocol version 1 and RSA or DSA keys for use by SSH protocol version 2.

infrastructure time-zone

Set the local time zone or set the system time zone.

networking-profiles neighbor-discovery-profile

Configure an ICMPV6 Neighbor Discovery profile.

MCC supports Secure Neighbor Discovery protocol (SEND) as described in RFC 3971/3972 for secure Neighbor Discovery (ND). To support this feature, MCC acts as a host and discovers its router (Router Discovery) and the subnet prefix.

MCC supports SEND at the IP interface level for the following Neighbor Discovery Protocol (NDP) messages: RS (Router Solicitation), RA (Router Advertisement), CPS (Certification Path Solicitation), CPA (Certification Path Advertisement), NS (Neighbor Solicitation), and NA (Neighbor Advertisement).

MCC connects to the router only if the router has a certification path to one of the trust anchors proposed by MCC. MCC also supports secure discovery of its on-link neighbors (Neighbor Discovery).

MCC supports a Cryptographically Generated Address (CGA) corresponding to the interface link local address prefix. This address is generated using the 64 bit prefix of the default link local address of the interface, RSA public key, and the configured security level. The CGA remains valid for as long as the interface is up.

MCC mitigates threats related to the NDP protocol using secure neighbor discovery options such as CGA, RSA, Timestamp, and Nonce.

Rivest, Shamir, Adleman (RSA) is an asymmetric key algorithm for the key-pair generation. The algorithm generates a private-public key pair for digital signatures.

MCC supports the SEND feature in full secure mode at the interface level and global level.

- When enabled at interface level – MCC process only NDP messages received with SEND payload options and drops NDP messages without SEND options.
- When enabled at the global level – The configuration applies to all interfaces but the SEND functionality is applied only for NDP messages that are sent on link local addresses.

Note: MCC does not support the SEND protocol as a router. In other words, MCC does not support RS (Router Solicitation) requests from the peer.

Configure the objects as follows:

Table 3-9 networking-profiles neighbor-discovery-profile

Object	Description
default-lifetime	The lifetime associated with the default router in units of seconds.
name*	Neighbor discover profile name.
number-unsolicited-ra	The number of unsolicited router advertisements the MCC would send when a UE attaches to the network. A value of 0 indicates to send continuously spaced by ra-interval.
ra-mtu	MTU for ipv6 Neighbor Discovery Router Advertisements.
reachable-time	The time, in milliseconds, a neighbor is considered reachable after receiving a reachability confirmation. A value of 0 indicates unspecified.
retransmit-timer	The time in seconds between retransmitted Neighbor Solicitation messages.
suppress-ra	Whether to suppress sending icmpv6 Router Advertisements.

restore-config

Restores a saved configuration. This command impacts service and results in a service restart. Select a configuration to restore from the list of saved configurations.

save-config

Execute the save-config command in exec mode. Saves the current configuration.

snmp

Execute the snmp command in config mode to configure the Simple Network Management Protocol configuration. Options include:

- community-string – SNMP community string
- trap-target – SNMP trap target
- v3 – Authentication and privacy. See [snmp v3](#).

snmp v3

The snmp v3 object supports SNMPv3 security for Authentication and Privacy:

- Authentication ensures that traps are read only by the intended recipient. As messages are created, they are given a special key based on the EngineID of the entity. The key is shared with the intended recipient and used to receive the message.
- Privacy encrypts the payload of the SNMP message to ensure that it cannot be read by unauthorized users. Any intercepted messages are filled with garbled characters and are unreadable. Privacy is useful in applications where SNMP messages must be routed over the Internet.

When an SNMPv3 is configured, MCC internally updates the following standard MIBs:

- SNMP-USER-BASED-SM-MIB (user-based security model)
- SNMP-VIEW-BASED-ACM-MIB (view-based access control model)
- SNMP-TARGET-MIB

Configure the objects as follow:

Table 3-10 snmp v3 community-entry

Option	Description
name	Enter a name.
engine-id*	Enter the engine ID.
user*	■ security-authentication-protocol – The value none is valid only when security-type is noAuthnoPriv. Options include: none, md5, sha. ■ security-name* – Enter a security name. ■ security-password* – Enter a security password. ■ security-privacy-protocol – The value none is valid only when security-type is noAuthnoPriv or authNoPriv. Options include: none, des, aes. ■ security-type* – noAuthnoPriv - Communication without authentication and privacy. – authNoPriv - Communication with authentication and without privacy. Protocols supported for authentication are MD5 and SHA. – authPriv - Communication with authentication and privacy. Protocols supported for authentication are MD5 and SHA. Protocols supported for privacy are DES and AES.

Show Commands

MCC supports the following system management show commands. For specific details and output, see the CLI and *Acuitas Performance Statistics*.

- `show image-management card-status <id> <slot> <cpu> <state>`
- `show image-management release-modifier-status <release-role> <release-version> <release-modifier>`
- `show image-management status <state> <id> <slot> <cpu>`
- `show infrastructure edr edr-bearer-stats <network-context> <name>`
- `show infrastructure edr edr-flow-stats <network-context> <name>`
- `show infrastructure edr edr-http-stats <network-context> <name>`
- `show infrastructure edr edr-pilot-stats <network-context> <name>`
- `show infrastructure edr edr-server-stats <network-context> <name>`
- `show infrastructure edr edr-session-stats <network-context> <name>`
- `show infrastructure ntp-server status <remote>`
- `show infrastructure peer-overload-control configuration <pacername> <instancename>`
- `show infrastructure peer-overload-control counters-detail <pacername> <instancename> <interval>`
- `show infrastructure peer-overload-control counters-detail-cumulative <pacername> <instancename>`
- `show infrastructure peer-overload-control counters-summary <pacername> <instancename> <interval>`
- `show infrastructure peer-overload-control counters-summary-cumulative <pacername> <instancename>`
- `show infrastructure peer-overload-control rates <pacername> <instancename> <interval>`

show running-config infrastructure

Display information as follows:

Table 3-11 show running-config infrastructure

Object	Displays
alarms	Alarms
cluster	Virtual Node management
extended-storage	Extended storage parameters
geographic-redundancy	Geographic Redundancy parameters
image-management	Image Management parameters
infrastructure	Infrastructure setting: ■ dynamic-scale ■ file-management ■ ip-protocols – Manage IP Network Protocols ■ network-context – Configure network context parameters ■ networking-profiles – Networking profiles. ■ performance-statistics-management – View Performance Statistics Management Status ■ security – Security configuration ■ service-construct – Service Constructs ■ services – Services ■ snmp – Simple Network Management Protocol configuration, community-strings and trap-target ■ user-access – User access commands ■ workflow – Configure workflow parameters ■ zone – Zone parameters

For example:

```
an3000-mcm-slot7cpu1# show running-config infrastructure
file-management profile compress-format raw
infrastructure file-management profile datarecord HTDR
purge-thr 100
purge-low-thr 0
file-rotate-size-bytes 10485760
file-transfer-enabled disabled
upload-frequency 1hour
upload-offset-secs 0
file-rotate-frequency 30mins
file-rotate-offset 0
```

```

purge-file-older-than 1day
purge-enabled         enabled
compress-enabled      enabled
compress-format       raw
logging
ntp-server
peer-overload-control
time-zone

```

Also see *Acuitas Performance Statistics*.

show infrastructure

Display the objects as follows:

Table 3-12 show infrastructure

Object	Displays
edr	EDR statistics.
file-management	Data file management.
ntp-server-status	NTP server statistics. This command may truncate the display of long IPv6 addresses.
peer-overload-control	The peer-overload-control parameters and pacer statistics.
ssh-host-keys-fingerprint	Keys for use by SSH protocol version 1 and RSA or DSA keys for use by SSH protocol version 2.

For example:

```

an3000-mcm-slot7cpu1# show infrastructure ntp-server-status
remote refid      st t when poll reach  delay  offset  jitter
=====
LOCAL(1)          .LOCL.          12 l   20   64  377    0.000    0.000    0.001
*10.32.0.254      205.233.73.201  3 u  802 1024 377    9.242   -1.354    0.205
an3000-mcm-slot7cpu1#

```

show image-management release-modifier-status

Display the release modifier version (for example, of a DPI signature file or a plugin file) applied to the current release version. Display the information as follows:

Table 3-13 show image-management release-modifier-status

Object	Displays
original	Display the original release modifier and revert to the original modifier image (listed as original-version in show image-management release-modifier-status)

Table 3-13 show image-management release-modifier-status

Object	Displays
rollback	Display the release modifier status and toggle to the last modifier image version (listed as rollback-version in show image-management release-modifier-status) Returns the NE to the previous release and configuration: ■ If the last operation was an upgrade, rollback activates the previous release and reboots the chassis. ■ If the last operation was a successful online upgrade, rollback activates the previous release and reboots the chassis. ■ If the last operation was a failed online upgrade, rollback activates the original release and reboots the chassis.
version	Display available version and specify a modifier image version to apply to the current release (show image-management available admin versions)

For example:

```
an3000-mcm-slot8cpu1# show image-management release-modifier-status
image-management release-modifier-status
RELEASE                                RELEASE  OPERATIONAL  ROLLBACK
ROLE      RELEASE VERSION  MODIFIER  VERSION      VERSION
ORIGINAL VERSION
-----
rollback   3.1.2.0-107.REL  dpisig    dpisig_2013-09-06
NotAvailable dpisig_2013-09-06
operational 3.1.2.0-138.REL  dpisig    dpisig_2013-09-06
NotAvailable dpisig_2013-09-06
prepared   3.1.2.1-138.REL.ENG dpisig    dpisig_2013-09-06
NotAvailable dpisig_2013-09-06
```

To perform the release modifier:

- 1 Copy the modifier image file (sig-*<date>*.tgz) to /public/images.
- 2 Enter the following command to perform the release modifier:

```
an3000-mcm-slot7cpu1(config)# image-management modifier <version>
<rollback> <original>
```

where *<version>* is the modifier image version to apply to the current release, *<rollback>* specifies to toggle to the last modifier image version, or *<original>* specifies to revert to the original modifier image.

To perform the rollback:

- 1 Enter the following command to perform a rollback to restore the system to the original state:

```
an3000-mcm-slot7cpu1(config)# image-management rollback force
```
- 2 Enter the following command to observe the rollback progress:

```
an3000-mcm-slot7cpu1(config)# show image-management progress
```
- 3 Enter the following command to check the progress of the completed or failed rollback:

```
an3000-mcm-slot7cpu1(config)# show image-management
last-activation-result.
```

show image-management

Display the current status of image-management parameters such as admin version, image-management status, activation result, online-upgrade-plan, and activation progress.

Note: A forced-sync may take between 30 and 90 minutes to complete.

Display the information as follows:

Table 3-14 show image-management

Object	Displays
available-admin-versions	Versions currently available for upgrade.
card-status	Each card image status.
force-sync-admin	Status of force sync admin (enabled/disabled).
last-activation-result	Last activation result (success/failure) since the last reboot.

Table 3-14 show image-management

Object	Displays
online-upgrade-plan	Online upgrade plan for the version. This option considers the metadata, the options, and the system state and displays how an online upgrade would proceed if issued at this time. Options include: ■ version – Version to upgrade to (show image-management available-admin-version) ■ drain – Drain timeout per node in seconds. Allows tasks to drain of work before being replaced. It can reduce the disruption caused by replacement, but may significantly extend the time to complete the upgrade ■ strategy – Reset strategy to control how many tasks are replaced at a time: cardreset, taskreset, toreset.
progress	Activation progress during upgrade or rollback. This object enables you to include or exclude certain lines and control pagination. Options include: ■ release-modifier-status – Display current release modifier status: operational-version, original-version, rollback-version. ■ upgrade-progress – This command is retained for backward compatibility. ■ status – Current image-management status. For example:

Use the following command to determine if a forced synchronization occurred:

```
an3000-mcm-slot7cpu1# show image-management upgrade-progress
```

Use the following command to display the status:

```
an3000-mcm-slot8cpu1# show image-management status
```

```
image-management status
```

				OPERATIONAL	ACTIVATION	
ROLLBACK	PREPARED					
STATE	ID	SLOT	CPU	ADMIN VERSION	VERSION	STATUS
VERSION			VERSION			

active	105	8	1	5.3.3.0-25.REL	5.3.3.0-25.REL	Idle
5.3.3.0-24.REL						NotAvailable

System Security

This section describes the system security objects and commands. The security function provides global firewall and IPsec services at a network-context level. For each service, you must enable security.

Enabling Security

Note: Enabling security on a network requires 4 GB of additional memory on one of the control plane VMs (CSM/CPM) and 3 GB of additional memory on each of the data plane VMs (ISM/SSM/WSM/IOM).

Enable security services on any network-context as follows:

- 1 Set the administrative state to **enabled** to enable security.
`an3000-mcm-slot7cpu1(config-network-context-GN)# security`
Possible completions:
`<administrative state {enabled | disabled}>[enabled]`
- 2 Continue with the following sections:
 - [Firewall Services \(page 4-2\)](#)
 - [IPsec Tunnel Services \(page 4-5\)](#)

Firewall Services

After you enable security on a network-context it is applied to all packets entering and exiting the network-context. The security rules configured on the network-context determine the action taken on each packet. By default, packets are dropped. In other words, MCC drops packets if they do not match any of the configured security rules.

security policy rule

MCC uses security policy rules to determine which packets are allowed or disallowed on a particular network-context. MCC enforces the following policy rules:

- All subsequent packets with the same 5-tuple are processed by this flow. MCC deletes the flow if it is idle for 30 seconds.
- MCC deletes the flow after 30 seconds so that re-evaluation occurs periodically if the flow is created by the default rule (drop).
- MCC deletes the flow and re-evaluation occurs if the flow is created by a user-configured rule and the rule is modified.

Configure the objects as follows:

Table 4-1 security policy rule (firewall)

Object	Description
action	Configure the rule action as drop, pass, or reject.
clone-reverse	Set to true to automatically clone this rule, reversing source/destination packet filters. Setting this value to true configures a bi-directional rule. (one rule from source to destination and one rule from destination to source). <i>Note: Affirmed Networks recommends using this option in most cases. When asymmetrical security is required, this option may be omitted.</i>
control-flow-filter	Control flow filter specification for the traffic selector. When the control-flow-filter references a remote server entity, RADIUS, Diameter, etc., the control-flow-filter is specified. The Diameter and RADIUS IP addresses and ports are taken from the referenced object. For server configurations that specify only the remote server, the source information may be directly entered or left open-ended, implying that all possible source addresses within the network-context and all possible source ports are used.
destination-ip-pool	IP pool used as the destination IP specification.
destination-packet-filter	Destination packet-filter specification.

Table 4-1 security policy rule (firewall)

Object	Description
flow-rate-limit	Number of flow creations matching this rule allowed per second. A default value of 0 (zero) indicates unlimited. This object applies flow rate limits to mitigate DDos attacks (and also protect from bad IP and TCP header flags). The flow-rate-limit is confined to connection creation only.
packet-rate-limit	A value of zero (default) in flow-rate-limit and packet-rate-limit fields disables the rate limits. The flow-rate-limit is confined to connection creation only. This object applies packet rate limits to mitigate DDos attacks (and also protect from bad IP and TCP header flags). Note: If tunnel is set to either to-tunnel or from-tunnel for this security policy rule, you must set flow-rate-limit and packet-rate-limit to 0.
priority	Evaluation priority for this rule. Overlapping rules must have unique priority. A lower number indicates a higher priority.
name*	Name of this rule
network-context*	Network context for this rule
source/destination-packet-filter	The source-packet-filter may be specified directly or may be specified implicitly by referencing a control-plane configuration (that is, a RADIUS remote server) or address pool. Since source/destination-packet-filter can overlap, associate the filter with a priority. The protocol is specified as the IANA IP protocol type and ports are specified in a range. IPv4 and IPv6 addresses are specified in a range, as a host address, or as a CIDR subnet. The option 255 indicates all protocols. ■ ip-start – Starting inclusive IP address for source/destination IP addresses. ■ ip-end – Ending inclusive IP address for source/destination IP addresses. ■ name* – Name of this packet-filter ■ port-start – Starting inclusive port for source/destination port. ■ port-end – Ending inclusive port for source/destination port. ■ protocol – source or destination ip protocol values (all: 255, ICMP: 1, TCP: 6, UDP: 17, SCTP: 132). Set to 255 for any protocol.
source/destination-ip-pool	The source/destination-ip-pool may reference a nat-pool or user-equipment-pool instead of a source IP address and/or destination IP address. The system expands all of the underlying subpools. If both source and destination point to subpools, then the expansion creates a full matrix.

Example Security Policy Rule Configuration

The following example configuration allows any packets between 100.100.100.0/24 and 200.200.200.0/24 but drops UDP traffic between 100.100.100.10 and 200.200.200.10.

```
an3000-mcm-slot7cpu1# show running-config security policy rule
management-context
```

```
security policy rule management-context drop-10
  priority 100
  action drop
  clone-reverse
  source-packet-filter spf
    protocol 17
    ip-start 100.100.100.10
    ip-end 100.100.100.10
  !
  destination-packet-filter dpf
    protocol 17
    ip-start 200.200.200.10
    ip-end 200.200.200.10
  !
!
security policy rule management-context pass-range
  priority 500
  action pass
  clone-reverse
  source-packet-filter spf
    protocol 255
    ip-start 100.100.100.0
    ip-end 100.100.100.255
  !
  destination-packet-filter dpf
    protocol 255
    ip-start 200.200.200.0
    ip-end 200.200.200.255
  !
!
```

IPsec Tunnel Services

A tunnel is a bidirectional IPsec SA with respect to an IKE peer. [Table 4-2](#) contains information pertaining to the level at which remote user authentication occurs; at the tunnel level or authentication domain level.

Note: The use of the term *tunnel* in this section differs from *transfer-mode* or *tunnel-mode-encapsulation*.

Use the information in [Table 4-2](#) as a guide when configuring IPsec security tunnels. Use the following guidelines:

- Initiators always require tunnel PSK and tunnel local ID.
- V1 main-mode responders require tunnel PSK and tunnel remote-id.
- V1 aggressive-mode and V2 responders require authentication remote-secret.
- V2-V1 fallback does not support the V1 aggressive mode option.

Table 4-2 IPsec Security Tunnel Configuration Guidelines

Specified					Required?				
Manual Key	IKE Version	V1-aggressive	Auto-initiate	Responder-only	Tunnel PSK	Tunnel Local ID	Tunnel Remote ID	Authentication Remote Secret	Special Considerations
Y	N/A	N/A	N/A	N/A	N	N	N	N	No IKE
N	V2	N/A	N	N	Y	Y	N	Y	V2 initiator and responder
-	-	-	N	Y	N	N	N	Y	V2 responder only
-	-	-	Y	N/A	Y	Y	N	N	V2 initiator only
-	V1	N	N	N	Y	Y	Y	N	V1 main mode initiator and responder
-	-	-	N	Y	Y	N	Y	N	V1 main mode responder only
-	-	-	Y	N/A	Y	Y	N	N	V1 main mode initiator only
-	V1	Y	N	N	Y	Y	N	Y	V1 aggressive mode initiator and responder

Table 4-2 IPsec Security Tunnel Configuration Guidelines

Specified					Required?				
-	-	-	N	Y	N	N	N	Y	V1 aggressive mode responder only
-	-	-	Y	N/A	Y	Y	N	N	V1 aggressive mode initiator only
-	V2-V1	N/A	N	N	Y	Y	Y	Y	V2 or V1 main mode initiator and responder
-	-	-	N	Y	N	N	Y	Y	V2 or V1 main mode responder only
-	-	-	Y	N/A	Y	Y	N	N	V2 or V1 main mode initiator only

About IP Security (IPsec)

IPsec, short for IP Security, is a set of protocols developed by the IETF to support secure exchange of packets at the IP layer. IPsec is deployed widely to implement Virtual Private Networks (VPNs).

IPsec supports two encryption modes:

- *Transport* – Encrypts only the data portion (payload) of each packet, but leaves the header untouched.
- *Tunnel* – Encrypts both the header and the payload. On the receiving side, an IPsec-compliant device decrypts each packet. Tunnel mode encapsulates the entire IP packet to provide a virtual "secure hop" between two gateways. Tunnel mode is used to form a traditional VPN, where the tunnel generally creates a secure tunnel across an un-trusted Internet.

For IPsec to work, the sending and receiving devices must share a public key. This is accomplished through a protocol known as Internet Security Association and Key Management Protocol/Oakley (ISAKMP/Oakley).

The system supports firewall capability, including ACL filtering and IPsec. The firewall is enabled/disabled on a network-context basis and is supported on all network-contexts, with the exception of the management-context.

The firewall filters all incoming traffic before it is sent to the network. Operators configure the firewall with policy rules and optionally, IPsec tunnels. The policy rules implement Layer 3/4 packet filtering and may be specified as follows:

- *source/destination-packet-filter* – The source/destination packet filter specification.
- *linked to control-flow-filter* (that is, RADIUS and Diameter) in order to specify the filter based on the IP addresses and ports utilized by the protocol interface
- *linked to source/destination-ip-pools* in order to specify the filter based on the IP address ranges utilized by the address pools.

Operators can configure policy rules to either block or pass traffic. Note that when security is enabled, traffic matching no rule is also blocked. Rules that are set to pass traffic may optionally be associated with an IPsec tunnel. The tunnel specifies the manner in which the matching traffic is ciphered and/or authenticated, and identifies the peer security gateway or host that performs the inverse operations. Tunnels may be signaled using Internet Key Exchange (IKE v1 and IKE v2). Affirmed also supports automatic fallback from IKE v2 to IKE v1. In addition, manual keying may be selected when IKE signaling is not desired. Manual keys require manual entry of the secret key material on both ends.

Adding IPsec VPNs

Use the following steps to add a new IPsec VPN. Do not enable security in the VPN network-context until a least one tunnel that references the VPN is configured. To add multiple VPNs in series or in parallel, repeat the following steps for each VPN.

- 1 Create the VPN network-context including all of its configuration, but leave the **security** setting disabled.
- 2 Create the IPsec tunnel (or tunnels) that will carry traffic for the VPN. The **tunneled-network-context** attribute of each tunnel must reference the VPN.
- 3 Create the IPsec rules in the VPN network-context.
- 4 Enable **security** in the VPN network-context.

To remove a VPN (reverse order of adding a VPN):

- 1 Disable **security** in the VPN network-context.
- 2 Delete the rule in the VPN.
- 3 Delete the tunnels referencing the VPN.
- 4 Delete the VPN network-context.

Configuring General IP Security Settings

To configure general security settings:

- 1 Set the **cookie-threshold** value. For IKEv2, this specifies the number of half-open SAs as a percentage of the maximum number of allowed tunnels, above which cookie-challenges are enabled. See [Configuring General IP Security Settings \(page 4-8\)](#). For example:

```
an3000-mcm-slot8cpu1(config)# security cookie-threshold <value>
```
- 2 Set the history buffer entries for IPsec and IKE SA negotiations/exchanges. See [security history \(page 4-12\)](#). For example:

```
an3000-mcm-slot8cpu1(config)# security history admin state [enabled  

|disabled] ike-sa-completion-depth <value> ike-sa-failure-depth  

<value> ipsec-sa-completion-depth <value> ipsec-sa-failure-depth  

<value>
```
- 3 Set the **hw-assist** parameter for Diffie-Hellman calculations. See [Configuring General IP Security Settings \(page 4-8\)](#). For example:

```
an3000-mcm-slot8cpu1(config)# security hw-assist
```

Configuring IPsec Objects

Configure the objects as follows:

Table 4-3 security (IPsec)

Object	Description
asynchronous-transactions	Set to true/false to allow SPM to set up transforms in parallel.
authentication	Security authentication domain configuration.
cookie-threshold	For IKEv2, specifies the number of half-open SAs as percentage of maximum allowed tunnels, above which cookie-challenges are enabled. This added security is to prevent Denial of Service (DoS) attacks. For IKEv2, the cookie-threshold value specifies the number of half-open SAs as a percentage of the maximum number of allowed tunnels, above which cookie-challenges are enabled.
debug	Security debug settings.
history	IKE and IPsec transaction history table configuration parameters.
hw-assist	Hardware assist enable control setting. <i>Note: This parameter applies only to the AN3000 hardware platform. If you are operating MCC on a virtual platform, ignore this parameter.</i> ■ Enabled – Specifies that all Diffie-Hellman calculations are performed using the hardware acceleration components of the Subscriber Services Module (SSM) on the AN3000 platform. ■ None (default) – Specifies that all Diffie-Hellman calculations are performed using software on the Content Services Module (CSM).
policy	Security policy.
return-routability-check	Options for MOBIKE return routability check.
secure-nd-cmn-cfg	MCC Secure NDP protocol common configuration.

security authentication-domain

Authentication domain configuration. Configure the objects as follows:

Table 4-4 security authentication-domain

Object	Description
eap-additional-methods	Supported EAP methods: EAP-MSCHAPv2, EAP-TLS
eap-gtc-request-text	When type is set to eap-gtc, text to be displayed to the user during the EAP-GTC exchange. If not entered, the default string will be Password.
eap-only	Allow authentication based only on EAP if requested by UE.
name*	Name of this domain object.
network-context*	Network context name.
suppress-certreq-to-initiator	Do not send CERTREQ to IKEv2 Initiator when EAP is used for authentication
type	Type of authentication domain (requires security be administratively toggled to take effect). Options include: pki-plus-eap eap-gtc pki

security authentication pki-certificate-management

The pki-certificate-management object provides support for X.509 certificates on ePDG.

For example:

```
an3000-mcm-slot7cpu1(config)# security authentication
pki-certificate-management <key-name> network-context
<network-context-name> file-name <file-name> file-type [binary|pem]
certificate-type [local|ca-trust|intermediate-trust] key-passphrase*
<string>
```

*The key-passphrase specifies the encryption type (such as X.509) for a private key.

The ePDG supports the following public key infrastructure (PKI) certificate types:

Table 4-5 security authentication pki-certificate-management

Object	Description
authentication-domain	Associated authentication domain - if not specified, the default is used. Options include pki-plus0eap, eap-gtc, and pki. The pki authentication-domain type allows non-ePDG applications to utilize PKI-based authentication.

Table 4-5 security authentication pki-certificate-management

Object	Description
certificate-type	Type of certificate. Options include: ■ local – A local certificate/private-key instance on the ePDG. The ePDG uses both the local certificate and private key to sign credentials in the IKE AUTH response. ■ ca-trust – A root-level certification authority (CA) instance on the ePDG. ■ intermediate-trust – An intermediate certificate that can be used as a proxy for the root certificate. The intermediate certificate can be used to sign the SSLs that customers install (vs. the root certificate) and to maintain the "chain of trust."
file-name*	Filename (without the extension) containing the certificate and private key (if local) or the ca certificate and an optional CRL (if ca-trust or intermediate-trust). Certificate files must have the extension .crt, private key files must have the extension .prv, CRL files must have the extension .crl
file-type*	Format of files.
key-passphrase	If private key file is encrypted, the passphrase used to decrypt the file. Supported encryption formats are: SSH proprietary, SSH client version 1 RSA, SSH client version 2 RSA or DSA, PKCS#8
name*	The remote-secret name.
network-context*	The network-context for this remote-secret.

security authentication remote-secret

Configure the pre-shared keys used to authenticate IKE peers. Configure the objects as follows:

Table 4-6 security authentication remote-secret

Object	Description
name*	The name of this remote-secret object.
network-context*	The network-context for this remote-secret.
secret*	

security debug

This command allows users to enable/disable IKE key-material logging on a per network-context basis. This option allows Operators to diagnose problems with ePDG sessions and is intended as a diagnostic tool only. See the *Affirmed Networks ePDG Administration Guide* for details.

Configure the objects as follows:

Table 4-7 security debug

Object	Description
key-logging ikev2	Use the no form of the command to disable key-logging. The command output is logged in the debug log at the Info level. In the debug log, enter the text that follows “ikev2_description_table” directly into Wireshark’s ikev2_description_table configuration file.

security history

When the security history feature is enabled, the IPsec processing module records each Internet Key Exchange (IKE) and IPsec security association negotiation in a set of history buffers that can be displayed via the CLI. Operators can individually set the depth of each history buffer between 0-1000 entries. Configure the objects as follows:

Table 4-8 security history

Object	Description
admin-state	Enable/disable IPsec history.
ike-sa-completion-depth	Specifies the history buffer depth (number of entries) for successful IKE security association (SA) exchanges. Range is 0-1000; default is 10.
ike-sa-failure-depth	Specifies the IKE SA failure history buffer depth (number of entries) for unsuccessful IKE security association (SA) exchanges. Range is 0-1000; default is 10.
ipsec-sa-completion-depth	Specifies the IPsec SA completion history buffer depth (number of entries) for successful IPsec security association (SA) negotiations. Range is 0-1000; default is 10.
ipsec-sa-failure-depth	Specifies the history buffer depth (number of entries) for unsuccessful IPsec security association (SA) negotiations. Range is 0-1000; default is 10.

For example:

ipsec-sa-completion-depth

Use the following CLI command to specify the history buffer depth for successful IPsec security associations (SA) negotiations.

```
an3000-mcm-slot7cpu1(config)# security history
ipsec-sa-completion-depth <value>
```

The system displays output similar to the following:

```
security history ipsec-negotiations BOSTON 1
timestamp          1403811491
mode               Responder,Tunnel
diffie-hellman     none
local-address      2001:470:8865:20:49::2%14
local-port         4500
local-id           "affirmed.com (fqdn)"
local-traffic-selector ipv6(2001:470:8865:20:49::3)
remote-address     2001:470:8865:20::49:1
remote-port        4500
remote-id           "tunnel1 (fqdn)"
remote-traffic-selector ipv6(2001:470:8865:ad::4)
transform          ESP
inbound-spi        85ea8e8d
outbound-spi       c39d583d
algorithm           "aes-cbc/128 - hmac-sha1-96"
local-lifetime      "3600 seconds"
an3000-mcm-slot7cpu1#
```

ipsec-failures

Use the following CLI command to specify the history buffer depth for unsuccessful IPsec SA negotiations.

```
an3000-mcm-slot7cpu1(config)# security history ipsec-sa-failure-depth
<value>
```

The system displays output similar to the following:

```
security history ipsec-failures BOSTON 2
timestamp          1403811514
mode               Responder
local-address      2001:470:8865:20:49::2%14
local-port         500
local-id           (null)
remote-address     2001:470:8865:20::49:1
remote-port        500
remote-id           (null)
error-message      "No proposal chosen"
error-id           14
error-reason       Unknown
```

ike-negotiations

Use the following CLI command to specify the history buffer depth for successful IKE exchanges.

```
an3000-mcm-slot7cpu1(config)# security history ike-completion-depth  
<value>
```

The system displays output similar to the following:

```
security history ike-negotiations BOSTON 1  
timestamp      1403811491  
mode           Responder  
ikeVersion      v2  
local-auth-method "Pre-shared key"  
remote-auth-method "Pre-shared key"  
ike-algorithms  "aes128-cbc, hmac-sha1, hmac-sha1-96"  
diffie-hellman  "group 5 (1536 bits)"  
local-address   2001:470:8865:20:49::2%14  
local-port      4500  
local-id        "affirmed.com (fqdn)"  
remote-address  2001:470:8865:20::49:1  
remote-port     4500  
remote-id       "tunnel1 (fqdn)"  
initiator-spi   "f26579a6 d9d4ab4b"  
responder-spi   "c9a19a79 0ea8fd20"  
local-lifetime  86400
```

ike-failures

Use the following CLI command to specify the history buffer depth for unsuccessful IKE exchanges.

```
an3000-mcm-slot7cpu1(config)# security history ike-failure-depth  
<value>
```

The system displays output similar to the following:

```
security history ike-failures BOSTON 2  
timestamp      1403811514  
mode           Responder  
ikeVersion      v2  
local-address   2001:470:8865:20:49::2%14  
local-port      500  
local-id        (null)  
remote-address  2001:470:8865:20::49:1  
remote-port     500  
remote-id       (null)  
error-message   "No proposal chosen"  
error-id        14  
error-reason    "Local ID mismatch"
```


security policy rule

A rule is a traffic filter that is associated to a tunnel.

Configure the policy objects with respect to a single network-context. Configure the following policies:

rule – security rule. See [security policy rule \(page 4-15\)](#) for details.

tunnel – security tunnel. See [security policy tunnel \(IPsec\) \(page 4-16\)](#) for details.

A remote peer is a remote security gateway or host to which the system forms an IKE SA. Configure the identity and authentication parameters of remote IKE gateways or hosts.

Note: *Packet-filter and ip-pool can be mixed in the same rule. For example, source-ip-pool and destination-packet-filter.*

Configure the objects as follows:

Table 4-9 security policy rule

Object	Description
action	Configure the rule action as pass, block, or reject. Additional fields must be used with pass/block rules and may not be used with IPsec rules (since IKE signaling passes the standard 5-tuple traffic selectors, only, to the peer). Operators can use the ICMP type generalized firewall rules.
clone-reverse	Set to true to automatically clone this rule, reversing source/destination and to-tunnel/from-tunnel. <i>Note: Affirmed Networks recommends using this option in most cases. When asymmetrical security is required, this option may be omitted.</i>
source/destination-packet-filter	The source-packet-filter may be specified directly or may be specified implicitly by referencing a control-plane configuration (that is, a RADIUS remote server) or address pool. Since source/destination-packet-filter can overlap, associate the filter with a priority. The protocol is specified as the IANA IP protocol type and ports are specified in a range. IPv4 and IPv6 addresses are specified in a range, as a host address, or as a CIDR subnet. The option 255 indicates all protocols.

Table 4-9 security policy rule

Object	Description
control-flow-filter	The control flow specification of the traffic-selector. When the control-flow-filter references a remote server entity, RADIUS, Diameter, etc., the control-flow-filter is specified. The Diameter and RADIUS IP addresses and ports are taken from the referenced object. For server configurations that specify only the remote server, the source information may be directly entered or left open-ended, implying that all possible source addresses within the network-context and all possible source ports are used.
source/destination-ip-pool	The source/destination-ip-pool may reference a nat-pool or user-equipment-pool instead of a source IP address and/or destination IP address. The system expands all of the underlying subpools. If both source and destination point to subpools, then the expansion creates a full matrix.
priority	Specify a priority between 1 and 4096. The default is 100. Set the security policy rules' priorities such that the more specific rule has a higher priority.

security policy tunnel (IPsec)

The IPsec traffic triggered tunnel feature enables Operators to configure on-demand IPsec tunnels. MCC establishes on-demand tunnels after determining that IPsec is required to transmit a packet. After an established tunnel remains inactive (no packets transmitted or received), MCC tears the tunnel down and all resources are released.

Note: To configure a traffic triggered tunnel, do not configure **responder-only** or **anti-replay** for options.

Note: Each security transform requires a different key length. Each tunnel must have a security rule to which it is associated and that rule must specify the direction of the traffic such as **to-tunnel** or **from-tunnel**. Divide the key size (in bits) by 4 to get the required number of hex digits. For example, a 128 bit key requires 32 hex digits.

For example:

```
an3000-mcm-slot7cpu1(config)# security policy rule 61 tunnelrule
action pass to-tunnel tunnel1 source-packet-filter tunnelfilter
ip-start 1.1.1.1 ip-prefix-length 16
```

Note: MCC supports a maximum of 15 proposals received in the IPsec tunnel initialization messages and re-key.

Configure the objects as follows:

Table 4-10 security policy tunnel (IPsec)

Object	Description
admin-state	Admin state of the security policy tunnel. This parameter is set to disabled by default. Set to enabled to delete all active/stale IPsec SAs associated with the security policy tunnel.
anonymous-idi	IKEv2 IDI value(s) for anonymous IKE AUTH (IMSI protection).
authentication-domain	Associated authentication domain - if not specified the default is used
dead-peer-detection	Time, in seconds, to detect dead IKE peer; 0 disables DPD.
epdg-specific-no-gw-found-error-code	IKEv2 error code used to indicate no GW is available to process request. Used by the ePDG GW service 0.. 65536(36).
ike-fragment-size	Set IKEv2 Fragmentation maximum fragment packet size (default 576 for IPv4, 1280 for IPv6).
ike-fragmentation	This feature allows IKEv2 to inter-operate with IKEv2 peers that support IKE Fragmentation. Per RFC 7383, IKE fragmentation adds a new encrypted fragment payload to IKEv2. This payload contains a segment from the encrypted data payload and each encrypted payload is encrypted separately. When decrypted, the segments are concatenated and processed as if they are a single, encrypted data payload. IKE fragmentation applies only to packets that would have been encrypted; it does not apply to IKE_SA_INIT messages. IKE fragmentation applies to all encrypted messages including rekey and delete messages. The fragment size should be less than the path MTU to avoid IP fragmentation. The fragment size is chosen independently by each peer and can use path MTU discovery. IKE fragmentation is used only if both peers advertise support for IKE fragmentation in a NOTIFY payload during IKE_SA_INIT. To enable IKE fragmentation for a tunnel and set an outgoing fragment packet size: <pre>an3000-mcm-slot8cpu1(config)# security policy tunnel NC-GI-1 name ike-fragmentation</pre> Possible completions: <pre>disabled enabled</pre> <pre>an3000-mcm-slot8cpu1(config)# security policy tunnel NC-GI-1 name ike-fragment-size</pre>
ike-groups	The allowed Diffie-Helman IKE groups ISAKPM phase 1 signaling for this tunnel (group-1 group-2 group-5 group-14 group-15 group-16 group-17 group-18).
ike-life-time	The lifetime of the phase 1 IKE SA, in seconds. (180 to 86400)

Table 4-10 security policy tunnel (IPsec)

Object	Description
ike-transform	IKE transform selections enumerated as <code><cipher><authentication><prf></code> The complete set of header type, cipher, and authentication algorithms. IPsec defines two encapsulation protocols: ESP (Encryption Security Protocol) and AH (Authentication Header). ESP includes both cipher and auth, but AH performs only auth. The transform naming convention is as follows: <code><encapsulation>-cipher algorithm>-cipher key size>-<auth algorithm>-authkey size>-<auth digest size></code> ■ auth* – Supported Integrity algorithms: md5 sha1 sha256 sha384 sha512 aes-xcbc ■ cipher* – Supported IKE encryption algorithms: 3des aes128 aes192 aes256 aesctr128 aesctr192 aesctr256 ■ name* ■ prf* – Supported Pseudorandom Function algorithms: md5 sha1 sha256 sha384 sha512 aes-xcbc <i>Note: The AES GCM performs both cipher and auth together.</i>
ike-v2-psk-select-source	The source from which a pre-shared key is selected for IKE V2 negotiations. ■ remote-identity (default) – The identity provided by the remote peer as its identity is used to select the pre-shared key from the authentication configuration. ■ local-identity – The identity provided by the remote peer as that of the MCC device is used to select the pre-shared key from the authentication configuration. This may be configured as email, fully-qualified-domain-name, and ip-address. ■ from-tunnel – The pre-shared key configured for the tunnel is selected. <i>Note: This control only affects the pre-shared key selection, and has no impact on the identity validation performed as part of the IKE V2 negotiation sequence.</i>
ike-version	The ike version for this tunnel (v1 v2 v2-fallback-v1).
ipsec-life-time	The lifetime of the phase 2 IPsec SA, in seconds (720 to 3600).

Table 4-10 security policy tunnel (IPsec)

Object	Description
ipsec-transform	IPsec transform selections enumerated as <code><cipher><authentication></code> Used to separate the transform configuration for IPsec-SA ■ auth – Supported Integrity algorithms: md5 sha1 sha256 sha384 sha512 aes-xcbc ■ cipher – Supported ESP encryption algorithms: 3des aes128 aes192 aes256 aesctr128 aesctr192 aesctr256 aes128gcm8 aes128gcm16 aes192gcm8 aes192gcm16 aes256gcm8 aes256gcm16 null ■ name* ■ prf* – Supported Pseudorandom Function algorithms: md5 sha1 sha256 sha384 sha512 aes-xcbc The AES-128 GCM and AES-256 GCM symmetric-key algorithms use Intel’s Advanced Encryption Standard (AES) New Instructions (AES-NI). These instructions use Intel hardware support to perform some of the complex processing, enabling faster execution of the AES algorithm. Symmetric-key algorithms use the same cryptographic keys for both encryption of plain text and decryption of cipher text. The keys represent a shared secret between two or more entities to maintain a private link for data security and confidentiality. AES-GCM can be used with IPsec Encapsulating Security Payload (ESP) for origin authenticity, integrity, and data confidentiality.
local-identity	The local IKE identity of this tunnel. enforce – Set to true or false to indicate whether to enforce that the remote identity matches what is signaled by the peer.
loopback-ip	The local loopback IP address to use for tunnel encapsulation and/or IKE negotiation. This loopback must be configured on the network-context prior to configuring it at the security tunnel.
manual-key	The manual key information for this tunnel. Each transform will have a different length of keys (authentication-in authentication-out cipher-in cipher-out spi-in spi-out). Note: When manual-key is specified, all IKE parameters become obsolete.
max-number-of-child-sas	Limit the number of child SAs per tunnel.
mobike	Enable or disable MOBIKE (for responder mode only).
name	Name of this tunnel.
network-context	Network context for this tunnel.

Table 4-10 security policy tunnel (IPsec)

Object	Description
options	The various options controlling this tunnel, such as: ikev1-aggressive-mode, long-sequence-numbers-only, anti-replay, responder-only, responder-uses-remote-secret, auto-initiate, transport-mode, short-sequence-numbers-only, ipsec-pre-frag. ipsec-pre-frag – Large packets (size after IPsec encryption/encapsulation > egress interface MTU) are fragmented before encryption. The resulting IP fragments are encrypted into individual ESP packets to be transmitted. This feature is useful in cases where NAT devices are unable to handle IP fragments. Post-fragmentation (fragmentation on ESP packets) results in fragment loss when passing through such NAT devices. This feature improves interoperability with other IPsec implementations. MCC and ePDG require this capability to inter-operate with a security gateway or function as a security gateway. This feature is supported on the MCC Enterprise VPN (or GRE-VPN), ePDG, and other IPsec applications. Pre-fragmentation applies to IPv4, IPv6, and a mix of IPv4/IPv6 (inner IPv4/outer IPv6 or inner IPv6/outer IPv4) IPsec tunnels and applies to IPsec tunnel mode only. If configured, IPsec attempts to fragment the packet's option. If not configured, fragmentation occurs at the physical interface (IPsec post-encapsulation). <i>Note: If IPsec fragmentation fails, MCC sends a Path Maximum Transmission Unit (PMTU) message to the packet originator.</i> ipsec pre-frag is supported on different applications as follows: ■ ePDG applications – IPsec pre-fragmentation only occurs on complete IP packets without any L2/L3 tunnel headers at the SWu interface. ■ MCC enterprise VPN applications – IPsec pre-fragmentation occurs on the complete IP packets at the MCC Gi interface. ■ IPsec WAG (IPsec egress direction) and MCC GRE-VPN application (IPsec egress direction) – Egress packets are encapsulated twice (GRE then IPsec encapsulation). GRE bypasses fragmentation if packets are targeted for a security-enabled network-context. GRE pushes the IPv6 fragmentation boundary to IPsec. IPsec pre-fragmentation occurs on the GRE payloads (the original IP packets), not on the GRE packets. If large packets do not require security protection, MCC falls back to GRE fragmentation.
pfs-groups	The allowed Diffie-Helman Perfect Forward Security (PFS) groups for this tunnel (group-01 group-02 group-05 group-14 group-15 group-16 group-17 group-18 group-19 group-20 group-21 none). Use none to support UEs that both require PFS and do not support PFS.
pre-shared key	Pre-shared key for IKE negotiation.

Table 4-10 security policy tunnel (IPsec)

Object	Description
remote-identity	The remote IKE identity for this tunnel. This may be configured as email, fully-qualified-domain-name, and ip-address. enforce – Set to true or false to indicate whether to enforce that the remote identity matches what is signaled by the peer.
remote-ip	The remote IP address to use and tunnel encapsulation and/or IKE negotiation.
second-authentication-domain	Authentication domain for second authentication exchange.
tunnel-alarm-on	Enable an alarm indication when IPsec tunnels are removed and established between the MCC and its remote IPsec peers. The tunnel-alarm-on option enables Operators to enable (true) this alarm on a per-tunnel basis. The default is off . MCC generates the following alarms if tunnel-alarm-on is set to true : TUNNEL-DOWN – All IPsec tunnels are removed between MCC and a specific IPsec peer. TUNNEL-UP – A new IPsec tunnel is established between MCC and a specific IPsec peer. See tunnel-alarm-on (page 4-22).
tunneled-network-context	The network context for which this tunnel will carry traffic.

For example:

IKE tunnel configuration:

```
an3000-mcm-slot7cpu1(config)# security policy tunnel GI tunnelname
ike-groups group1 ike-life-time 28800 ike-version v1 ipsec-life-time
3600 loopback-network-context GI loopback-ip dataloop options
auto-initiate,ikev1-aggressive-mode pre-shared-key ipsec remote-ip
1.1.1.1 transform esp-3des-192-sha1-160
```

Manual key tunnel configuration:

```
an3000-mcm-slot7cpu1(config)# security policy tunnel GI manualtunnel
remote-ip 1.1.1.1 loopback-network-context GI loopback-ip dataloop
transform esp-aes-128-md5-128 manual-key spi-in 256 spi-out 1000
authentication-in aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa authentication-out
bbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb cipher-in
1111111122222222aaaaaaaaacccccccc cipher-out
1111111122222222aaaaaaaaacccccccc
```

Traffic triggered tunnel configuration:

```
an3000-mcm-slot7cpu1(config)# security policy tunnel GI tunnelname
ike-groups group1 ike-life-time 28800 ike-version v1 ipsec-life-time
3600 loopback-network-context GI loopback-ip dataloop options
"",ikev1-aggressive-mode pre-shared-key ipsec remote-ip 1.1.1.1
transform esp-3des-192-sha1-160
```

tunnel-alarm-on

After a cluster reboot, the tunnel alarm behavior is the same as when a new policy tunnel is created with the **tunnel-alarm-on** option set to **true**.

New and existing policy tunnels behave as follows when the **tunnel-alarm-on** option is set to **true**. For details, see *Affirmed Networks SNMP MIB and Alarm Reference Guide*.

New Policy Tunnel

MCC generates the *TUNNEL-UP* alarm when the first IPsec tunnel is established on this policy tunnel.

MCC generates the *TUNNEL-DOWN* alarm when all IPsec tunnels are removed from this policy tunnel and when no IPsec tunnel exists on this policy tunnel within 30 seconds of creating the policy tunnel.

Existing Policy Tunnel

When the **tunnel-alarm-on** option is changed from **false** to **true** on an existing policy tunnel, MCC generates the *TUNNEL-UP* alarm.

MCC generates the *TUNNEL-DOWN* alarm after all IPsec tunnels are removed and when no IPsec tunnel exists on this policy tunnel within 30 seconds of the configuration change.

CSM Failover

Typically, a failover of the CSM (on which SPM is running) does not trigger the tunnel alarm because IPsec tunnels (if any) persist over failovers. However, if an IPsec tunnel failure or IPsec tunnel re-establishment occurs due to a CSM failover, MCC generates the *TUNNEL-DOWN* alarm during (or after) CSM failover if existing IPsec tunnels are not preserved. MCC generates the *TUNNEL-DOWN* alarm if no IPsec tunnel exists within 30 seconds after the failover, regardless if IPsec tunnels are established on a policy tunnel prior to the CSM failover. MCC generates the *TUNNEL-UP* alarm if the IPsec tunnel is re-established on the new active CSM.

Show Security Commands

Display security statistics and security rules. Also displays security transforms resulting from IKE and manual tunnels.

history – Display IKE and IPsec transaction history table configuration parameters.

ike-stats – Display IKE statistics.

instance-status – Security instance status table.

modexp-stats – Display MODEXP statistics.

redundancy-stats – Display redundancy statistics.

service-status – Security service status table.

statistics – Display network-context and rule security rules and security transforms resulting from IKE and manual tunnels.

MCC supports the following security show commands. For specific details and output, see the CLI and *Acuitas Performance Statistics*.

- `show security history ike-failures <network-context> <id>`
- `show security history ike-negotiations <network-context> <id>`
- `show security history ipsec-failures <network-context> <id>`
- `show security history ipsec-negotiations <network-context> <id>`
- `show security ike-stats <service-name>`
- `show security instance-status <network-context> <service-name>`
- `show security modexp-stats <service-name>`
- `show security redundancy-stats <service-name>`
- `show security sa-data <network-context>`
- `show security sa-data <remote-ip>`
- `show security service-status <network-context>`
- `show security statistics cpu-level data-stats <slot> <cpuid>`
- `show security statistics data-stats <dummyIndex>`
- `show security statistics debug epdg <network-context>`
- `show security statistics network-context <name>`
- `show security statistics rule-stats <network-context> <name>`
- `show security statistics tunnel <network-context> <svc-id> <index>`

Managing Clusters and Nodes

This section outlines the cluster configuration objects and CLI commands used for virtual node management. For additional details, see the Mobile Content Cloud Deployments guides for your platform.

This section describes the following:

- building-block-strategy – Strategy used to create VM personalities. See [cluster \(page 5-2\)](#).
- graceful-restart – MCC gracefully (online) restarts every nodes in the cluster. See [graceful-restart \(page 5-9\)](#).
- data-plane-protection – Automatic protection switching of inter-node data plane (1+1, requires two distinct vlans for inter-node data connectivity protection). See [node \(page 5-12\)](#).
- gnp-sub-pool – GNP subpool range used for internal routing addresses. Configure the CIDR prefix length of the GNP sub-pool range and the start and end value for IP address [v4] of the GNP of the GNP sub-pool range. See [node \(page 5-12\)](#).
- max-asms – Maximum number of ASMs in the cluster (to cap dynamic ASM expansion in the cluster). Changing this value will disrupt in-progress connections. See [node \(page 5-12\)](#).
- node – Card config/status. See [node \(page 5-12\)](#).

an3000-init

Recommission the cluster, clear the config, and reboot the node.

```
an3000-mcm-slot7cpu1# an3000-init
```

This command will recommission the cluster, clear the config, and reboot. Do you want to proceed? [yes,no]

cluster

In **exec** mode, the cluster object enables Operators to perform the following virtual node management functions:

Table 5-1 cluster (exec mode)

Object	Description
audit-config	Audit the system configuration. Options include: ■ director – This options initiates a global audit of the system to show if the configuration is consistent from persistent storage to the application. MCC displays this information in the audit-summary and audit-detail tables using the CLI show command. ■ repair – resends a row if it is missing or inconsistent on the agent as well as send a delete that exists on the agent but not in NMLA.
clear-faults	Clear all the fault files on this chassis.
graceful-restart	Restart the set processes or nodes, in order to clear the ActionRequest. Use the graceful-restart CLI command to gracefully reboot a cluster sequentially in order to preserve established sessions and data. Operators can also use the graceful-restart CLI command to restart selected nodes that are asserting a defined list of reason codes. See graceful-restart (page 5-9)
mcmswitchover	Perform a switchover from the primary MCM to the secondary MCM. The secondary MCM becomes the new primary MCM while the former primary MCM reboots to assume the role of secondary. The CLI prevents a switchover if the secondary MCM's processes are not all operational. The force option allows Operators to override this functionality. Affirmed Networks recommends performing this procedure during a maintenance window. See mcmswitchover (page 5-3).
node	Configure node or node status. Options include: ■ diskcache – Disk Cache. Check the file system for errors or format the disk. ■ processes – Process status. See mcmswitchover (page 5-3). ■ reboot – Reset the slot hard soft. ■ vm-name –

Table 5-1 cluster (exec mode)

reboot	Perform a hard or soft reboot on the node.
--------	--

mcmsswitchover

The MCM module is the Master Controller Module for a cluster of payload nodes (that is, CSM/SSM). The MCM is responsible for a number of cluster-level tasks such as distributing topology information, notification of nodes joining/leaving the cluster, ensuring all nodes are operational with the intended software versions, detecting failure events for tasks/nodes, and initiating associated task redundancy failover events. The MCM controller performs processing that is vital to the functionality of the cluster, therefore, a secondary or standby MCM can be deployed to sync with the primary MCM in order to assume the role of primary in the event of an MCM failure. A well-known virtual IP address is assigned to the primary MCM.

In an MCC VM/cloud environment, signaling between nodes in a cluster is accomplished using TCP/IP messages over the control plane or base network. The MCC periodically exchanges heartbeat messages among nodes to determine the operational/health status. In the event of a primary MCM failure, the secondary MCM detects the heartbeat loss and assumes the role of primary. However, if the cause of the heartbeat failure is due to the underlying control plane failure or segmentation, a split-brain scenario may occur, leading to two MCMs in the primary state. In this scenario, each primary MCM may be connected to a different set of payload nodes and may have independently determined other nodes as failed and therefore performed associated task failover events. If the control plane heals, the MCMs reconnect and perform arbitration to determine the primary MCM. This may result in restarting tasks/nodes in order to coalesce cluster state with the current primary MCM. To avoid split brain, deploy clusters with control plane redundancy.

The following guidelines apply to an MCM switchover:

- The MCC does not support unpausing a VM.
- The MCM redundancy feature is intended to detect MCM faults and failover a secondary MCM to primary without impacting service. The cause of such faults may include any of the following: hardware failure, disk error, or kernel of software error.
- The **cluster <node id> mcmsswitchover** command is used as a graceful MCM failure. A reboot of the former primary MCM is performed in order to exercise boot and file system checks, as well as reload the kernel to ensure that the MCM node is healthy and thus capable of assuming the role of primary MCM.

processes

Perform the following actions on a process:

- `clearrestartdebug` – Restart this application in normal mode after debugging.
- `faults`– Kill a task and generate gdb fault file.
- `debug` – Enter debug command.
- `force-restart` – Force restart this application. Note, this will cause all apps in the process to restart.
- `maintenance` – Cause this application to stop accepting new work and shed existing work.
- `ping` – Ping a TOS task to check for a response. If a response is received within the timeout period, the system returns a result success message. If a response is not received, the system returns a result error message.
- `restart` – Restart this application. Note, this will cause all apps in the process to restart. This allows the task to save work within a timeout period (15-30 seconds) before rebooting. If the timeout period is reached before work is saved, the task is rebooted without saving work.
- `restartdebug` – Restart this application in debug mode.

In **config** mode, the cluster object enables Operators to perform the following virtual node management functions:

Table 5-2 cluster (config mode)

Object	Description
asm-io-level	Allows the optimization of performance of the ASM VMs based on the node's projected traffic throughput. Configure the number of cores that will be used for I/O on an ASM VM. This configuration applies to all ASM VMs in the cluster. For example: <pre>an3000-mcm-slot7cpu1(config)# cluster 83 asm-io-level (<unsignedInt, 1 .. 3>)</pre> 1 — expected throughput on a node is 10 Gbps or less. 2 — expected throughput on a node is 25 Gbps or less. 3 — expected throughput on a node exceeds 25 Gbps. To view the core allocation: <pre>show cluster core-layout</pre> <i>Note:</i> Changes to this value require an ASM VM reboot. See graceful-restart (page 5-9). This CLI command takes effect after a VM reboot. Verify that there are no outstanding <i>anNodeRebootRequired</i> alarms.
building-block-strategy	Strategy used to create VM personalities. See building-block-strategy (page 5-10).

Table 5-2 cluster (config mode)

data-plane-protection	Automatic protection switching of inter-node data plane (1+1, requires two distinct vlans for inter-node data connectivity protection). The entire cluster must use either L3 (Layer-3) or L2 (Layer-2) mode. MCC supports L2 (default) and L3 IP addressable (IPv4/IPv6) data fabric communication over east/west interfaces. In virtualized deployments, the fabric network is the customer's data-center network. Since proprietary ether types are not allowed, this feature allows for an IPv4/IPv6 encapsulation of data plane packets. When selecting L3 as the data plane protocol, a configured data fabric IP address for each SSM/CSM node is required, and these addresses must belong to a common IP subnet. Note that in either mode (L2 or L3), an optional VLAN tag can be configured for the data fabric ports. ■ admin-state – Administrative state of automatic protection switching ■ detached-node-interval – Number of seconds a node must be detached from the data plane before initiating a reset operation mode protected (monitoring paths A and B) or unprotected (monitoring path A only) ■ mode – Protected (monitoring paths A and B) or unprotected (monitoring path A only). See Increasing Traffic Capacity Over East/West Links (page 5-8). ■ protocol-A – protocol type for path A: Layer 2 (MAC) or Layer 3 (IP) L2 L3 ■ protocol-B – protocol type for path B: Layer 2 (MAC) or Layer 3 (IP) L2 L3 ■ reset-detached-nodes – Enables a node to initiate a reset operation when it becomes detached from the data plane ■ vlan-id-A – vlan Id of external vlan 1 ■ vlan-id-B – vlan Id of external vlan 2
gnp-sub-pool	GNP subpool range used for internal routing addresses. Configure the CIDR prefix length of the GNP sub-pool range and the start and end value for IPv4 address of the GNP of the GNP sub-pool range
heap-check	Heap Checker
max-asms	Maximum number of ASMs the cluster can have. Note that changing this value will disrupt in-progress connections

Table 5-2 cluster (config mode)

max-nat-workflow-modules	Maximum number of workflow modules supported by NAT. Previously, when a cluster booted, the port range was allocated into 15 blocks and the MCM assigned one of the 15 port blocks to an SSM when it joined the system. This parameter specifies to carve the port range into a specific number of blocks. To efficiently use the address space defined in your NAT pools, Affirmed Networks recommends setting this parameter to coincide with the number of SSMs/WSMs in your cluster. In other words, in a cluster with only three SSMs/WSMs, set this parameter to 3 even though this parameter defaults to 15.
max-packet-size	Maximum packet size. The default is 2048. You must reboot the cluster after modifying this value. This value must be at least as large as the MTU/MRU value you want to use for the IP interfaces. See the <i>Affirmed Networks Gateway Administration Guide</i>.
memory-error-detection	Enable or disable memory error detection
node	Card config/status such as admin-state, diskcache, dry-up-period, personality, ramdisk, and type. See node (page 5-12) .
ssm-io-level	Allows the optimization of performance of the SSM, IOM, WSM, and ISM VMs based on the node's projected traffic throughput. Configure the number of cores to be used for I/O on SSM, IOM, WSM, and ISM VMs. This configuration applies to all SSM, IOM, WSM, and ISM VMs in the cluster. For example: <pre>an3000-mcm-slot7cpu1(config)# cluster 83 ssm-io-level (<unsignedInt, 1 .. 3>)</pre> 1 – expected throughput on a node is 10 Gbps or less. 2 – expected throughput on a node is 25 Gbps or less. 3 – expected throughput on a node exceeds 25 Gbps. To view the core allocation: <pre>show cluster core-layout</pre> Note: Changes to this value require an SSM, IOM, WSM, and ISM VM reboot. See graceful-restart (page 5-9). This CLI command takes effect after a VM reboot. Verify that there are no outstanding <i>anNodeRebootRequired</i> alarms.

Table 5-2 cluster (config mode)

vcpu-allocation-profile	The vcpu-allocation-profile object specifies the vCPU allocation profile for ASM and ISM VMs on a cluster-wide basis. Configure a value between 1-10. MCC supports dynamic proxy task count. The task count depends on the processor size and VM configuration options in the vcpu-allocation-profile. <i>Note: This setting depends on your call model. Consult the TAC prior to setting this value.</i> ■ ASM – This profile allows control of CPU resource allocation between packet I/O and proxy services. Configure a higher value to provide more resources for packet I/O. ■ ISM – This profile allows control of CPU resource allocation between workflow processing and proxy services. Configure a higher value to provide more resources for proxy services. To view the core allocation: show cluster core-layout <i>Note: Changes to this value require an ASM/ISM VM reboot. See graceful-restart (page 5-9). This CLI command takes effect after a VM reboot. Verify that there are no outstanding anNodeRebootRequired alarms.</i>
-------------------------	--

Increasing Traffic Capacity Over East/West Links

Currently, East/West (EW) links are configured as active and standby, where the standby link supports redundancy. This feature enables Operators to configure both EW links, path A and path B, as active-active, which doubles the traffic capacity. To support the increased traffic, one additional SSM CPU core is required (one SSM core per EW link) which must be applied when provisioning the SSM node. In active-active mode, CSM/SSM load-balancing distributes traffic over both EW links.

When EW links lose connectivity, traffic fails over to the remaining active link but may cause the link to become oversubscribed. You can configure nodes that lose greater than 50% connectivity with other nodes in the cluster on a single EW link over a specified interval to be automatically removed from service using the **detached-on failure-paths-A-or-B** parameter and by enabling **reset-detached-nodes**. Alternatively, configure the **detached-on failure-paths-A-and-B** parameter to remove a node from service only if connectivity of both EW links is less than 50%. Nodes that are removed from service are automatically restored to service when connectivity of EW link(s) is greater than 50%. This configuration applies to both node types, CSM and SSM.

Use the following command to enable active-active mode and detached node failure handling for connectivity failure on either path A or B, or connectivity failure on both paths A and B:

```
an3000-mcm-slot8cpu1(config)# cluster <id> data-plane-protection mode
unprotected-paths-A+B detached-on
[failure-paths-A-or-B|failure-paths-A-and-B] reset-detached-nodes
[enabled|disabled]
```

Use the following command to configure the time in seconds that a node must be detached from the data plane before it is removed from service. The default value is 300.

```
an3000-mcm-slot8cpu1(config)# cluster <id> data-plane-protection mode
unprotected-paths-A+B detached-node-interval
```

Possible completions:

```
<unsignedInt, 30 .. 3600>
```

graceful-restart

Execute this command in **exec** mode. If no arguments are entered, MCC gracefully (online) restarts every nodes in the cluster. The maximum restart time for a 32-node system is approximately two hours. This feature supports the following four operational modes:

- **abort** – The restart-reason to be aborted.
- **max-groups** – Maximum number of protection groups.
- **node** – node number. Gracefully restarts only that node. Note: This is a controlled shutdown/restart with a drain phase and is significantly less disruptive than `cluster <> node <> reboot`.
- **reason** – The restart-reason the graceful-restart acts upon. If a reason is not specified, MCC gracefully restarts all nodes that are declaring any of the defined reasons. The restart time depends on the number of affected nodes. If a reason is specified, MCC gracefully restarts all nodes that are declaring that specific reason. The restart time depends on the number of affected nodes.

The graceful-restart process may take several minutes to complete. Rather than waiting for this process to complete, Operators can issue the **cluster <> graceful-restart abort** command to terminate the graceful state machine after it finishes draining and restarting the current group of nodes.

Use the following command to display the graceful-restart status:

```
an3000-mcm-slot7cpu1# show cluster graceful-restart-status
```

Possible completions:

```
| <cr>
```

```
an3000-mcm-slot7cpu1# show cluster graceful-restart-status
```

```
result success
```

```
reason OK
```

```
graceful-restart-status:
```

```
Idle
```

To resolve the alarm state, either manually reboot the ASM/CSM node or issue one of the following commands to gracefully restart all TosFpFlowAgt processes in the cluster:

- To restart the entire cluster gracefully, enter:
#cluster cluster_number graceful-restart
- To restart all of the nodes that require a restart, enter:
cluster cluster_number graceful-restart reason <reason>
- To restart only those processes that require a restart due to <event-id> on the specified node, enter:
cluster cluster_number graceful-restart node <node-number> event <event-id>

building-block-strategy

Execute this command in **config** mode. Node personality defines which services are started on a given node in a cluster. A node can be configured to use a specific personality, each of which runs a predefined collection of functionality sets. This feature allows Operators to run specialized card/nodes with the exact services they want and increases MCC performance.

The personalities available for a node are filtered based on the Building Block Strategy specified for the cluster. See [About Node Personalities \(page 5-15\)](#). To configure personalities, first specify a *Building Block Strategy* for the cluster. The strategy specifies the supported combinations of node personalities to build the cluster. The Building Block Strategies are:

- **compact Strategy** — For traditional MCM, CSM, and SSM nodes.
Contents:
 - MCM (Management Control Module)
 - SSM (Subscriber Services Module)
 - CSM (Control/Content Services Module)

- **advanced-services-separation Strategy** — Optimized for small clusters (less than 12 nodes). This strategy separates the advanced services (such as Multi Protocol Proxy, Cache, and Video Transcoding) from the Call-Control Management services (PGW, SGW, and GGSN). Systems that do not provide advanced services *should not* deploy the ASM personality node in this strategy. Systems that do provide advanced services do need to deploy the ASM personality node.

Contents:

- MCM (Management Control Module)
- SSM (Subscriber Services Module)
- ASM (Advanced Services Module – Content services)
- CPM (Control Plane Module)

- **control-services-separation Strategy:** This strategy separates the centralized and distributed aspects of the Call-Control Management services. If either the advanced services (such as Multi Protocol Proxy, Cache, and Video Transcoding) or the Call-Control Management services (PGW, SGW, and GGSN) increase over time, the cluster can be expanded/scaled to support the growth.

Contents:

- MCM (Management Control Module)
- ASM (Advanced Services Module – Content services)
- SSM (Subscriber Services Module)
- CCM (Centralized Control Module)
- DCM (Distributed Control Module)

- **specialized Strategy** — Optimized for large clusters (more than 12 nodes). In addition to the separation provided by advanced-service-separation, this strategy separates the I/O and the workflow services on the SSM node. In addition, it separates the centralized and distributed aspects of the Call-Control Management services. Systems that do not provide advanced services do not need to deploy the ASM personality node in this strategy. Systems that do provide advanced services do need to deploy the ASM personality node. Applies to cluster nodes only.

Contents:

- MCM (Management Control Module)
- IOM (I/O Module)
- WSM (Workflow Services Module)
- ASM (Advanced Services Module – Content services)
- CCM (Centralized Control Module)
- DCM (Distributed Control Module)

- **integrated-services Strategy** — Centralizes the execution of data forwarding services (workflow and I/O) and advanced services (such as Multi Protocol Proxy, Cache, and Video Transcoding) in one VM and Call-Control Management services (PGW, SGW, and GGSN) in another VM. As a result, the cluster can be expanded/scaled to independently support growth of gateway call control services, workflow, I/O, and content services. The ISM provides a one-stop processing node for data forwarding services and content services.

Contents:

- MCM (Management Control Module)
- CPM (Control Plane Module)
- ISM (Integrated Services Module: Content and Subscriber Services)

- **workflow-services separation Strategy (supports ePDG only)** — Combines the required (separate) building blocks into one building-block strategy to build the ePDG VMs (MCM, CPM, IOM, and WSM). This strategy streamlines the ePDG VM creation process and enables VNF Manager to build the cluster directly.

Contents:

- MCM (Management Control Module)
- CPM (Control Plane Module)
- IOM (I/O Module)
- WSM (Workflow Services Module)

node

Note: The MCC allocates redundant resources on nodes that share the least fate with nodes where the active resource is running. This feature maximizes the chances that a secondary is able to take over when an active resource encounters an outage. The physical location properties of a node (such as site, rack, shelf, blade, and node ID) are used to determine the fate-sharing costs between the primary and secondary pair.

Note: Operators must configure the site, rack, shelf, and blade parameters for all payload VMs during initial commissioning. If you later change any of these values, you must reboot the cluster for proper fate-sharing to take effect. Operators configure these physical location properties using the following command:

```
an3000-mcm-slot7cpu1(config)# cluster 14 node 5
```

Possible completions:

blade Physical blade where VM is hosted

rack Physical rack where VM is hosted

shelf Physical shelf where VM is hosted

site Physical site where VM is hosted

Configure the objects as follows:

Table 5-3 cluster node

Object	Description
admin-state	Administrative state of this card. Options include: ■ enabled – Card is providing full service. Tasks are active and running on the card. The overall capacity of system goes up by the capacity of this card. ■ disabled – Card is out of service. Tasks are not running on the card and the card is powered down. The tasks for this card are no longer visible from the CLI. This state of this card is similar to that of a card that is not yet provisioned. The overall capacity of the system goes down by the capacity of this card. See the <i>Software Release Notes for the Mobile Content Cloud</i> for important maintenance and system capacity information prior to using this command. ■ maintenance – Card has stopped processing new service and is gradually transferring existing services elsewhere. The tasks on this card are allotted time (specified by the dry-up-period) to transfer existing services. This card cannot serve as a backup for services provided by other cards. The standby tasks on this card are moved elsewhere. This card cannot serve as a provider of any global scope services, therefore any global scope tasks that are active on this card are switched over to their backups. Any standby global scope tasks from this card are moved to other enabled cards. The global scope services are provided with a very short dry-up-period before switching over. See the <i>Software Release Notes for the Mobile Content Cloud</i> for important maintenance and system capacity information prior to using this command.
blade	Physical blade where VM is hosted.
data-fabric-ip-address-A	Data fabric IP Address port A MCC supports both L2 East/West (EW) and L3 EW. When you upgrade a cluster to a release supporting L2/L3 EW, the cluster automatically defaults to the existing L2 solution. When operating in L2 mode, existing cluster-level parameters are used. L3 EW supports fabric redundancy and each node has two fabric interfaces, each configured for different L3 subnets. MCC continually monitors fabric connectivity after the node is fully operational. MCC removes a node from service if fabric connectivity is lost or degraded.
data-fabric-ip-address-B	Data fabric IP Address for redundant path B
data-fabric-mtu-A	MTU on data fabric port A
data-fabric-mtu-B	MTU on data fabric port B
data-fabric-prefix-length-A	Prefix length of the data fabric IP address port A
data-fabric-prefix-length-B	Prefix length of the data fabric IP address port B

Table 5-3 cluster node

Object	Description
diskcache	Disk cache
dry-up-period	Number of seconds allowed for process to dry up (shed work) before going idle
id	1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128
ip-address	Management IP address of hardware
personality	Defines which services are started on a given node in a cluster. A node can be configured to use a specific personality, each of which runs a predefined collection of functionality sets. This feature allows Operators to run specialized card/nodes with the exact services they want and increases MCC performance. Also see About Node Personalities (page 5-15) .
rack	Physical rack where VM is hosted
ramdisk	Store Charging Data Records (CDR) and NAT Binding Records (NBR) to volatile storage for v-MCM node types. The MCC performs a vMCM memory check when 512 M or higher ramdisk is requested. The total amount of vMCM memory required is 8 G plus the ramdisk memory. A vMCM with 8 G supports 128/256 M ram disk. For example: <pre>cluster <id> node <id> ramdisk size 256M admin-state enabled</pre> where size can be 128M, 256M, 512M, 1G, 2G, 4G, 8G, 16G (512M and greater require a minimum RAM of (8GB + size) on the v-MCM). Once enabled on the primary MCM, subsequent changes to RAMDISK size or admin-state require a switchover or reboot to take effect. Use show cluster <id> node <id> disk to view the RAMDISK. See (page 5-29). For NBR records, configure a port NBR interface service profile of type ramdatarecord. <i>Note: CDR/NBR records are stored on RAMDISK on the primary MCM. In the case of MCM failure or switchover, any records that have not shipped to the remote server (since the last transfer) are lost. To minimize loss of files, set the upload-frequency (page 3-12) to the minimum value of 1 minute.</i>
shelf	Physical shelf where VM is hosted

Table 5-3 cluster node

Object	Description
site	Physical site where VM is hosted
timeout	Maximum time in milliseconds allowed to detect card failure, typically a loss of power. If not configured, the default of 8 seconds is used.
type	■ CSM: Control/Content Services Module ■ MCM: Management Configuration Module ■ SSM: Subscriber Services Module ■ v-CSM: Virtual Control/Content Services Module ■ v-MCM: Virtual Management Configuration Module ■ v-SSM: Virtual Subscriber Services Module
vm-ip-address	Management IP address of VM
vm-name	Name of VM

About Node Personalities

When you configure a node for the cluster, select the node type and node personality.

- Node type:
 - CSM: Control/Content Services Module
 - MCM: Management Configuration Module
 - SSM: Subscriber Services Module
 - vCSM: Virtual Control/Content Services Module
 - vMCM: Virtual Management Configuration Module
 - vSSM: Virtual Subscriber Services Module
- Node Personality: The personalities available for a node are filtered based on the Building Block Strategy specified for the cluster.

The various personalities are described in the following table.

Table 5-4 Personalities

Personality	Description
CCM: Centralized Control Module <i>Note: Do not set Admin State to disabled for CCM personality.</i>	<i>Applies to node type vCSM</i> Runs the centralized tasks needed for call-control functionality and routing protocols.
DCM: Distributed Control Module	<i>Applies to node type vCSM</i> Runs the distributed tasks needed for call-control functionality (mainly the PGW, SGW, and RADIUS tasks).
CPM: Control Plane Module <i>Note: Do not set Admin State to disabled for CPM Personality,</i>	<i>Applies to node type vCSM</i> Runs all tasks needed for call-control functionality.
IOM: I/O Module <i>Note: Do not set Admin State to disabled for IOM personality.</i>	<i>Applies to node type vSSM</i> Is responsible for receiving packets into the MCC and sending the packets out.
WSM: Workflow Services Module	<i>Applies to node type vSSM</i> Provides workflow services.
SSM: Subscriber Services Module	<i>Applies to node type vSSM</i> Runs all services provided by IOM and WSM.
ASM: Advanced Services Module	<i>Applies to node type vCSM</i> Runs the tasks needed for providing advanced services like Multi Protocol Proxy, Cache, WAP GW, etc.
CSM: Control/Content Services Module	<i>Applies to node type vCSM</i> Runs all services provided by CPM and ASM.
ISM: Integrated Services Module	<i>Applies to node type vSSM</i> Runs content and subscriber services

Value-Added Services Guidelines

When deploying Value-Added Services (VAS), use the following VM provisioning guidelines:

- Use one of the following cluster building block strategies:
 - specialized (preferred)
 - advanced-services-separation (small deployments only)
 - compact (small deployments only)
- Provision a sufficient number of CSM VMs as required for your network deployment:
 - use CCM, DCM, and ASM personalities for specialized cluster building block strategy
 - use CPM and ASM personalities for advanced-content-separation cluster building block strategy
 - use default v-CSM personality for compact cluster building block strategy
- Designate a dedicated CPU to each ASM by setting the processor affinity.
- Set the number of ASM virtual cores to the number of CPU cores minus 10%, reserving the remainder for the VM hypervisor. For example:
 - 18 on CPU with 20 virtual cores
 - 22 on CPU with 24 virtual cores

When deploying CPU intensive functionality, the minimum number of virtual cores is 18.

- Provision memory for ASMs or CSM VMs running Value-Added Services:
 - Minimum of 55GB for typical deployments
 - Minimum of 32GB evaluation deployments with a limited number (up to a few hundred) of subscribers
- Performance and throughput at any vCPU provisioning level (including 16 vCPUs) depends on traffic profiles and call models. Between this assignment and the setting in ASM optimization level, you can scale performance and throughput based on the network services you provide in your network deployment.

Scale In Limitations

Scale In have the following limitations:

- On Scale In of a DCM, some sessions that do not passively get closed (on the gateway that is being removed) will be closed, disrupting the traffic belonging to those sessions.
- On Scale In of an ASM, some HTTP connections that do not close during the dry-up period will be closed. The UE can re-establish a new connection.
- When a WSM node is disabled as part of dynamic scaling, the grid removes the range of FSAPIs that were earlier assigned to the card. The gateway migrates all of the sessions using FSAPIs of this card to another active SSM (or WSM) node. As a result, workflow sessions are deleted on an old FSAPI and a new workflow session are created with the new FSAPI. The following limitations arise with this implementation:
 - During the session migration from an old FSAPI to a new FSAPI, data may be lost in the timing window between when the session is deleted on the old FSAPI and recreated on the new FSAPI.
 - If Proxy is enabled, all of the connection data information maintained by wfdivert module will be lost and the corresponding HTTP transactions will timeout. The same limitation applies to all of the Content services that use this module (SIP, WAP Gateway, SMTP Proxy).
 - If NAT is enabled, all of the active connections (not just TCP connections) will be lost. They must be reestablished after the workflow session is recreated on the new WSM. The same limitation applies to all the supported ALGs, as the flow state is not synchronized with a standby (SIP, RTSP, FTP, TFTP ALGs).
 - When an NE is deployed as a PGW or GGSN, the reassembly of fragmented packets that originate from UE is performed in the workflow manager module. As part of dynamic contraction, when a session is migrated to another SSM (WSM), any fragments that have not yet completed reassembly will be lost (because after FSAPI migration, the workflow for this session is anchored on a different WSM.)
 - The grid can choose to use WSM modules as backup for FSAPIs assigned to other cards. When a WSM is Scaled Out, the grid designates one of the remaining cards to work as a backup for those FSAPIs for which the WSM being disabled was serving as backup. Thus, the backup for certain FSAPIs will move from one node to another during both dynamic WSM expansion and contraction. This condition exposes a timing window if (even before the workflow sessions on the backup FSAPIs are fully set up) the SSM(WSM) with the active FSAPI either reboot or crash. Data will be lost during this timing window. This condition is expected to be transient, as normal operation is expected to resume as soon as the workflow session is created on the SSM(WSM).

data-path dpi

In config mode, Operators can configure cdc classification, decode dns, configure a dns script, and configure dpi tethering. Also see [Tethering Detection Overview \(page 5-21\)](#).

Configure the objects as follows:

Table 5-5 data-path dpi

Object	Description
cdc-classification	Enable or disable cdc classification. To enhance release of DPI Signatures, MCC implements the Custom Defined Classification (CDC) feature with the main DPI Detection Engine (PACE2). PACE2 supports adding custom defined classifications to extend detection without requiring a new PACE2 Detection library. When you enable CDC Support, MCC is able to reduce the turnaround time to support new Application Detection. By default, this parameter is disabled. Changes to this setting take effect upon next reboot.
decode-dns	This feature allows a subscriber to perform DNS queries for specific free-rated applications and sites even when a subscriber runs out of credit during the session or is already out of credit upon PDN session creation. A subscriber can access the free-rated applications and sites if the DNS queries/replies match the configured domains. Set to enabled to enable DPI for DNS packets. This object defaults to disabled. Note: If you change the status from enabled to disabled or disabled to enabled, you must perform an SSM reboot. Note: This feature supports only DNS query and response packets; DNS on TCP is not supported. Configure the high priority offline rating-group with white listed domains configured in attached service rules, along with DNS packet filters and the online rating-group (with comparatively lower priority than offline-rating-group) to capture all DNS traffic.
decode-first-http-transaction-only	For HTTP persistent connections, a single TCP connection is used to send multiple HTTP requests/responses. Because the persistent connection is not closed, all packets are subjected to DPI decoding, which can cause an increase in CPU usage. Enable this object to avoid a CPU usage spike. ■ enabled (default) – On persistent HTTP connections, as soon as fastpath detects the first server response header, the connection is closed (at L7 level) and subsequent HTTP transactions are not sent to the DPI engine for decoding. This setting can cause classification errors, as DPI examines only the first transaction. ■ disabled – On persistent HTTP connections, packets of all HTTP transactions are sent to the DPI engine for decoding.

Table 5-5 data-path dpi

Object	Description
dpiscript	Configure the dpiscript-related parameters. See the <i>Affirmed Networks DPI Script User Guide</i> for information on configuring and creating a DPI script and collecting and analyzing pcaps. ■ script-classification – Set to enabled or disabled (default). In the disabled state, user-defined signatures are loaded on the system but the classification/detection phase does not consider these signatures. ■ http-classification – Set to enabled or disabled for dpiscript http support. This parameter takes effect only when script-classification is enabled. ■ https-classification – Set to enabled or disabled for dpiscript https support. This parameter takes effect only when script-classification is enabled.
tethering-detection-algorithm	Control the tethering detection algorithm. ■ gquic-user-agent – Admin state for running tethering detection algorithm to detect Pdanet+ Apps. The default is disabled. Changes to this setting take effect upon an SSM reboot. ■ tcpip-fingerprinting – Set to Enable to run the tethering detection algorithm using TCP-IP fingerprinting. By default, this parameter is disabled. Changes to this setting take effect upon next reboot. ■ ttl – Admin state for running tethering detection algorithm using ttl. ■ ttl-connection-tracking – Admin state for running tethering detection algorithm using ttl by taking connection tracking into consideration. ■ user-agent – Admin state for running tethering detection algorithm using user-agent
tethering-quality-index	Enter a quality index value to set the value for the tethering quality (confidence) index to mark a connection as tethered. A connection is marked as tethered only if this reported quality index is greater than or equal to the detection algorithm QI value (configured using the data-path dpi tethering-quality-index value command.
tethering-quality-index-result-weight	Set the weight value (percentage) for different tethering method's quality index.

Tethering Detection Overview

Affirmed Networks supports the following tethering detection protocols:

- NAT detection based on subscriber OS detection (User Agent (UA) based tethering detection). See [User-Agent Tethering Detection \(page 5-21\)](#).
- Time-to-Live (TTL). See [TTL DPI Detection \(page 5-21\)](#).
- TCP/IP fingerprinting. See [TCP/IP Fingerprinting Detection \(page 5-23\)](#).

User-Agent Tethering Detection

Subscriber OS detection is a combination of patterns and heuristics and involves the HTTP user agent, protocols used only for a certain OS, and information from specific protocols.

Affirmed Networks uses the following DPI detection algorithms:

- NAT detection results to find the primary device's OS type
- OS detection results to find a flow's subscriber OS type
 - If the two OS types are different, then the flow is marked as tethered
 - If the two OS types are the same, then the flow is marked as primary

TTL DPI Detection

Affirmed Networks uses the following methods in the Time-to-Live (TTL) DPI detection algorithm to determine if a subscriber is tethering. TTL is a counter which is decremented every time the packet is forwarded. If the counter is one less than what the network expects, it indicates that it has been forwarded once, presumably via a tether.

- **Method 1** – Using the following well-known default OS TTL for detection: Window (128), Android/Linux/MAC OS (64), and special (255):
 - When MCC recognizes the packet as TTL + 1 or TTL+2, it marks the flow as tethered and determines that the subscriber is tethering.
- **Method 2** – Using TTL delta = 2 for detection and maintaining a bitset of all known TTLs - [0,255]:
 - If a packet's TTL – 1 or TTL-2 is a known TTL of the bitset, then it is a primary flow.
 - If a packet's TTL + 1 or TTL + 2 is a known TTL, then it is a tethered flow and the subscriber is tethering.

■ **Method 3** – Using DPI OS/version + TTL for detection:

If a flow's OS type is different from the main operating system, then it is a tethered flow and the subscriber is tethering.

- If the OS types are the same, the main operating system version is unknown, and the packet TTL is well-known or TTL-1 is known from the bitset, then MCC sets the main operating system's version to the flow's OS version.
- If the OS types are the same and OS versions are different, MCC marks the flow as tethered and determines that the subscriber is tethering.

■ **Method 4** – Restart tethering detection state:

A subscriber's tethering state should be regularly checked to determine if the device is turning on/off tethering at will. MCC uses the time stamp variable *last_tethered_pkt_received_time* to remember the latest update for the tethered traffic. If no tethered traffic for this subscriber is recorded over a configured threshold, then MCC changes the subscriber tethering state to *not_yet_detected* and initiates the workflow reclassification and the tethering detection process.

TCP/IP Fingerprinting Detection

Different computer Operating Systems (for example, Android, iOS, Windows, Mac OSX, Linux, etc.) set up their TCP/IP stacks with different default values and settings (for example, the initial packet size, initial TTL, window size, etc.). The combination of these values can give a *fingerprint* that can be used to identify what operating system is running on the originating device.

TCP/IP fingerprinting, also known as TCP stack fingerprinting, is the analysis of data fields in a TCP/IP packet to identify the various configuration attributes of a networked device. The information that can be learned from a TCP fingerprint includes the type of device from which the packet originated and the operating system that it is running.

The following table shows the operating system coverage.

Version	Windows		Linux/Android		Mac OS/iOS
	≤ version 7	≥ version 8	< kernel 4.1.0	≥ kernel 4.1.0	
TCP	IP fragmentation ID	Not supported	TCP time stamp	Not supported	TCP time stamp
UDP	IP fragmentation ID	Not supported	Not supported	Not supported	Not supported

TCP-based Fingerprinting for Detection of NAT

DPI uses the TCP time stamp fields in the TCP header to detect TCP packets per-client and determine whether NAT exists. This method is supported on Unix, Linux, MacOS/iOS, and Android systems (not for windows).

IP-based Fingerprinting for Detection of NAT

DPI uses identification fields in the IP header for fingerprinting to detect IP packets per-client and determine whether NAT exists. This method is supported on Windows systems up to Windows 7 (Windows 8 and 10 are not supported).

Tethering Quality Index Value

DPI delivers a quality index of tethering detection which gives the confidence value on the tethering decision (EDRs).

Operators configure the data-path, dpi, and tethering-quality-index value. MCC takes this value into consideration while marking a connection as tethered (by comparing this configured value with the dpi delivered value). This value can be used to reduce false positives.

EDRs

Following tethering-related fields are sent in flow EDRs.

- tetheredFlow
- tetheringEntitled
- tetheringAlgorithm
- qualityIndex

data-path workflow

In exec mode, Operators can display cluster and node-level statistics. See *Acuitas Performance Statistics* for details.

extended-storage

In exec mode, Operators can set the following extended-storage parameters:

- *create* – Partition and format the extended storage disk. See [Creating the Extended Storage Disk \(page 5-25\)](#).
- *delete* – Unmount and delete the extended storage disk. See [Deleting the Extended Storage Disk \(page 5-26\)](#).
- *reset* – Reset the extended storage state machine after a failure in order to retry

Creating the Extended Storage Disk

Use the following example to create the extended storage disk:

- 1 Issue the following command to verify the settings on disk 7 on each MCM:

```
show cluster <number> node <node-id> disk
```

The settings should be:

```
admin-state = unused
oper-status = down
disk-size = at least 100 GB
disk-error = No
```

- 2 Issue the following command to create the disk:

```
extended-storage create
```

- 3 At the confirmation prompt, enter **yes**.

The command should return:

```
result success
reason OK
```

- 4 Issue the following command to verify that the create process is running:

```
show extended-storage show-progress
```

- 5 Continue to issue the **show extended-storage progress** command until the create process completes.

When the process completes, the system displays:

```
DISK CREATE - State Machine Completed Successfully
```

- 6 Issue the following command to verify the settings on disk 7 on each MCM:

```
show cluster <number> node <node-id> disk
```

The settings should be:

```
admin-state = enabled
oper-status = up
All size fields contain data
disk-error = No
```

Deleting the Extended Storage Disk

If you want to replace a current extended storage partition with a new larger partition, use the **extended-storage delete** command. The command moves the current storage data back to the `/boot/partitions/storage` partition.

Use the following example to delete the extended storage disk:

- 1 Issue the following command to verify the settings on disk 7 on each MCM:

```
show cluster <number> node <node-id> disk
```

Use the following settings:

```
admin-state = enabled
```

```
oper-status = up
```

```
All size fields contains data
```

```
disk-error = No
```

Note: *IMPORTANT: In the process of deleting extended-storage, the system moves all data on the extended-storage partition (diskID 7) back to the original storage partition (diskID 2). If the disk-usage on diskID 7 is greater than the available storage on diskID 2 (that is, disk-size of diskID 2 minus disk-usage of diskID 2), the delete process fails and MCC generates an alarm indicating the lack of free space on diskID 2.*

Therefore, you should check the amount of free space available on diskID 2 prior to attempting the delete process. If diskID 2 lacks sufficient space for the contents of diskID 7, you must manually remove a sufficient amount of data off of diskID 7 to bring it under the diskID 2 free space threshold.

- 2 At the confirmation prompt, enter **yes**. The system displays:

```
result success
```

```
reason OK
```

- 3 Issue the following command to verify that the delete process is running:

```
show extended-storage progress
```

- 4 Continue to issue the **show extended-storage progress** command until the delete process completes. When the delete process completes, the system displays:

```
DISK DELETE - State Machine Completed Successfully
```

- 5 Issue the following command to verify the settings on disk 7 on each MCM:

```
show cluster <number> node <node-id> disk
```

Use the following settings:

```
admin-state = unused
```

```
oper-status = down
```

```
disk-size = at least 100 GB
```

```
disk-error = No
```

Show Commands

MCC supports the following cluster and node show commands. For specific details and output, see the CLI and *Acuitas Performance Statistics*. This section includes some examples.

```
an3000-mcm-slot7cpu1# show cluster
```

Possible completions:

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graceful-restart-status Display graceful-restart status

| Output modifiers

<cr>

Possible match completions:

audit-detail Configuration audit detail

audit-summary Configuration audit detail

control-plane-protection Automatic protection switching of
inter-node control data plane (1+1, requires two distinct vlans for
inter-node data connectivity protection)

core-layout View Configured Core Layout Of Nodes

data-plane-protection Automatic protection switching of inter-node
data plane (1+1, requires two distinct vlans for inter-node data
connectivity protection)

node Card config/status

statistics Chassis statistics

summary Chassis Summary

tnd-peer-stats TND peer counts

tnd-total-stats get total tnd stats for an ssm

show cluster <id> audit-detail

Display configuration audit details such as the process name, NMLA agent name (that uniquely identifies each NML agent within a TOS task), error description, and the incorrect xpath. It also includes a flag indicating if an attempt was made to repair the error (based on the repair option).

show cluster <id> audit-summary

Display configuration audit summary such as the number of agents connected, number of agents that have registered audit callbacks, number of rows with errors, and the time of the last audit.

show cluster <id> control-plane-protection status <node-id>

- *status* – Displays the data plane redundancy table including the bitmap of reachable nodes over port A and B, elapsed time since switchover, cause of the switchover, and number of switchovers.
- *node-id* – Source Node Number
- *using-port* – Port used: A, B, A+B. Indicates the source node is using either entirely redundant path A, B, or combination of A and B to reach other nodes. When MCM controller 7 is primary, the preferred (or default) redundant data path is A, and when MCM controller 8 is primary, the preferred path is B. If both ports A + B are in use, it implies a connectivity health check has failed on the preferred path.
- *num-switchovers* – Number of times the source node has changed a path selection for at least one other node.
- *bitmap-reachable-nodes-port-A* – Indicates a bit map (in hexadecimal) of other nodes that are reachable over port A. For example, for the reported value 0x000000000037, the hexadecimal value 37 is 00110111 in binary. The lowest order bit (rightmost) refers to node 1, next bit slot 2, and so on. The value 00110111 implies nodes 1,2,3,5,6 are reachable over port A.
- *bitmap-reachable-nodes-port-B* – Indicates a bit map (in hexadecimal) of other nodes that are reachable over redundant port B. For example, for the reported value 0x000000000037, the hexadecimal value 37 is 00110111 in binary. The lowest order bit (rightmost) refers to node 1, next bit slot 2, and so on. The value 00110111 implies nodes 1,2,3,5,6 are reachable over port B.
- *most-recent-switchover-cause* – A date time stamp and cause for the most recent change in path selection (which is based on keep-alive connectivity health checks). For example:
- 2015-05-04T14:34:55-0400 Switching to path:B map: ffffffff, (A=eth1.700, B=eth3.800), primary MCM path B
- Because the primary MCM controller is mcm-8, the preferred path is port B. All known nodes are reachable over path B. The map variable (hexadecimal 0xffffffff) indicates all 1's in binary, implying all nodes can be reached using port B. Value '0' in the map indicates redundant port A for the corresponding node.

show cluster <id> control-plane-protection keepalive <node-id> <peer-node-id>

- *keepalive* – port-A-most-recent-rx-ms (Last Rx port A) and port-B-most-recent-rx-ms (Last Rx port B).
- *node-id* – Source Node that is receiving keep-alives.
- *peer-node-id* – Source Node is receiving keep-alives from this peer (that is, the sender).
- *port-A-most-recent-rx-ms* – The elapsed milliseconds since a keep-alive was received from the peer.
- *port-A-most-recent-rx+tx-ms* – The elapsed milliseconds since a keep-alive was received from the peer where the peer is indicating that it is receiving keep-alives from the source node (that is, a keep-alive indicating connectivity in both directions is successful).
- *5-minute-history-percent-rx-port-A* – The percentage time the link has been receiving keep-alives from the peer over the past 5-minute interval.
- *5-minute-history-percent-rx+tx-port-A* – The percentage time the link has been receiving keep-alive from the peer over the past 5-minute interval, where peer is receiving from the source node.
- *port-B-most-recent-rx-ms* – The elapsed milliseconds since a keep-alive was received from the peer.
- *port-B-most-recent-rx+tx-ms* – The elapsed milliseconds since a keep-alive was received from the peer where the peer is indicating that it is receiving keep-alives from the source node (that is, a keep-alive indicating connectivity in both directions is successful).
- *5-minute-history-percent-rx-port-B* – The percentage time the link has been receiving keep-alives from the peer over the past 5-minute interval.
- *5-minute-history-percent-rx+tx-port-B* – Thhe percentage time the link has been receiving keep-alive from the peer over the past 5-minute interval, where peer is receiving from the source node.

Table 5-6 show cluster control-plane-protection

Object	Parameter	Description
Keep Alive	Node ID	Source Node that is receiving keep-alives.
	Peer Node ID	Source Node is receiving keep-alives from this peer (that is, the sender).
	Port A Most Recent Rx MS	The elapsed milliseconds since a keep-alive was received from the peer.
	Port A Most Recent Rx+Tx MS	The elapsed milliseconds since a keep-alive was received from the peer where the peer is indicating that it is receiving keep-alives from the source node (that is, a keep-alive indicating connectivity in both directions is successful).
	5 Minute History Percent Rx Port A	The percentage time the link has been receiving keep-alives from the peer over the past 5-minute interval.
	5 Minute History Percent Rx+Tx Port A	The percentage time the link has been receiving keep-alive from the peer over the past 5-minute interval, where peer is receiving from the source node.
	Port B Most Recent Rx MS	The elapsed milliseconds since a keep-alive was received from the peer.
	Port B Most Recent Rx+Tx MS	The elapsed milliseconds since a keep-alive was received from the peer where the peer is indicating that it is receiving keep-alives from the source node (that is, a keep-alive indicating connectivity in both directions is successful).
	5 Minute History Percent Rx Port B	The percentage time the link has been receiving keep-alives from the peer over the past 5-minute interval.
	5 Minute History Percent Rx+Tx Port B	The percentage time the link has been receiving keep-alive from the peer over the past 5-minute interval, where peer is receiving from the source node.

Table 5-6 show cluster control-plane-protection

Object	Parameter	Description
Status <i>Indicates the reachability of remote nodes over each of the fabric ports</i>	Node ID	Source Node Number
	Using Port	Port used: A, B, A+B Indicates the source node is using either entirely redundant path A, B, or combination of A and B to reach other nodes. When MCM controller 7 is primary, the preferred (or default) redundant data path is A, and when MCM controller 8 is primary, the preferred path is B. If both ports A + B are in use, it implies a connectivity health check has failed on the preferred path.
	Num Switchovers	Number of times the source node has changed a path selection for at least one other node.
	Bitmap Reachable Nodes Port A	Indicates a bit map (in hexadecimal) of other nodes that are reachable over port A. For example, for the reported value 0x000000000037, the hexadecimal value 37 is 00110111 in binary. The lowest order bit (rightmost) refers to node 1, next bit slot 2, and so on. The value 00110111 implies nodes 1,2,3,5,6 are reachable over port A.
	Bitmap Reachable Nodes Port B	Indicates a bit map (in hexadecimal) of other nodes that are reachable over redundant port B. For example, for the reported value 0x000000000037, the hexadecimal value 37 is 00110111 in binary. The lowest order bit (rightmost) refers to node 1, next bit slot 2, and so on. The value 00110111 implies nodes 1,2,3,5,6 are reachable over port B.
	Most Recent Switchover Cause	A date time stamp and cause for the most recent change in path selection (which is based on keep-alive connectivity health checks). For example: 2015-05-04T14:34:55-0400 Switching to path:B map: ffffffff, (A=eth1.700, B=eth3.800), primary MCM path B Because the primary MCM controller is mcm-8, the preferred path is port B. All known nodes are reachable over path B. The map variable (hexadecimal 0xffffffff) indicates all 1's in binary, implying all nodes can be reached using port B. Value '0' in the map indicates redundant port A for the corresponding node.

show cluster <id> data-plane-status

The output format for this command may be inconsistent or not display properly but the system displays the correct information.

```
an3000-mcm-slot8cpu1# show cluster 6 data-plane-protection
data-plane-protection status
ELAPSED
TIME SINCE BITMAP BITMAP
NODE USING NUM SWITCHOVER REACHABLE REACHABLE
ID PORT SWITCHOVERS MS NODES PORT A NODES PORT B MOST RECENT SWITCHOVER
CAUSE
-----
1 port-A 0 0 0xa000000000003 0xa000000000003 Initialize to:A
(eth1.88), standby dev=eth3.888
2 port-A 0 0 0xa000000000003 0xa000000000003 Initialize to:A
(eth1.88), standby dev=eth5.888
46 port-A 0 0 0xa000000000003 0xa000000000003 Initialize to:A
(eth1.88), standby dev=eth3.888
48 port-A 0 0 0xa000000000003 0xa000000000003 Initialize to:A
(eth1.88), standby dev=eth5.888
```

show cluster <id> graceful-restart-status

```
an3000-mcm-slot7cpu1# show cluster 93 graceful-restart-status |
result success
reason OK
graceful-restart-status:
Idle
```

show cluster <id> node <id>

To display all of the processes that require a restart due to a configuration mis-match, enter one of the following commands. The event-id for the graceful-restart is indicated by the restart-type. If event-id is not specified, the system displays all processes on the node that require an administrative action.

```
# show cluster <cluster_num> node <node-id> requested-restarts
<reason>
```

cluster_num: Cluster Number

node-id: Node number

event-id: Id of the event to show all the processes on the requested node that have raised the event event-id.

Enum value

```
an3000-mcm-slot8cpu1# show cluster 22 node task requested-restarts
node
```

ID	RESTART REASON	TASK NAME	RESTART TYPE
2	isol-cpus-mismatch	ImAgent_22_2_1-ImAgent_2	node-restart

The following example displays all processes that require a reboot in the cluster.

```
an3000-mcm-slot8cpu1# show cluster 22 node task requested-restarts
node
```

The following command displays processes that require a reboot on the requested node.

```
an3000-mcm-slot8cpu1# show cluster 22 node 2 task requested-restarts
```

The following command displays all processes on the cluster that require a reboot because of isol-cpus-mismatch.

```
an3000-mcm-slot8cpu1# show cluster 22 node 2 task requested-restarts
isol-cpus-mismatch
```

show cluster <id> summary

Displays information about each node, including admin state, HA state, and uptime. In the output, an **asterisk** at the end of the release number indicates that one or more tasks on the node is not yet operationally up. Depending on the configuration, some tasks can take longer than others to come up. When all nodes are operationally up, the cluster is fully operational.

For example:

SLOT	CPU					ADMIN		CPU	
ID	NUMBER	NUMBER	TYPE	PERSONALITY	MODEL	STATE	HA STATE	STATE	
CPU	UPTIME		RELEASE						
54	1	1	v-csm	CSM	vCSM	enabled	active	up	
D:000 H:01 M:50 S:13 9.2.0.0-896.REL*									

IP Protocols and Routing

This section describes the IP protocols and routing objects and commands.

This section describes how to configure the Border Gateway Protocol (BGP) and Bidirectional Forwarding Detection (BFD) IP protocol settings. Use the `ip-protocols` object to configure the following parameters:

- [ip-data-path](#) (page 6-2)
- [ip-protocols policy](#) (page 6-7)
- [ip-protocols bfd](#) (page 6-11)
- [ip-protocols bgp router](#) (page 6-12)
- [ip-protocols bgp-vpn router](#) (page 6-21)
- [ip-protocols routing-options](#) (page 6-22)
- [ip-protocols log-nsm](#) (page 6-23)
- [ip-protocols nat deterministic](#) (page 6-23)
- [ip-protocols ospfv2](#) (page 6-24)
- [ip-protocols ospfv3](#) (page 6-30)
- [ip-protocols tcp](#) (page 6-33)

ip-data-path

UE-IP based ECMP for subscriber traffic.

By default, MCC ECMP-based subscriber traffic is load balanced using 5-tuple hashing. Flows are hashed based on selected IP packet fields (source address, destination address, source port, destination port, IP version). The ip-data-path feature enables Operators to configure the load-balancing mode and enable scaling in Fastpath.

Configure the objects as follows:

Table 6-1 ip-data-path

Object	Description
allow-nonip-traffic	By default, MCC does not allow non-ip traffic resulting in the IOM dropping all non-ip traffic. Set to true to allow non-ip traffic.
ecmp-load-balance-mode	Configure the ECMP load-balancing mode. Options include: 5-tuple and ue-ip-address.
fast-aging-time	Workflow connection default fast-aging-time in seconds.
fast-aging-time-proxy	Workflow proxy connection fast-aging-time in seconds.
max-heap-paging-size	Maximum number of heap SKB page messages that can be queued in WSM.
max-tcp-reorder-buffers	Maximum number of TCP reorder buffers per SSM/WSM.
max-tcp-reorder-buffers-per-connection	Maximum number of TCP reorder buffers per connection.
stateless-wsm-mode	Enables scaling in Fastpath. This feature adds support for a local database to increase PGW capacity. See stateless-wsm-module - Increasing PGW Capacity (page 6-3) .
ue-ip-address	1-tuple based hashing. MCC uses the UE source IP address for upstream traffic (Gi) and the UE destination IP address for downstream traffic (Gn).

stateless-wsm-module - Increasing PGW Capacity

MCC supports the use of a local database to handle subscriber transactions, thereby reducing the need to use MCC capacity. When there is no subscriber activity, subscribers are moved to the local database. When subscriber activity occurs, MCC moves the subscriber out of the database and handles the events and messages. After a period of inactivity, the subscriber is again moved to the database.

Configuration commands control when a subscriber is eligible to move to the database. The local database increases session capacity per-CPM VM. Operators use CLI commands and filters to display which subscribers are in memory or in the database.

The following table lists the VM sizing requirements to support 3M sessions.

VM Type	vCPU	RAM
MCM1	8	16
SSM1	8	64
CPM1	8	64
MCM2	8	16
SSM2	8	64
CPM2	8	64

Note: The VM memory may require an adjustment based on the system load within the same total available memory.

Limitations

This feature has the following limitations:

- Paging messages are not stored as part of compressed sessions and are therefore flushed. If the session has any paging data pending, then the session is not moved to the database. Also, if Fastpath crashes, the queued packet is lost.
- Extensive workflow debug counters are not stored (only partial information is retained)
- Higher packet loss occurs during WSM switchover
- GGSN calls are not moved to the database
- Only 1 Gi/Gn data packet is stored per session
- Support for SGW calls only is not supported
- Do not use the **zone default gateway subscriber clear imsi** or **zone default gateway subscriber count imsi** commands to clear subscribers when the scaling feature is enabled. Instead, use the **subscriber clear** and **subscriber count** commands.
- If the subscriber is in the database, then the **show sub pdn-session <pdn-session-id>** and **show sub pdn-bearer <pdn-session-id> <ebi>** commands will not display any data statistics where the PDN session ID belongs to the SGW. The MCC correctly displays all other details.

Configuring Subscriber Inactivity

Since the scaling feature works upon subscriber inactivity, the inactivity-timer-duration must be configured for the subscriber. See Affirmed Networks Gateway Administration Guide for details.

The inactivity-time-duration enables Operators to detect and potentially take action on subscribers that are inactive. Inactive is defined as no user data or control messages in the uplink or downlink direction for all PDN sessions and all bearers.

- SGWs – The inactivity-timer-duration is defined in the gateway-common profile.
- PGWs and GGSNs – The inactivity-timer-duration is defined in the gateway apn object if the value is non-zero, otherwise the value is defined in the gateway-common profile.

Enabling Scaling

The **stateless-wsm-mode** command enables scaling in fastpath. After adding the configuration, if the packet is received in fastpath and the session is not found, then packets without fastpath sessions are handled and the session is fetched from the database to memory. If the **stateless-wsm-mode** CLI command is not issued, then the datapath drops the packet.

The default value is **false** indicating that IoT optimization is disabled in fastpath.

```
an3000-mcm-slot7cpu1(config)# ip-data-path stateless-wsm-mode true
```

The **push-to-db-threshold** command enables the threshold in the control path. This command applies to per-SGW and per-PGW processes. When this threshold is crossed, calls are moved to the database. If this value is set to **0**, then sessions are not moved to the local database. If this value is set to **1**, then all sessions are moved to the local database. For example, if **push-to-db-threshold** is configured as **2**, then session 2 and subsequent sessions are moved to the database.

If any of the following conditions exist, the session is not moved to the database:

SGW

- Session is in extended paging state
- Subscriber is created for indirect tunnel
- Subscriber has multiple PDNs.
- Sessions without dedicated bearers
- Subscriber is not yet inactive

PGW

- Subscriber has dedicated bearers
- Subscriber session has any of the Gx/Gy re-joining timers running
- Subscriber session has active procedures running
- Subscriber has allocated address using DHCP
- Subscriber is not yet inactive

For example:

```
an3000-mcm-slot7cpu1(config)# zone default gateway sgw MYSGW
an3000-mcm-slot7cpu1(config-sgw-MYSGW)# push-to-db-threshold 2
```

```
an3000-mcm-slot7cpu1(config-zone-default)# gateway pgw MYPGW
an3000-mcm-slot7cpu1(config-pgw-MYPGW)# push-to-db-threshold 2
```

Note: This limit is per-PDN and per-SGW task. For example, if the number is *X*, it indicates that *X-1* sessions are kept in memory. After sessions are added, additional sessions are moved to the database.

ip-protocols policy

Configure the objects as follows:

Table 6-2 ip-protocols policy

Object	Description
access-list	A list of addresses to match for a route filter. Use this command with the standard ranges to configure an access list for filtering packets and to configure the action, address, and mask entry for the access-list
access-list-word	A list of addresses to match for a route. Use this command to configure an access-list name to filter packets and to configure the action, address, and prefix-length for the named access-list. This command controls the transmission of packets on an interface and restricts the contents of routing updates. The switch stops checking the access list after a match occurs. The priority of an access-list is based on the order in which the access-list command was configured. For example: ■ If the user configures the Prefix ACL as deny, the label does not advertise to any peer. ■ If the user configures the Prefix ACL as no-match then it applies the next advert-list. This is interpreted as continue. ■ If the user configures the Prefix ACL as permit and there is a peer ACL, then the label advertises to all peers permitted by the peer ACL. ■ If the user configures the Prefix ACL as permit but the peer prefix is none then the label advertises to all peers.
aspath-access-list	Specify an aspath-access list expression and sequence number.
comm-list	Configure a community list <1-99> to match BGP standard community attributes.
comm-list-ext	Configure an extended community list <1-99> to match BGP extended community attributes.
comm-list-word	Configure a named community list <1-99> to match BGP standard community attributes.
comm-list-word-ext	Configure a named extended community List <WORD> to match BGP extended community attributes.
prefix-list	Configure the sequence number and IPv4 address for this IPv4 prefix list.
prefix-list-ipv6	Configure the sequence number and IPv6 address for this IPv6 prefix list.

Table 6-2 ip-protocols policy

Object	Description
route-map	Create a route map. Use this command to enter the route-map mode and to permit or deny match/set operations. This command controls and modifies routing information to allow redistribution of routes. It has a list of match and set commands to which it is associated. The match commands specify the conditions under which redistribution is allowed, and the set commands specify the particular redistribution actions to be performed if the criteria enforced by match commands are met. Route maps are used for detailed control over route distribution between routing processes. Route maps also allow policy routing, and might route packets to a different route than the obvious shortest path. The route-map object supports the following additional parameters: ■ match-as-path – BGP AS path list match ■ match-interface – Define the interface match criterion. This command specifies the next-hop interface name of a route to be matched. ■ match-ip-address – Specify the match address of the route. If a match exists for the specified IP address, and permit is specified, the route is redistributed or controlled, as specified by the set action. If the match criteria are met, and deny is specified then the route is not redistributed or controlled. If the match criteria are not met, the route is neither accepted nor forwarded, irrespective of the permit or deny specifications. The route specified by the policies might not be the same as specified by the routing protocols. Setting policies enable packets to take different routes, depending on their length or content. Packet forwarding based on configured policies overrides packet forwarding specified in routing tables. ■ match-ip-address-prefix-list – BGP prefix list match by name. ■ match-origin – Match BGP origin attribute ■ match-route-type-external – Match the specified external route type. AS-external LSA is either type-1 or type-2. External type-1 matches only type 1 external routes and external type-2 matches only type 2 external routes. ■ match-tag – Match the specified tag value. ■ set-as-path-prepend – Prepend string for a BGP AS-path attribute. If multiple AS-Numbers are to be pre-pended they must be separated by a space and quoted, for example: "700 800". <1-65535>;AS-Number ■ set-atomic-aggregate – Set atomic aggregate BGP attribute ■ set-ip-next-hop – Set the specified next-hop value. ■ set-ipv6-next-hop – Set ipv6 address of the next hop ■ set-level – Set where to import route. ■ set-local-preference – Set BGP local preference path attribute

Table 6-2 ip-protocols policy

Object	Description
route-map continued ...	■ match-ip-next-hop – Specify a next-hop address to be matched in a route-map. If a match exists for the specified next-hop address, and permit is specified, the route is redistributed or controlled as specified by the set action. If the match criteria are met, and deny is specified, the route is not redistributed or controlled. If the match criteria are not met, the route is neither accepted nor forwarded, irrespective of the permit or deny specifications. The route specified by the policies might not be the same as the route specified by the routing protocols. Setting policies enable packets to take different routes depending on their length or content. Packet forwarding based on configured policies overrides packet forwarding specified in routing tables. ■ match-ipv6-address-prefix-list – BGP IPV6 prefix list match by name ■ match-ipv6-next-hop – Match ipv6 address of the next hop ■ match-metric – Match a metric of a route. If a match exists for the specified metric, and permit is specified, the route is redistributed or controlled as specified by the set action. If the match criteria are met, and deny is specified, the route is not redistributed or controlled. If the match criteria are not met, the route is neither accepted nor forwarded, irrespective of permit or deny specifications. The route specified by the policies might not be the same a the route specified by the routing protocols. Setting policies enable packets to take different routes depending on their length or content. Packet forwarding based on configured policies overrides packet forwarding specified in routing tables. ■ set-metric – Set metric of route ■ set-metric-type – Set type of metric for destination routing protocol ■ set-origin – Set BGP origin code ■ set-originator-id – Set the ip address of the originator of the route ■ set-tag – Set tag value for destination routing protocol ■ set-vpn4-next-hop – Set the next hop address for vpnv4 routes ■ set-weight – Set BGP weight for routing table

Table 6-2 ip-protocols policy

Object	Description
route-map continued ...	■ set-metric – Set a metric value for a route and influence external neighbors about the preferred path into an Autonomous System (AS). The preferred path is the one with a lower metric value. A router compares metrics for paths from neighbors in the same ASs. To use this command, you must first have a match clause. Match and set commands set the conditions for redistributing routes from one routing protocol to another. The match command specifies the match criteria under which redistribution is allowed for the current route-map. The set command specifies the set redistribution actions to be performed, if the match criteria are met. If the packets do not match any of the defined criteria, they are routed through the normal routing process. ■ set-metric-type – Set the metric type for the destination routing protocol. Select a type to be either Type-1 or Type-2 in the AS-external-LSA when the route-map matches the condition. ■ set-tag – Set a specified tag value. The Tag parameter is the route tag that is labeled by another routing protocol (BGP or other IGP when redistributing), because AS-external-LSA has a route-tag field in its LSAs. In addition, when using route-map, the software can tag the LSAs with the appropriate tag value. Sometimes the tag matches with using route-map, and sometimes, the value may be used by another application.

ip-protocols bfd

Configure the objects as follows:

Table 6-3 ip-protocols bfd

Object	Description
bfd-timer-profile	Configure the BFD timer parameters.
common send-bfd-session-down-alarms	When enabled, the system generates this alarm when an existing or new BFD session goes down. This alarm automatically clears when the session restores. A gratuitous clear is performed the first time a BFD session is restored, even if there are no active BFD session down alarms. This alarm has a Critical severity. By default, this parameter is set to disabled.
instance	Configure and manage BFD sessions within a network-context. One instance is supported per network-context. See instance (page 6-11). The instance object supports the following options: ■ bdf-auth-profile – Configure an authentication key and a chain of authentication keys. ■ bfd-session – Enable BFD sessions for all IPv4/IPv6 static routes on all interfaces. ■ bfd-interface – Enable the BFD protocol at the interface and routing protocol levels. ■ bfd-multihop-peer – Configure BFD multi-hop peer timer and authentication values. ■ key-chain – Configure a chain of authentication keys.

instance

The instance object is used to configure and manage BFD sessions within a network-context. One instance is supported per network-context.

The Bidirectional Forwarding Detection (BFD) network protocol is used to detect forwarding path failures between two peer routers. BFD peers are not required to be directly connected. BFD provides a low-overhead fault detection mechanism for all media types, encapsulations, topologies, and routing protocols.

Routing protocols, such as BGP and OSPF can use BFD to provide fast peer failure detection. BFD contributes to reduced overall network convergence time by sending rapid failure detection notices to the routing protocols in the local router to initiate the routing table recalculation process.

Note: This system supports BFD for static routes and BGP only.

In networks that deploy L2/L3 switches, link status is not always indicative to the health of the forwarding plane between two peers. Unlike BGP, static routes have no mechanism to detect a failed next-hop that is connected to an L2/L3 switch, thereby creating a situation where traffic is routed to a next-hop that is no longer available. To avoid this situation, Operators can utilize BFD for static routes. BFD can create sessions for all static routes that use a particular interface, or BFD can be configured for an individual static route.

The Affirmed Networks BFD feature adheres to the following specifications:

- *RFC 5880 –Bidirectional Forwarding Detection*
- *RFC 5881 – Bidirectional Forwarding Detection (BFD) for IPv4 and IPv6 (Single Hop)*

ip-protocols bgp router

The Border Gateway Protocol (BGP) features enable you to add BGP4 routers and manage the following features:

- Enable logging and filter activity
- Manage IPv4 and IPv6 unicast address family features on a neighbor peering
- Enable [ip-protocols bgp-vpn router \(page 6-21\)](#)

MCC allows BGP to advertise Martian routes created for UE Pools and loopbacks. You can use route maps to suppress Martian routes. Martian addresses are addresses reserved by IANA for special use. The following addresses are reserved for private networks and treated as Martian routes.

10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16, 169.254.0.0/16

A BGP community is a group of destinations that share a common property. Community information is included as a path attribute in BGP update messages. This information identifies community members and enables you to perform actions on a group without having to elaborate upon each member.

BGP supports two types of community attributes:

- *Standard* community access lists only match standard community attributes. Standard community lists are used to specify well-known communities and community numbers.
- *Extended* community access lists only match extended community attributes. Extended community lists are used to filter communities using a regular expression.

You can use standard and extended community attributes to trigger routing decisions, such as acceptance, rejection, preference, or redistribution.

Multi-instance BGP supports the following features and functions:

- IP-VPN based on MPLS/BGP (RFC 4364)
- Provide PE functionality:
 - Multiple VRFs (VPNs with IPv4/v6 address family)
 - VPN labeled routes distribution via MP-BGP
 - VPN VRFs are mapped to network-contexts (populated with UE pools prefixes)
- Gn network reachability to other Operators (inter-AS)

The system supports multiple BGP instances but only a single instance is capable of supporting IP-VPN (MP-eBGP). See [About max-paths \(eBGP multipath\) \(page 6-21\)](#).

MPLS/BGP provides the capability to process labeled packets as follows:

- MPLS labeled packet forwarding for VPN traffic
- Support for packet forwarding according to RFC 4364 (inter-as option B):
 - Packet headers contain one label (VPN label) and VPN IP header
 - One label is pushed on outgoing packet
 - One label is popped for incoming packet
 - Ethernet header indicates MPLS packet type

Single-Hop Border Gateway Protocol

The MCC supports IPv4 and IPv6 SH-BGP within a shared subnet. MCC is dually homed to two routers for redundancy and each router belongs to a distinct shared subnet. MCC uses a common VLAN 1000.

For example:

- Router 1 belongs to 1.1.1.0/24
- Router 2 belongs to 2.2.2.0/24

MCC's routing table accounts for route updates and withdrawals from multiple neighbors with same remote address. For example, if peering session 1 fails (but other sessions remain active), the routing table does not lose the route to 10.10.10.0/24.

You can configure multiple neighbors with the same remote address but you must configure a different **update-source** address for each neighbor. MCC uses the **update-source** address as the local address for the neighbor. For example:

```
ip-protocols bgp router neighbor BGP-vth-2-1-gi

    neighbor-addr 107.243.141.14

    remote-as      64903

    timers keep-alive 30

    timers hold-time 90

    update-source phy-interface GI_2_1_5

    fall-over-bfd single-hop

!

ip-protocols bgp router neighbor BGP-vth-3-1-gi

    neighbor-addr 107.243.141.14

    remote-as      64903

    timers keep-alive 30

    timers hold-time 90

    update-source phy-interface GI_3_1_5

    fall-over-bfd single-hop
```

Multi-Active Mode with Single-Hop Shared Subnet

MCC supports a multi-active mode with single hop shared-subnet configuration. This release supports multiple BGP peering to the same remote address for SH-BGP with shared subnets. Each peer is configured to use a different local address. When MCC receives a BGP connection request, it performs the appropriate peer lookup. The next hop interface for the peer is derived from the appropriate interface matching its local address.

Enhanced BGP and NSM to handle multiple routes with the same next hop, but different next hop interfaces.

When a BGP alarm is activated or deactivated for a peer, MCC looks up the peer matching the local and remote address.

This configuration supports:

- Multiple, Single-Hop BGP sessions in a shared subnet
- Multiple SH-BGP sessions per-line card with the same remote address. Although, the typical configuration is one SH-BGP session per slot.
- Single-Hop BFD session to protect the SH-BGP sessions
- Single-Hop BGP shared subnet with/without multi active routing
- IPv4 and IPv6 BGP sessions and MP-BGP
- TAM support with shared subnet

router

BGP router supports the following:

- IPv4 and IPv6 address family options and manage ipv4/ipv6 unicast address family features on a neighbor peering
- filtering to inbound and/or outbound update messages carried by this peering by specifying a filter direction and list-name, and route-map-name
- equal cost multi-path routing support. Configure the maximum number of EBGp/IBGP ECMP routes allowed to a destination
- 20 IPv4/IPv6 next-hops using BGP multi-path routing for each network-context route table.

Configure and manage BGP routing including always-compare-med, default-local-preference, extended-asn-cap, local-as*, network-context*, router-id*

Configure the objects as follows:

Table 6-4 ip-protocols bgp router

Object	Description
aggregate-nexthop-check	Aggregate a route into a "summary" route only if the nexthop of the route is the same as the nexthop of the summary route. Applies to both IPv4 and IPv6 unicast address families. Disabled by default. Cannot be modified once BGP router has been created.
always-compare-med	Use MED in path selection when comparing paths from different ASs. Disabled by default.
best-path	Configure best path selection algorithm such as as-path-ignore, compare-routerid, dont-compare-originator-id, dont-tie-break-on-age.
bgp-logging	Enable all logging or enable logging for specific areas
bgp-timers	Configure keep-alive and hold-timers for all peers.
clear all hard	Clear routes by resetting peerings.
clear all soft	Clear routes without resetting peerings.
default-local-preference	Set the default local preference value. Default = 100.
extended-asn-cap	Enable 4-octet ASN support. Disabled by default. Cannot be modified once BGP router has been created.
graceful-restart	Configure restart BGP graceful options such as fakeKey*, restart-time, stalepath-time

Table 6-4 ip-protocols bgp router

Object	Description
ipv4/6-ucast-af	Enable BGP peers to carry routing information belonging to the ipv4/v6 Unicast address family. ■ capabilities – Configure the capabilities of this peering ■ default-originate – Used to announce a default route to BGP peer(s). This route is used by the peer router to establish reachability to the MCC ■ distribute-list – Apply access list filtering to in(bound) and/or out(bound) update messages carried by this peering ■ filter-list – Apply AS path access list filtering to in(bound) and/or out(bound) update messages carried by this peering ■ maximum-prefix – Configure the maximum number of ipv4-ucast prefixes accepted on this peering. If that number (max-value) is reached the peering will be shut down and an "error" event will be logged. However, if the "warning-only" option is also specified, only the "error" event will be logged and the peering will remain . A "threshold" option is available to log "warning" events once the number of prefixes received exceeds the "threshold" but not the max-value. ■ next-hop-self – Set router as next-hop for all routes advertised to peer. ■ prefix-list – Apply prefix list filtering to in(bound) and/or out(bound) update messages carried by this peering ■ route-map – Apply route-map filtering to in(bound) and/or out(bound) update messages carried by this peering ■ soft-reconfiguration-inbound – Enable local storage of received updates. If enabled, it stores incoming update information to RIB-IN for a specific peer. This also enables soft clearing (without tearing down the session) of sessions. This is useful for re-evaluating the inbound policy on the same set of incoming updates.
ipv4/6-ucast-af aggregate-address	Configure the route aggregation policy. By default, the summarized routes continue to be advertised and the summary route has the "atomic aggregate" attribute.
ipv4/6-ucast-af clear dampening	Clear reset all dampened bgp routes or specific bgp routes in both dampened and history status.
ipv4/6-ucast-af clear flap-statistics	Clear the flap count and history duration for all the prefixes.

Table 6-4 ip-protocols bgp router

Object	Description
ipv4/6-ucast-af dampening rDampening	Configure dampening options such as max-suppress-time, reachability-half-life, reuse-limit, suppress-limit, unreachable-half-life. BGP route flapping describes the situation in which BGP systems send an excessive number of update messages to advertise network reachability information. BGP flap damping is a method of reducing the number of update messages sent between BGP peers, thereby reducing the load on these peers, without adversely affecting the route convergence time for stable routes. Dampening is supported for both IPv4 and IPv6 addresses. ■ max-suppress-time – The longest amount of time in minutes (1..255) that a route can be suppressed. This value must be greater than or equal to the unreachable-half-life value. The default is 60 minutes. ■ reachability-half-life – A minute value (1..45) used to describe the amount of time that must elapse to reduce the penalty by one half. When a BGP route flaps, it is assigned a penalty. The penalty is decreased by half after each reachability-half-life period. The default is 15 minutes. ■ reuse-limit – A minute value (1..20000) that is compared with the penalty. If the penalty is less than the reuse limit, a suppressed route that is up will no longer be suppressed. The value must be less than or equal to the max-suppress-time value. When the route flapping penalty decreases and falls below this reuse-limit, the route is unsuppressed. The default is 750 minutes. ■ suppress-limit – A minute value (1..20000) to specify the suppress limit. A route is suppressed when its penalty exceeds this limit. The value must be greater than or equal to the reuse-limit value. The default is 2000 minutes. ■ unreachable-half-life – A minute value (1..45) used to describe the time for penalty. This value must be greater than or equal to the reachability-half-life value. The default is 15 minutes.
ipv4/6-ucast-af neighbor maximum-prefix	Specify the maximum number of ipv4/6 ucast prefixes accepted on this peering. If that number (maximum-prefix) is reached, peering is shut down and an "error" event is logged. However, if the "warning-only" option is also specified, only the "error" event is logged and the peering remains. A "threshold" option is available to log "warning" events once the number of prefixes received exceeds the "threshold" but not the max-value. Specify the max-value, threshold, and warning-only options.
ipv4/6-ucast-af redistribute	Specify the protocol and route-map-tag for the redistribution policy configuration.
local-as	Autonomous System Number for this BGP router. Cannot be modified once BGP router has been created.

Table 6-4 ip-protocols bgp router

Object	Description
neighbor	■ advertisement-interval – Set the minimum interval (0..600 seconds) between routing updates sent on this peering. A value of "0" means an update will be sent immediately after a routing change. Default value is 5 seconds on IBGP peerings and 30 seconds on EBGP peerings. ■ capabilities – Configure the capabilities of this peering. – route-refresh – Advertise support of route-refresh ■ clear-hard – Clear routes by resetting peering. ■ clear-soft – Clear routes without resetting peering. – both – Clear advertised and received routes. – in – Clear only received routes. – out – Clear only advertised routes. ■ connection-retry-time – Time between attempts to connect to peer. Default is 120 secs. ■ ebgp-multihop – Enable multi-hop EBGP peering and set maximum TTL. Default is 255. ■ fall-over-bfd – Enable Bi-directional Forwarding Detection. Configure single-hop or multi-hop. ■ local-as – Local Autonomous System Number (ASN) for this peering. Both the router ASN and this ASN will be prepended to Path Attributes advertised on this peering. <i>WARNING: The behavior of this setting is modified by the local-as-no-prepend setting. See the help for that parameter for more detail.</i> – If local-as is configured for a peering and local-as-no-prepend is not set, both the router ASN and the peering ASN are prepended to path attributes advertised on this peering. – If local-as is configured for a peering and local-as-no-prepend is set, only the peering ASN is prepended to path attributes advertised on this peering, (the router ASN is suppressed). – If local-as is not configured for a peering, the system ignores the local-as-no-prepend setting and prepends only the router ASN path attributes advertised on this peering. ■ local-as-no-prepend – When a local-as value is set on a peering using neighbor <name> local-as <ASN>, setting neighbor <name> local-as-no-prepend prevents the router's ASN from being included in the path attributes for prefixes advertised by the peering. ■ neighbor-addr* – IP address of neighbor. Cannot be modified once neighbor has been created.

Table 6-4 ip-protocols bgp router

Object	Description
neighbor (continued) ...	■ password – Enable MD5 authentication on peering and set password. BGP supports MD5-based authentication on IPv4 and IPv6 peering as specified in RFC 2385. The password object allows the MCC to connect to a peer router via the BGP protocol using an agreed upon password. The password object applies to eBGP/iBGP, direct/multi-hop peering sessions and uses a shared password model. In other words, the same password must be configured for both peers. A password mismatch causes an authentication failure. See password (page 6-20). ■ peering-name* – Name of peering. ■ remote-as* – Autonomous System of remote neighbor. Cannot be modified once neighbor has been created. ■ timers – Configure keep-alive and hold-time for peering. – hold-time – Hold timer [0, 3..65535] secs. – keep-alive – Keep-alive timer [0..65535] secs. ■ update-source – Specify the interface that will act as the source of this peering session. For SH-BGP, you can configure multiple neighbors with the same remote address but you must configure a different update-source address for each neighbor. MCC uses the update-source address as the local address for the neighbor.
router-id	Changing the router-id will reset peers.

password

Authentication applies to the TCP connection associated with BGP session. If authentication is configured, a peering session is established if authentication succeeds. If a failure occurs, MCC drops the TCP packets and the session is torn down if keep-alive is configured.

To enable BGP MD5 authentication for a BGP peer, configure the password as follows:

If special characters or spaces are used within the password, they must be enclosed in quotes. For example:

```
ip-protocols bgp-vpn router BOS neighbor cisco7200 password "other one"
```

*peering-name** – Name of the peering.

*remote-as** – Autonomous System of remote neighbor. Cannot be modified once neighbor has been created.

ip-protocols bgp-vpn router

Manage BGP4 routing with VPN support.

To configure dampening parameters, see [ipv4/6-ucast-af dampening rDampening \(page 6-18\)](#). To configure router parameters, see [router \(page 6-15\)](#) and [OSPFv2 Router Options \(page 6-26\)](#).

Configure the objects as follows:

Table 6-5 ip-protocols bgp-vpn router

Object	Description
max-paths (eBGP multipath)	The eBGP multipath feature allows BGP to submit multiple, equal-cost paths to the IP routing manager. This feature allows load balancing of traffic when equal-cost paths exist. If one path fails, traffic moves to the remaining path(s). MCC supports 128 equal-cost multi-path routes for non-MPLS. See About max-paths (eBGP multipath) (page 6-21). ■ <i>ebgp</i> – Maximum number of EBGp ECMP routes allowed to a destination ■ <i>ibgp</i> – Maximum number of IBGP ECMP routes allowed to a destination

About max-paths (eBGP multipath)

Normally, BGP installs only a single route to a given destination network, even if multiple equal-cost paths exist. BGP Equal-Cost Multi-Path (ECMP) routing allows multiple equal-cost paths to be installed in the routing table when they exist. Traffic is then load-shared over those multiple paths. ECMP potentially offers substantial increases in bandwidth by load-balancing traffic over multiple paths.

Operators can configure an ECMP environment containing all Internal Border Gateway Protocol (IBGP) peers, all (External Border Gateway Protocol) EBGp peers, or a mix of both. Configure the maximum number of EBGp ECMP routes and the maximum number of IBGP ECMP routes allowed to a destination. MCC supports 128 equal-cost multi-path routes for non-MPLS.

Note: BGP's view of equal cost is from the Autonomous System level; it does not consider low-level characteristics such as link speed.

ip-protocols routing-options

Set routing options.

Configure the objects as follows:

Table 6-6 ip-protocols routing-options

Object	Description
geored-rmg-deglitch-interval	RMG failure deglitch interval (seconds). Also see service-construct geo-redundancy-server in the <i>Affirmed Networks Gateway Administration Guide</i> . Also see the <i>Affirmed Networks ePDG Administration Guide</i> .
multi-active-routing-mode	Enable or disable multi-active-routing-mode. See Multi Active Routing Mode in the <i>Affirmed Networks Gateway Administration Guide</i> .

Multi Active Routing Mode

Multi-active routing mode provides distributed routing control as a form of high availability (HA) for routing. Multi-active routing mode provides faster geo-redundancy failover time compared to the BGP graceful-restart routing control failure recovery (default behavior). Initially only BGP is supported.

Configuration is applied concurrently on all CSMs. The routing processes are active concurrently on CSMs for all network-contexts. BGP peering is distributed across all BGP processes. The assignment of BGP peers to CSMs is performed automatically during configuration.

In the event of a CSM failure, the BGP peers that are assigned to the failed CSM will be down, causing a partial loss of routing information. BGP peers will be reassigned to different CSMs if the current assigned CSM is administratively disabled.

Multi-Active routing high-availability (HA) mode of operation provides support to distribute BGP peers to the various admin-enabled CCM/CSMs, so that the MCC is capable of continuing through single CCM/CSM failures. Disable the CSM that is operationally down and MCC re-assigns its hosted peers to other viable CSMs.

In the Multi-Active Routing HA architecture, BGP sessions are distributed to the pool of admin-enabled CCMs / CSMs in the cluster to maximize diversity so that loss of one CSM does not cause an entire network context or BGP transport type (such as IPv6 Single Hop) to fail completely.

MCC invokes the CSM-selection algorithm whenever:

- A peer is added
- A CSM is admin-enabled – some BGP peering sessions are selected for relocation to this new CSM in an effort to continually maximize CSM diversity
- A CSM is admin-disabled and its hosted peers (rendered down as a result) are relocated to other admin-enabled CSMs

ip-protocols log-nsm

Enable NSM logging.

ip-protocols nat deterministic

Save deterministic nat-map to a file corresponding to a given network-context and nat-pool. Execute this command in **exec** mode.

The following command uses the nat-pool name and network-context name as input. After MCC executes this command, it saves a deterministic nat-map in CSV format for a given NAT pool into a file with a name *<nat-pool-name><timestamp>.csv* at location */var/corefiles/diags/*. For more information on nat pools, see *Affirmed Networks Gateway Administration Guide*.

For example:

```
ip-protocols nat deterministic save-map network-context  
<network-context-name> nat-pool <nat-pool-name>
```

The follow is an example of a CSV file.

UE-IP-NC	UE-POOL	UE-IP-SUB-POOL	UE-IP-PREFIX-LEN	UE-IP	NAT-IP-NC	NAT-POOL	NAT-IP-SUB-POOL	NAT-IP-PREFIX-LEN	NAT-IP	NAT-PORT-START	NAT-PORT-END
GI	LOCALPOOL	LOCALSUBPOOL_V4	0	10.53.203.6	GI	NAT-POOL	NIPLOCAL	0	10.53.204.26	1024	1087
GI	LOCALPOOL	LOCALSUBPOOL_V4	0	10.53.203.7	GI	NAT-POOL	NIPLOCAL	0	10.53.204.26	1088	1151
GI	LOCALPOOL	LOCALSUBPOOL_V4	0	10.53.203.8	GI	NAT-POOL	NIPLOCAL	0	10.53.204.26	1152	1215
GI	LOCALPOOL	LOCALSUBPOOL_V4	0	10.53.203.9	GI	NAT-POOL	NIPLOCAL	0	10.53.204.26	1216	1279

ip-protocols ospfv2

Manage OSPFv2 routers. The *Open Shortest Path First* (OSPF) link-state routing protocol enables dynamic routing between the system and other network elements. OSPF simplifies the routing configuration and eliminates the need to configure additional network elements provisioned with the specific IP addresses. The system supports OSPFv2 for IPv4 and OSPFv3 for IPv6 networks. For information on OSPFv3, see [ip-protocols ospfv3 \(page 6-30\)](#).

Configure the objects as follows:

Table 6-7 ip-protocols ospfv2

Object	Description
restart	Restart all OSPF routers gracefully.
ospf-logging	Configure logging for all OSPF routers. Configure the following logging options: ■ log-all – Enable all logging. ■ log-events – Enable event logging for the following events: ABR, ASBR, LSA, NSSA, OS interaction, router, virtual link. ■ log-if-fsm – Enable Interface FSM logging. ■ log-lsa – Enable LSA logging for the following: flooding, generation, installation, maximum age (seconds), refresh. ■ log-nbr-fsm – Enable Neighbor FSM logging. ■ log-nsm – Enable networking services logging for interface and redistribute. ■ log-packet – Enable packet logging for the following packets: database, detailed information, hello packets, LSA acknowledgement, link state requests, link state updates, received packets, sent packets. ■ log-route – Enable route logging for inter-area route calculation, external route calculation, route installation, and SPF calculation.

Table 6-7 ip-protocols ospfv2

Object	Description
restart-response	Configure the OSPF router response to a neighbor restart. The OSPF graceful restart feature enables an OSPF router to stay on the forwarding path while its OSPF software is restarted. This feature helps minimize the negative affects on routing as a result of an unplanned outage or restart. When graceful restart is enabled, the router or helper temporarily maintains routing information in its database. Use the restart-response command to enable OSPF graceful restart (or restart signaling). If a router is not restart-enabled, it cannot enter graceful restart mode and act as a helper. Configure the following restart-response parameters: ■ <i>helper</i> – <i>max-grace-period</i> – Specify the max-grace-period for the helper (between 1 and 1800 seconds). – <i>never</i> – Never act as a graceful-restart helper. Enter the routerid of the neighbor which should never act as helper. – <i>only-reloads</i> – Only act as helper on software reloads. – <i>only-upgrades</i> – Only act as helper on software upgrades. ■ <i>routerid</i> – Router ID of neighbor that never acts as helper.
router	Configure and manage OSPF routers.

OSPFv2 Router Options

Configure and manage ospfv2 routers using the following options:

Table 6-8 ip-protocols OSPFv2 Router Options

Option	Description
area	OSPF area parameters.
auto-cost-reference-bandwidth	Control how OSPF calculates default metrics for the interface. By default, OSPF calculates the OSPF metric for an interface by dividing the reference bandwidth by the interface bandwidth. The default value for the reference bandwidth is 100 Mbps. The auto-cost command is used to differentiate high bandwidth links. For multiple links with high bandwidth, specify a larger reference bandwidth value to differentiate cost on those links.
bfd-all-interfaces	Enable BFD for OSPF neighbors on this interface. Enabling this command enables BFD to be used to detect the reachability of OSPF neighbors and to notify OSPF when that reachability is lost. This mechanism is used to expedite route convergence, rather than relying on OSPF's hello and dead timers to determine reachability.
capability-opaque	Enable Opaque LSA capability. Values: false true (default).
capability-restart	Enable OSPF graceful restart (or restart signaling). If a router is not restart-enabled, it cannot enter graceful restart mode and act as a helper. See restart-response (page 6-25).
compatible-rfc1583	Restore the method used to calculate summary route costs per RFC. Prior to RFC 2328, OSPF was compliant with RFC 1583, which specified method for calculating the metric for summary routes based on the minimum metric of the component paths available. RFC 2328 specifies a method for calculating metrics based on maximum cost. With this change, it is possible that all of the ABRs in an area might not be upgraded to the new code at the same time. The compatible RFC 1583 command addresses this issue and allows the selective disabling of compatibility with RFC 2328. Values: false true (default).
database	Show the OSPF database.
default-metric	Set default metric values for the OSPF routing protocol. A default metric facilitates redistributing routes even with incompatible metrics. If the metrics do not convert, the default metric provides an alternative and enables the redistribution to continue. The default-metric command is used to cause the current routing protocol to use the same metric value for all redistributed routes. Use this command in conjunction with redistribute.
distribute-list-in	Filter networks in routing updates. Use this command when distributing other routing protocols into the OSPF routing table.
distribute-list-out	Filter networks in routing updates. Use this command when distributing other routing protocols out of the OSPF routing table.

Table 6-8 ip-protocols OSPFv2 Router Options

Option	Description
host	Configure a stub host entry belonging to a particular area. Using this command, you can advertise specific host routes in the router-LSA as stub link. Since stub host belongs to the specified router, specifying cost is not important.
interface	Configure OSPF settings for an IP Interface. Set the priority to 0 to guarantee the neighbor adjacency to come with neighbor being the DR.
max-concurrent-dd	Set the limit for the maximum number of Database Descriptors (DD) that can be processed concurrently. This command is useful when a router's performance is affected from simultaneously bringing up several OSPF adjacencies. This command limits the maximum number of DD exchanges that can occur concurrently per OSPF instance, thus allowing for all of the adjacencies to come up.
loopback-ip	Configure OSPF settings for an IP loop.
network-ip	Enable OSPF routing with a specified Area ID (and optionally an instance ID) on interfaces with IP addresses that match the specified network address. OSPF routing can be enabled on a per IPv4 subnet basis. The network address can be defined using the prefix length or a wildcard mask comprised of consecutive 0s as network bits and consecutive 1s as host bits.
ospf-abr-type	Set an OSPF Area Border Router (ABR) type. Specifying the ABR type allows better functioning between different implementations. This command is useful in a multi-vendor environment. The following are the different ABR types: ■ Cisco – This router is considered an ABR if it has more than one area ly attached and one of them is the backbone area. ■ Standard – This router is considered an ABR if it has more than one area ly attached to it. ■ IBM – This router is considered an ABR if it has more than one area ly attached and the backbone area is configured. In this case, the configured backbone need not be ly connected. ■ Shortcut – The Shortcut ABR improves over the Standard ABR behavior by modifying the calculation of inter-area routes. It is allowed to install inter-area routes through non-backbone areas if the non-backbone path is better, thus providing a shortcut through these areas. To prevent routing loops, the inter-area routes are re-advertised only if they are associated with the backbone area.

Table 6-8 ip-protocols OSPFv2 Router Options

Option	Description
passive-interface	Suppress sending hello packets on all interfaces, or on a specified interface. The passive-interface command is used to configure OSPF on simplex Ethernet interfaces. Since the simplex interfaces represent only one network segment between two devices, configure the transmitting interface as a passive interface. This ensures that OSPF does not send hello packets for the transmitting interface. Both the devices can see each other via the hello packet generated for the receiving interface. ■ Using the passive-interface command without the optional parameters puts all interfaces into passive mode. ■ Using the no passive-interface command without the optional parameters removes all interfaces from passive mode.
redistribute	Redistribute routes from other routing protocols, static routes, and kernel routes into an ospf routing table. This command injects routes, learned from other routing protocols, into the OSPF domain to generate AS-external-LSAs. <i>static</i> – IP subpools and IP loopback addresses are advertised by dynamic routing protocols such as OSPF and BGP.
router-id	Specify a router ID for the OSPF process. Configure each router with a unique router-id. In an OSPF router process which has neighbors, a new router-id is used at the next reload or when you start the OSPF manually.

OSPFv2 Interface Options

Configure the following OSPFv2 options for an IP interface:

Table 6-9 ip-protocols OSPFv2 Interface Options

Option	Description
area	A.B.C.D; Area ID in IPv4 address format.
cost	Specify the cost of link-state metric in a router-LSA. The interface cost indicates the overhead required to send packets across a certain interface. This cost is stated in the Router-LSA's link. The cost is inversely proportional to the bandwidth of an interface. By default, the cost of an interface is calculated based on the bandwidth ($10^8 / \text{bandwidth}$); use this command to set the cost manually.
bfd-enable	Enable the BFD option for OSPF neighbors on an interface. This command overrides the router-level bfd-all-interfaces command (see Table 6-8 on page 6-26).

Table 6-9 ip-protocols OSPFv2 Interface Options

Option	Description
dead-interval	Set the interval during which no hello packets are received and after which a neighbor is declared dead. If you configure this command and specify the IP address of the interface and want to remove the configuration, use the no parameter with the specified IP address (no ip ospf dead-interval A.B.C.D). The dead-interval is the amount of time that the router waits to receive an OSPF hello packet from the neighbor before declaring the neighbor down. This value is advertised in the router's hello packets. It must be a multiple of hello-interval and be the same for all routers on a specific network.
display-route-single-line	Configure show OSPF route to display in a single line.
hello-interval	Specify the interval between hello packets. The hello-interval is advertised in the hello packets. Configure the same hello-interval for all routers on a specific network. A shorter hello interval ensures faster detection of topological changes but results in more routing traffic.
mtu-ignore	Configure OSPF so that it does not check the MTU size during DD (Database Description) exchange. By default, during DD exchange process, OSPF checks the MTU size described in DD packets received from the neighbor. If the MTU size does not match the interface MTU, the neighbor adjacency is not established. Using this command makes OSPF ignore this check and allows establishing of adjacency regardless of MTU size in the DD packet.
network	Configure the OSPFv3 network type on an interface.
priority	Set the priority to help determine the OSPF Designated Router (DR) for a network. If two routers attempt to become the DR, the router with the higher router priority becomes the DR. If the router priority is the same for two routers, the router with the higher router ID takes precedence. Only routers with nonzero router priority values are eligible to become the designated or backup designated router. Configure router priority for multiaccess networks only and not for point-to-point networks.
retransmit-interval	Specify the time between link-state advertisement (LSA) retransmissions for adjacencies belonging to the interface. After sending an LSA to a neighbor, the router keeps the LSA until it receives an acknowledgement. In case the router does not receive an acknowledgement during the set time (the retransmit interval value) it retransmits the LSA. Set the retransmission interval value conservatively to avoid needless retransmission. The interval should be greater than the expected round-trip delay between two routers.

Table 6-9 ip-protocols OSPFv2 Interface Options

Option	Description
tag	OSPFv3 process tag.
transmit-delay	Set the estimated time it takes to transmit a link-state-update packet on the interface. The transmit delay value adds a specified time to the age field of an update. If the delay is not added, the time in which the LSA transmits over the link is not considered. This command is especially useful for low speed links. Add transmission and propagation delays when setting the transmit delay value.

ip-protocols ospfv3

Manage OSPFv3 routers. The *Open Shortest Path First* (OSPF) link-state routing protocol enables dynamic routing between the system and other network elements. OSPF simplifies the routing configuration and eliminates the need to configure additional network elements provisioned with the specific IP addresses. The system supports OSPFv2 for IPv4 and OSPFv3 for IPv6 networks. For information on OSPFv2, see [ip-protocols ospfv2 \(page 6-24\)](#).

Configure the objects as follows:

Table 6-10 ip-protocols ospfv3

Object	Description
bfd-all-interfaces	Enable BFD for OSPF neighbors on this interface. Enabling this command enables BFD to be used to detect the reachability of OSPF neighbors and to notify OSPF when that reachability is lost. This mechanism is used to expedite route convergence, rather than relying on OSPF's hello and dead timers to determine reachability.

OSPFv3 Router Options

Configure and manage OSPFv3 routers using the following options:

Table 6-11 ip-protocols OSPFv3 Router Options

Option	Description
area	OSPF area parameters.
auto-cost-reference-bandwidth	Control how OSPF calculates default metrics for the interface. By default, OSPF calculates the OSPF metric for an interface by dividing the reference bandwidth by the interface bandwidth. The default value for the reference bandwidth is 100 Mbps. The auto-cost command is used to differentiate high bandwidth links. For multiple links with high bandwidth, specify a larger reference bandwidth value to differentiate cost on those links.

Table 6-11 ip-protocols OSPFv3 Router Options

Option	Description
capability	Enable router capabilities.
database	Display the OSPFv3 LS database.
db-summary-opt	Enable database summary optimization.
default-metric	Set default metric values for the OSPF routing protocol. A default metric facilitates redistributing routes even with incompatible metrics. If the metrics do not convert, the default metric provides an alternative and enables the redistribution to continue. The default-metric command is used to cause the current routing protocol to use the same metric value for all redistributed routes. Use this command in conjunction with redistribute.
max-concurrent-dd	Set the limit for the maximum number of Database Descriptors (DD) that can be processed concurrently. This command is useful when a router's performance is affected from simultaneously bringing up several OSPF adjacencies. This command limits the maximum number of DD exchanges that can occur concurrently per OSPF instance, thus allowing for all of the adjacencies to come up.
passive-interface	Suppress sending hello packets on all interfaces, or on a specified interface. The passive-interface command is used to configure OSPF on simplex Ethernet interfaces. Since the simplex interfaces represent only one network segment between two devices, configure the transmitting interface as a passive interface. This ensures that OSPF does not send hello packets for the transmitting interface. Both the devices can see each other via the hello packet generated for the receiving interface. ■ Using the passive-interface command without the optional parameters puts all interfaces into passive mode. ■ Using the no passive-interface command without the optional parameters removes all interfaces from passive mode.
redistribute	Redistribute routes from other routing protocols, static routes, and kernel routes into an ospf routing table. This command injects routes, learned from other routing protocols, into the OSPF domain to generate AS-external-LSAs. <i>static</i> – IP subpools and IP loopback addresses are advertised by dynamic routing protocols such as OSPF and BGP.
router-id	Specify a router ID for the OSPF process. Configure each router with a unique router-id. In an OSPF router process which has neighbors, a new router-id is used at the next reload or when you start the OSPF manually.
sptimers	Adjust SPF exponential timers. Enter a value between 0..2147483647 for the minimum (500) and maximum (50000) delay in milliseconds between SPF calculations.

OSPFv3 Interface Options

Configure the following OSPFv3 options for an IP interface:

Table 6-12 ip-protocols OSPFv3 Interface Options

Option	Description
area	A.B.C.D; Area ID in IPv4 address format.
bfd-enable	Enable BFD for OSPFv3 neighbors on this interface.
cost	Specify the cost of link-state metric in a router-LSA. The interface cost indicates the overhead required to send packets across a certain interface. This cost is stated in the Router-LSA's link. The cost is inversely proportional to the bandwidth of an interface. By default, the cost of an interface is calculated based on the bandwidth ($10^8 / \text{bandwidth}$); use this command to set the cost manually.
dead-interval	Set the interval during which no hello packets are received and after which a neighbor is declared dead. If you configure this command and specify the IP address of the interface and want to remove the configuration, use the no parameter with the specified IP address (no ip ospf dead-interval A.B.C.D). The dead-interval is the amount of time that the router waits to receive an OSPF hello packet from the neighbor before declaring the neighbor down. This value is advertised in the router's hello packets. It must be a multiple of hello-interval and be the same for all routers on a specific network.
hello-interval	Specify the interval between hello packets. The hello-interval is advertised in the hello packets. Configure the same hello-interval for all routers on a specific network. A shorter hello interval ensures faster detection of topological changes but results in more routing traffic.
mtu-ignore	Configure OSPF so that it does not check the MTU size during DD (Database Description) exchange. By default, during DD exchange process, OSPF checks the MTU size described in DD packets received from the neighbor. If the MTU size does not match the interface MTU, the neighbor adjacency is not established. Using this command makes OSPF ignore this check and allows establishing of adjacency regardless of MTU size in the DD packet.
network	Configure the OSPFv3 network type on an interface.

Table 6-12 ip-protocols OSPFv3 Interface Options

Option	Description
priority	Set the priority to help determine the OSPF Designated Router (DR) for a network. If two routers attempt to become the DR, the router with the higher router priority becomes the DR. If the router priority is the same for two routers, the router with the higher router ID takes precedence. Only routers with nonzero router priority values are eligible to become the designated or backup designated router. Configure router priority for multiaccess networks only and not for point-to-point networks.
retransmit-interval	Specify the time between link-state advertisement (LSA) retransmissions for adjacencies belonging to the interface. After sending an LSA to a neighbor, the router keeps the LSA until it receives an acknowledgement. In case the router does not receive an acknowledgement during the set time (the retransmit interval value) it retransmits the LSA. Set the retransmission interval value conservatively to avoid needless retransmission. The interval should be greater than the expected round-trip delay between two routers.
transmit-delay	Set the estimated time it takes to transmit a link-state-update packet on the interface. The transmit delay value adds a specified time to the age field of an update. If the delay is not added, the time in which the LSA transmits over the link is not considered. This command is especially useful for low speed links. Add transmission and propagation delays when setting the transmit delay value.

ip-protocols tcp

Manage the kernel TCP configuration. See the *Affirmed Networks Value Added Services Guide*.

Show Commands

MCC supports the following IP protocol show commands. For specific details and output, see the CLI and *Acuitas Performance Statistics*.

- `show ip-protocols bfd instance <network-context>
all-sessions-summary <ifname> <peeraddr>`
- `show ip-protocols bfd instance <network-context> interfaces <entry>`
- `show ip-protocols bfd instance <network-context> process-status
<entry>`
- `show ip-protocols bfd instance <network-context> session-details
<ifname> <peeraddr>`
- `show ip-protocols bfd instance <network-context> sessions <entry>`
- `show ip-protocols bgp router <network-context> ipv4-ucast-af
dampening <action>`
- `show ip-protocols bgp router <network-context> ipv4-ucast-af
dampening-multi-active <action> <slot-id>`
- `show ip-protocols bgp router <network-context> ipv4-ucast-af
route-detail <prefix> <prefix-length [0..32]>`
- `show ip-protocols bgp router <network-context> ipv4-ucast-af
route-detail-multi-active <prefix> <prefix-length [0..32]>
<slot-id>`
- `show ip-protocols bgp router <network-context> ipv4-ucast-af
route-table <slot-id>`
- `show ip-protocols bgp router <network-context> ipv6-ucast-af
dampening <action>`
- `show ip-protocols bgp router <network-context> ipv6-ucast-af
dampening-multi-active <action> <slot-id>`
- `show ip-protocols bgp router <network-context> ipv6-ucast-af
route-detail <prefix> <prefix-length>`
- `show ip-protocols bgp router <network-context> ipv6-ucast-af
route-detail-multi-active <prefix> <prefix-length> <slot-id>`
- `show ip-protocols bgp router <network-context> ipv6-ucast-af
route-table <slot-id>`
- `show ip-protocols bgp router <network-context> neighbor
<peering-name> detail <slot-id>`
- `show ip-protocols bgp router <network-context> summary <slot-id>`
- `show ip-protocols bgp-vpn router <network-context> ipv4-ucast-af
dampening <action>`
- `show ip-protocols bgp-vpn router <network-context> ipv4-ucast-af
dampening-multi-active <action> <slot-id>`

- `show ip-protocols bgp-vpn router <network-context> ipv4-ucast-af route-detail <prefix> <prefix-length [0..32]>`
- `show ip-protocols bgp-vpn router <network-context> ipv4-ucast-af route-detail-multi-active <prefix> <prefix-length [0..32]> <slot-id>`
- `show ip-protocols bgp-vpn router <network-context> ipv4-ucast-af route-table <slot-id>`
`show ip-protocols bgp-vpn router <network-context> ipv6-ucast-af dampening <action>`
- `show ip-protocols bgp-vpn router <network-context> ipv6-ucast-af dampening-multi-active <action> <slot-id>`
- `show ip-protocols bgp-vpn router <network-context> ipv6-ucast-af route-detail <prefix> <prefix-length>`
- `show ip-protocols bgp-vpn router <network-context> ipv6-ucast-af route-detail-multi-active <prefix> <prefix-length> <slot-id>`
- `show ip-protocols bgp-vpn router <network-context> ipv6-ucast-af route-table <slot-id>`
- `show ip-protocols bgp-vpn router <network-context> neighbor <peering-name> detail <slot-id>`
- `show ip-protocols bgp-vpn router <network-context> summary <slot-id>`
- `show ip-protocols bgp-vpn router <network-context> vpnv4-ucast-af route-detail <prefix> <prefix-length [0..32]>`
- `show ip-protocols bgp-vpn router <network-context> vpnv4-ucast-af route-detail-multi-active <prefix> <prefix-length [0..32]> <slot-id>`
- `show ip-protocols bgp-vpn router <network-context> vpnv4-ucast-af route-table <slot-id>`
- `show ip-protocols bgp-vpn router <network-context> vpnv6-ucast-af route-table <slot-id>`
- `show ip-protocols bgp-vpn router <network-context> vrf <network-context> ipv4-ucast-af route-table <slot-id>`
- `show ip-protocols bgp-vpn router <network-context> vrf <network-context> ipv6-ucast-af route-table <slot-id>`
- `show ip-protocols nat deterministic nat-to-ue-binding <nat-network-context> <nat-ip> <nat-port>`
- `show ip-protocols nat deterministic ue-to-nat-binding <ue-network-context> <ue-ip>`
- `show ip-protocols nat module-statistics`
- `show ip-protocols nat <guaranteed|deterministic> nat-pool-statistics <network-context-name> <nat-pool-name>>`

- `show ip-protocols nat <guaranteed|deterministic>
sub-pool-statistics <network-context-name <nat-pool-name
<nat-subpool-name>>`
- `show ip-protocols ospfv2 router <network-context> database
asbr-summary <entry>`
- `show ip-protocols ospfv2 router <network-context> database brief
<entry>`
- `show ip-protocols ospfv2 router <network-context> database external
<entry>`
- `show ip-protocols ospfv2 router <network-context> database network
<entry>`
- `show ip-protocols ospfv2 router <network-context> database nssa
<entry>`
- `show ip-protocols ospfv2 router <network-context> database
opaque-area <entry>`
- `show ip-protocols ospfv2 router <network-context> database
opaque-as <entry>`
- `show ip-protocols ospfv2 router <network-context> database
opaque-link <entry>`
- `show ip-protocols ospfv2 router <network-context> database router
<entry>`
- `show ip-protocols ospfv2 router <network-context> database summary
<entry>`
- `show ip-protocols ospfv2 router <network-context> interface <entry>`
- `show ip-protocols ospfv2 router <network-context> neighbor <entry>`
- `show ip-protocols ospfv2 router <network-context> route <entry>`
- `show ip-protocols ospfv2 router <network-context> summary <entry>`
- `show ip-protocols ospfv3 router <network-context> database external
<entry>`
- `show ip-protocols ospfv3 router <network-context> database grace
<entry>`
- `show ip-protocols ospfv3 router <network-context> database
inter-prefix <entry>`
- `show ip-protocols ospfv3 router <network-context> database
inter-router <entry>`
- `show ip-protocols ospfv3 router <network-context> database
intra-prefix <entry>`
- `show ip-protocols ospfv3 router <network-context> database link
<entry>`

- `show ip-protocols ospfv3 router <network-context> database max-age <entry>`
- `show ip-protocols ospfv3 router <network-context> database network <entry>`
- `show ip-protocols ospfv3 router <network-context> database nssa <entry>`
- `show ip-protocols ospfv3 router <network-context> database router <entry>`
- `show ip-protocols ospfv3 router <network-context> database self-originate <entry>`
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- `show ip-protocols ospfv3 router <network-context> interface <entry>`
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- `show ip-protocols ospfv3 router <network-context> route <entry>`
- `show ip-protocols ospfv3 router <network-context> summary <entry>`
- `show ip-protocols ospfv3 router <network-context> topology <entry>`
- `show ip-protocols ospfv3 router database external`
- `show ip-protocols ospfv3 router database grace`
- `show ip-protocols ospfv3 router database inter-prefix`
- `show ip-protocols ospfv3 router database inter-router`
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- `show ip-protocols ospfv3 router interface`
- `show ip-protocols ospfv3 router neighbor`
- `show ip-protocols ospfv3 router neighbor-detail`
- `show ip-protocols ospfv3 router route`

- `show ip-protocols ospfv3 router summary`
- `show ip-protocols ospfv3 router topology`

Monitoring the System

This section describes how to maintain and monitor the system. Also see the *Affirmed Networks Troubleshooting Guide*.

This section describes how to monitor a subscriber, and display alarms, system uptime, and logged on users. It also describes how to clear a configuration and collect debugging information for the TAC. All of the objects and CLI commands in this section are executed at the exec level.

alarms

Display alarms. If no alarms are currently , no data is displayed.

```
an3000-mcm-slot8cpu1# alarms active 43
```

Possible completions:

```
430363 430364 430367 430374 430375 430376 430379 430380
430381 430382 430385 430386 430387 430388 430391 430392
431209 431424 431478 431485 431563 431572 431573 431578
431579 431582 431770 431784 431872 431903 431911
```

See the *Affirmed Networks SNMP MIB and Alarm Reference Guide* for a list of alarms supported on the system.

arp

Display ARP cache information. This parameter enables Operators to print ARP cache entries. This command has the logic to read and display the ARP cache of each of the CSM30 blades installed in the system. Issue this command to obtain the L-3 (that is, the IP address) to L-2 (that is, the Ethernet MAC Address) address mapping.

```
an3000-mcm-slot7cpu1# show arp 1
```

```
arp
```

```
NETWORK
```

```
CARD CPU CONTEXT IP ADDRESS MAC ADDRESS FLAGS
```

```
-----
-----
```

```
1 1 Gi 10.49.1.1 00:26:f1:26:c1:00 reachable
1 1 Gi 10.49.1.56 00:0c:29:ed:f3:e6 reachable
1 1 Gi 10.49.1.57 00:0c:29:9f:aa:d2 reachable
1 1 Gi 10.49.1.58 00:0c:29:81:65:42 reachable
1 1 Gi 224.0.0.5 01:00:5e:00:00:05 noarp
1 1 Gi 224.0.0.6 01:00:5e:00:00:06 noarp
1 1 Gi fe80::226:f1ff:fe26:c100 00:26:f1:26:c1:00 stale
1 1 Gi ff02::1 33:33:00:00:00:01 noarp
1 1 Gi ff02::16 33:33:00:00:00:16 noarp
1 1 Gi ff02::2 33:33:00:00:00:02 noarp
1 1 Gi ff02::5 33:33:00:00:00:05 noarp
1 1 Gi ff02::6 33:33:00:00:00:06 noarp
```

calea

Display Communications Assistance for Law Enforcement Act (CALEA) status information. For more information, see the *Lawful Intercept Module Specification and Provisioning Guide*.

clear-config

Clear current configuration and restart the system.

Note: This command impacts service and results in a service restart.

configuration

Display the configuration options (in config mode).

an3000-mcm-slot7cpu1(config)# **show configuration**

Possible completions:

alarms	
cluster	Virtual Node management
data-path	MCC Fastpath
extended-storage	Ext-storage parameters
fault-management	Fault Management
geographic-redundancy	Configure Geographic Redundancy parameters
image-management	Image Management parameters
infrastructure	
ip-data-path	MCC ipDataPath
ip-protocols	Manage IP Network Protocols
license	license commands
network-context	Configure network context parameters
networking-profiles	Networking Profiles
performance-statistics-management	View Performance Statistics Management Status
security	SECURITY configuration
service-construct	Service Constructs
services	AN Gateway Services
snmp	Simple Network Management Protocol configuration
user-access	user access commands
vtap	MCC vTap functionality
workflow	Configure workflow parameters
zone	Zone parameters

commit	Display configuration changes
diff	Show a parameter

<code>merge</code>	Contents of merged configuration
<code>rollback</code>	Display rollback changes
<code>running</code>	Contents of running configuration
<code>this</code>	Show a parameter

content-cache evict

Evict content from memory, disk, or both or evict URL.

data-records

Display data record statistics.

detail

Display detailed statistics for each source (that is, tos task) of data records.

- *recordsAttempted* – Data records attempts made
- *recordsFailed* – Number of records for which send to agent failed
- *recordsSucceeded* – Number of records for which send to agent succeeded
- *recordsWriteFailed* – Number of records for which write failed
- *recordsWritten* – Number of records written to disk

For example:

```
an3000-mcm-slot7cpu1# show data-records detail
data-records detail
RECORDS
RECORDS   RECORDS   RECORDS   RECORDS   WRITE
AGENT                                SOURCE                                ATTEMPTED   FAILED   SUCCEEDED   WRITTEN   FAILED
-----
DataRecordAgentMCM_DataRecordAgentMCM_0 HttpProxy_6_1_1_0_HttpProxy_0      268773      0        268773      268773      0
DataRecordAgentMCM_DataRecordAgentMCM_0 HttpProxy_6_1_1_10                273879      0        273879      273879      0
DataRecordAgentMCM_DataRecordAgentMCM_0 HttpProxy_6_1_1_12_HttpProxy_12    268807      0        268807      268807      0
DataRecordAgentMCM_DataRecordAgentMCM_0 HttpProxy_6_1_1_14_HttpProxy_14    272907      0        272907      272907      0
```

statistics

Display data record statistics.

recordsFailed – Total records failed

recordsWritten – Total number of records written to disk

For example:

```
an3000-mcm-slot7cpu1# show data-records statistics data-records
statistics
```

	RECORDS WRITTEN	RECORDS FAILED
AGENT		

DataRecordAgentMCM_DataRecordAgentMCM_0	38862017	0

date

Display the system date and time.

```
an3000-mcm-slot7cpu1# date
Mon Sep 18 11:52:01 EDT 2017
```

extended-storage

Display extended storage progress and current status.

progress

- *append* – Append output text to a file
- *begin* – Begin with the line that matches
- *count* – Count the number of lines in the output
- *exclude* – Exclude lines that match
- *include* – Include lines that match
- *linnum* – Enumerate lines in the output
- *more* – Paginate output
- *nomore* – Suppress pagination
- *save* – Save output text to a file
- *until* – End with the line that matches

status

- *admin-state* – Admin State
- *current-operation* – Current disk operation
- *current-state* – Current disk operation state
- *disk-partition* – Current ext-storage disk partition
- *oper-status* – Operational Status

fault-management

Display a historical list of actions taken by fault-management director.

fault-management clear

Clear all entries.

fault-management throttle-limits

Fault management throttle limits. Set the event-limit. Set the number of events per alarm type forwarded per time period (0 disables throttling).

filecommand

Run limited filesystem commands over restricted directories, move or remove files, and verify the integrity of the copy (using scp). Available Commands:

```
ls [directory]
cp source destination
mv source destination
rm file
md5sum file
scp [options] source destination
touch/more/tail/cat filename
```


Affirmed Networks recommends using the following procedure to directly filecommand scp the file from its external source to /public/content/tls/servercerts/. while logged in as any admin group user.

```

admin connected from 10.1.1.123 using ssh on an3000-mcm-slot7cpu1
an3000-mcm-slot7cpu1# filecommand scp <username>@<hostname or ip>:/<dir>/<filename> /public/content/tls/servercerts/.
scp <username>@<hostname or ip>:/<dir>/<filename> /public/content/tls/servercerts/.
Warning: Permanently added '10.1.1.123' (RSA) to the list of known hosts.
<username>@<hostname or ip>'s password:
<filename> 100% 14 0.0KB/s 00:00
an3000-mcm-slot7cpu1# filecommand ls /public/content/tls/servercerts/.
ls /public/content/tls/servercerts/.
total 4
- 1 14 Apr 18 20:15 <filename>
an3000-mcm-slot7cpu1#

```

Operators can also copy information to the following directory:

```

Images           : /public/images
Logs             : /public/log
Configuration    : /public/config
faultfiles       : /public/faultfiles
TechSupportInfo files : /public/techSupportInfo
Content filters,etc... : /public/content
Home Directory   : /home/users/admin

```

The command output can be subjected to search, saved to a file, matched with a pattern, etc with the following operations:

- *append* – Append output text to a file.
- *begin* – Begin with the line that matches. The begin operation is similar to grep.
- *count* – Count the number of lines in the output.
- *exclude* – Exclude lines that match.
- *include* – Include lines that match.
- *linnum* – Enumerate lines in the output.
- *more* – Paginate output.
- *nomore* – Suppress pagination.
- *save* – Save output text to a file.
- *until* – End with the line that matches.

For example:

```
filecommand cat /var/log/giri | begin hi
```

history

Display a history of recent CLI commands. The default is 100. For example:

```
an3000-mcm-slot7cpu1# show history
```

monitor subscriber

You can use a single command to monitor a single subscriber based on the IMEI, IMSI or MSISDN, and at the same time store traces in the file so that you can observe all related subscriber activities. When you stop capturing the subscriber activities, it is possible to capture the control plane trace in PCAP format. If you are monitoring a subscriber on the ePDG, a log file is also generated for IKE sessions.

Use the following command to initiate/halt a subscriber trace:

```
monitor-subscriber [imei|imsi|msisdn] <id-value> [start|stop]
```

The **start** command generates a protocol logging policy with a log file name consisting of the type of filter field (imei, imsi, or msisdn) and the id-value you entered. For the ePDG, an IKE debug logging policy is also created, and the log file name consists of the type of filter field (imei, imsi, or msisdn), the id-value you entered, and the suffix **ikev2_decryption_table**.

The **stop** command removes the logging policies that were created by the **start** command.

The following conditions apply:

- A maximum of four simultaneous traces can be performed per cluster.
- The number of logging policies created by the **monitor-subscriber** command are counted towards the total MCC limit of 13 logging policies.
- Enabling subscriber tracing is supported for users who possess configuration privileges.
- Subscriber tracing is persistent and survives switchovers and upgrades/downgrades.
- Subscriber tracing applies only to new sessions, not for the existing sessions that were started prior to tracing being enabled.

Configuration Example

For example, the command **monitor-subscriber imsi 204029001100013 start** produces the following configuration:

```
infrastructure logging event-log policy IMSI_204029001100013_
  admin-state enabled
  log-type protocol
  protocol-logging-filter-field imsi
    filter-value 204029001100013
  !
!
sinks filesink
  type file
  file-name IMSI_204029001100013_
  file-format pcapng
```

If you are monitoring a subscriber on the ePDG, the following configuration is also produced for IKE debug logging:

```
infrastructure logging event-log policy
IMSI_204029001100013_ikev2__decryption_table
  admin-state enabled
  log-type debug
  filter-field component
    filter-value SecPolicyMgrIKEv2Key
    exclude false
  !
!
filter-field imsi
  filter-value 204029001100013
  exclude false
  !
!
sinks filesink
  type file
  file-name IMSI_204029001100013__ikev2_decryption_table
  file-format ascii
  display-field displaystring
    field-type raw
  !
```

save-tech-support-info

Collect detailed debugging information about the system (such as performance statistics management files, fault files, and logs), packages the file in one tarball and zips the tarball. This command collects information for both MCMs and creates two zipped tarballs. To display the optional arguments, use **-help** or **?**.

```
an3000-mcm-slot7cpu1# save-tech-support-info -h
```

```
Usage: an3000Dump.py [options]
```

Options:

```
-h, --help show this help message and exit
```

```
--runlocalonly collect the information only on local node and not on peer
```

```
--targetdir=DIR path to directory which will contain the info and then will be tarred-zipped
```

```
--nocores do not collect core files
```

```
--nofaults do not collect fault files
```

```
--nologs do not collect log files
```

```
--noroutinginfo
```

```
--numdays=NUMDAYS only collect files that have been modified in the last NUMDAYS
```

```
--numhours=NUMHOURS only collect files that have been modified in the last NUMHOURS
```

```
--datarecords collect datarecord files
```

```
--psm collect psm files
```

Use the following example to set specific parameters:

```
an3000-mcm-slot7cpu1# save-tech-support-info
```

```
--numhours=M
```

system

Display the system uptime in number of seconds. For example:

```
an3000-mcm-slot7cpu1# show system node uptime
```

SYSTEM	NODE	CPU	
ID	NUMBER	NUMBER	UPTIME
93	1	1	349804
93	5	1	349804
93	7	1	349921
93	8	1	0

who

Display currently logged on users. An asterisk appears next to the user who executed the command.

```
an3000-mcm-slot7cpu1# who
```

Possible completions:

```
| <cr>
```

```
an3000-mcm-slot7cpu1# who
```

Session	User	Context	From	Proto	Date	Mode
*19771	admin	cli	169.254.93.71	ssh	12:14:01	operational
991	admin	cli	169.254.93.71	ssh	2017-09-14	operational
26	emsadmin	netconf	10.6.1.149	ssh	2017-09-14	operational
21	emsadmin	netconf	10.32.1.147	ssh	2017-09-14	operational

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